

Infectious Diseases

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GENERAL PRINCIPLES OF MICROBIAL PATHOGENESIS

Despite the availability of effective vaccines and antibiotics, infectious diseases remain a major health problem throughout the world. In the United States and other high-resource countries, infectious diseases are particularly important causes of death among older adults and in people who are immunocompromised or who have debilitating chronic diseases. In low-resource nations, inadequate access to medical care and malnutrition contribute to a heavy burden of infectious diseases. In these regions of the world, 6 of the 10 leading causes of death are infectious diseases. Tragically, most of these deaths occur in children.

Microbial Entry and Dissemination

Most infectious diseases are caused by pathogenic organisms that exhibit a wide range of virulence. These organisms are acquired from a variety of sources, including people, animals, insect vectors, and the environment. We will start our review of infectious disease at the beginning of the process, the establishment of a beachhead in the host, and then discuss dissemination and transmission of infection, before turning to specific infections.

Routes of Entry of Microbes

Microbes can enter the host by breaching epithelial surfaces, or through inhalation, ingestion, or sexual transmission (Table 8.1). In general, respiratory, gastrointestinal, and genitourinary tract infections in otherwise healthy persons are caused by virulent microorganisms with the ability to adhere to or penetrate intact mucosal epithelium. By contrast, skin infections are mainly caused by organisms that enter the skin through injuries.

Skin. Most skin infections are initiated by injury of the epidermis, such as trauma, burns or medical interventions. The intact keratinized epidermis protects against infection by serving as a mechanical barrier, having a low pH, and by producing antimicrobial fatty acids and defensins, small peptides that are toxic to bacteria. Disruption of the epithelial barrier is the most frequent predisposing factor for most skin infections. Some pathogens penetrate the skin via an insect (*vector*) or animal bite. *Schistosoma* larvae traverse unbroken skin by releasing enzymes that dissolve the adhesive proteins that hold keratinocytes together.

Gastrointestinal Tract. Gastrointestinal tract infections may occur when local defenses are circumvented by a pathogen, or when they

TABLE 8.1 Routes of Microbial Infection

Site	Major Local Defense(s)	Basis for Failure of Local Defense	Pathogens (Examples)
Skin	Epidermal barrier	Mechanical defects (punctures, burns, ulcers) Needle sticks Arthropod and animal bites Direct penetration	<i>Staphylococcus aureus</i> , <i>Pseudomonas aeruginosa</i> Human immunodeficiency virus, hepatitis B and C viruses Yellow fever, plague, Lyme disease, malaria, rabies <i>Schistosoma</i> spp.
Gastrointestinal tract	Epithelial barrier Acidic secretions Peristalsis Bile and pancreatic enzymes Normal intestinal microbiota	Attachment and local proliferation of microbes Attachment and local invasion of microbes Uptake through M cells Acid-resistant cysts and eggs Obstruction, ileus, postsurgical adhesions Resistant microbial external coats Broad-spectrum antibiotic use	<i>Vibrio cholerae</i> , <i>Giardia duodenalis</i> <i>Shigella</i> spp., <i>Salmonella</i> spp., <i>Campylobacter</i> spp. Poliovirus, <i>Shigella</i> spp., <i>Salmonella</i> spp. Many protozoa and helminths Mixed aerobic and anaerobic bacteria (<i>Escherichia coli</i> , <i>Bacteroides</i> spp.) Hepatitis A, rotavirus, norovirus <i>Clostridioides difficile</i>
Respiratory tract	Mucociliary clearance Resident alveolar macrophages	Attachment and local proliferation of microbes Ciliary paralysis by toxins Resistance to killing by phagocytes	Influenza viruses <i>Haemophilus influenzae</i> , <i>Mycoplasma pneumoniae</i> , <i>Bordetella pertussis</i> <i>Mycobacterium tuberculosis</i>
Urogenital tract	Urination Normal vaginal microbiota Intact epidermal/epithelial barrier	Obstruction, microbial attachment, and local proliferation Antibiotic use Microbial attachment and local proliferation Direct infection, local invasion Local trauma	<i>Escherichia coli</i> <i>Candida albicans</i> <i>Neisseria gonorrhoeae</i> Herpes viruses, syphilis Various sexually transmitted infections (e.g., human papillomavirus)

are so weakened that even normal flora produce disease. Most gastrointestinal pathogens are transmitted by food or drink contaminated with fecal material; when hygiene fails, diarrheal disease becomes rampant. The gastrointestinal tract has several local defenses, including gastric acid, superficial mucus, secretory immunoglobulin A (IgA) and antimicrobial defensins. Of these, acidic gastric secretions are particularly important because they are highly effective at killing certain organisms. Neutralizing the stomach acid of healthy volunteers increased the infectivity of *Vibrio cholerae* by 10,000-fold. Peristalsis can clear organisms, preventing their local overgrowth. The gut microbiota competitively inhibits colonization and overgrowth by potential pathogens, such as *Clostridioides difficile*. Many common gastrointestinal pathogens are resistant to local defenses. For example, norovirus is resistant to acid, bile, and pancreatic enzymes due, in part, to the lack of a lipid envelope. Intestinal protozoa and helminths transmitted as cysts or eggs, respectively, have acid-resistant outer coats.

Respiratory Tract. Microorganisms are inhaled daily, mainly in dust or aerosol particles. Microorganisms in large particles are trapped in the mucociliary layer of the upper respiratory tract and transported by ciliary action to the back of the throat, where they are swallowed. Particles smaller than 5 microns are carried into the alveoli, where they are phagocytosed by leukocytes.

The microorganisms that infect the healthy respiratory tract evade local defenses through several different mechanisms. Some respiratory viruses, such as influenza, attach to, enter, and replicate within epithelial cells in the respiratory tract. Certain bacterial respiratory pathogens, including *Bordetella pertussis*, release toxins that enhance their ability to establish an infection by impairing ciliary activity. Another important mechanism of establishing respiratory infection is resistance to killing following phagocytosis. *Mycobacterium tuberculosis* gains a foothold in alveoli by surviving within the phagolysosomes of macrophages.

Other organisms establish disease when local or systemic defenses are impaired. Defective respiratory mucociliary clearance caused by influenza, mechanical ventilation, smoking, or cystic fibrosis sets the stage for bacterial infections. Many other infectious agents, such as *Pneumocystis jirovecii* and *Aspergillus* spp., cause respiratory infections primarily in the setting of systemic immunodeficiency.

Urogenital Tract. Urine normally contains small numbers of low-virulence bacteria. The urinary tract is protected from infection by regular emptying during micturition. Urinary tract pathogens (e.g., *Escherichia coli*) almost always gain access via the urethra and must be able to adhere to urothelium to avoid being washed away. **Obstruction of urinary flow or reflux of urine is a major factor in susceptibility to urinary tract infections.** Females have more than 10 times as many urinary tract infections as males because of the shorter urethra than in males, making females more susceptible to entry of bacteria from the rectum.

From puberty until menopause, the vagina is protected from pathogens by lactobacilli, which produce lactic acid, producing a low pH environment that suppresses the growth of pathogens.

Spread and Dissemination of Microbes Within the Body

Although some disease-causing microorganisms remain localized to the initial site of infection, others have the capacity to invade tissues and spread to distant sites via the lymphatics, the blood, or the nerves (Fig. 8.1). The most common and efficient mode of microbial dissemination is through the bloodstream, by which the organism can reach all organs. As discussed in Chapter 4, disseminated pathogens may produce

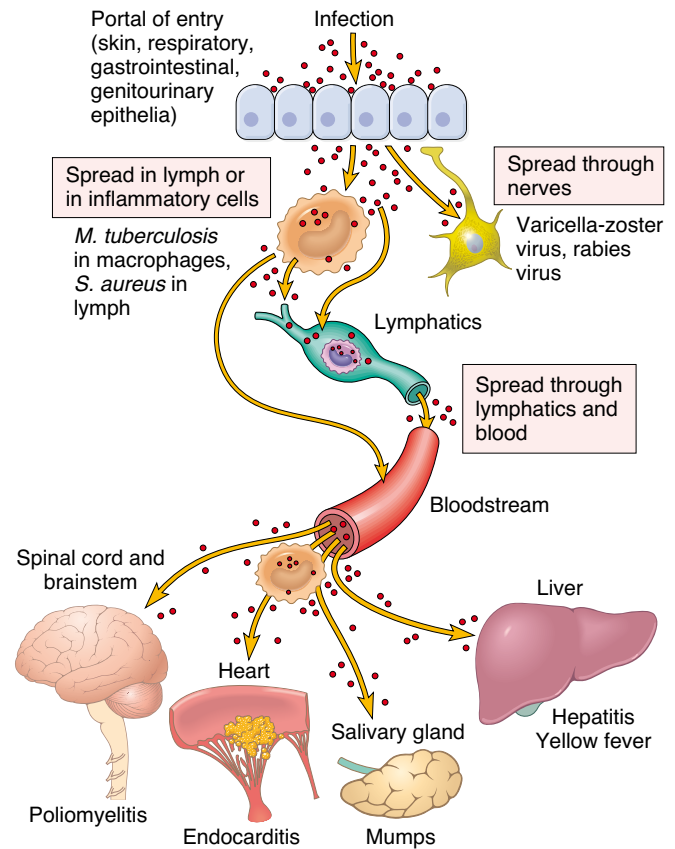


Fig. 8.1 Routes of entry and dissemination of microbes. To enter the body, microbes penetrate the epithelial or mucosal barriers. Infection may remain localized at the site of entry or spread to other sites in the body. Most common microbes (selected examples are shown) spread through the lymphatics or bloodstream (either freely or within inflammatory cells). However, certain viruses and bacterial toxins may also travel through nerves. (Modified from Mims CA. *The Pathogenesis of Infectious Disease*. 4th ed. Academic Press; 1996.)

sepsis that manifests as fever, low blood pressure, and coagulopathies and may progress to organ failure and death. In other instances, the major signs of spread of the infection are related to seeding of tissues by pathogens. Other routes of spread include contiguous spread by pathogens which secrete enzymes to break down tissues, lymphatic spread to the blood, and spread to the central nervous system (CNS) via axonal transport.

Release From the Body and Transmission of Microbes

Microbes use a variety of exit strategies to ensure their transmission from one host to the next. Some pathogens are released for only brief periods of time or periodically during disease flares, but others are shed for long periods by asymptomatic hosts.

Most pathogens are transmitted from person-to-person by respiratory, fecal-oral, or sexual routes, or by vertical transmission from mother to fetus or newborn.

- Viruses and bacteria spread by the respiratory route are aerosolized in droplets and released into the air. Some respiratory pathogens, including influenza viruses, are spread in droplets that travel no more than 3 feet from the source, but others, including *M. tuberculosis*, spread in particles that can travel longer distances.
- Most enteric pathogens are spread by the fecal-oral route, that is, by ingestion of stool-contaminated water or food. Water-borne

pathogens involved in epidemic outbreaks that are spread in this fashion include hepatitis A and E viruses, poliovirus, rotavirus, *Vibrio cholera*, *Shigella* spp., and *Salmonella enterica*. Some parasitic helminths shed eggs in stool that subsequently hatch as larvae that are capable of penetrating the skin of the next host.

- Sexual transmission entails intimate or mucosal contact and is responsible for spread of a wide variety of pathogens, including viruses (e.g., herpes simplex virus [HSV], human immunodeficiency virus [HIV], human papillomavirus [HPV]); bacteria (*Treponema pallidum*, *Neisseria gonorrhoeae*); protozoa (*Trichomonas vaginalis*); and arthropods (*Phthirus pubis*, lice).
- Vertical transmission of infectious agents from mother to fetus or newborn child is a common mode of transmission of pathogens and may occur in several ways.
 - Placental-fetal transmission is most likely to occur when the mother becomes infected with a pathogen during pregnancy. Some resulting infections, such as rubella, interfere with fetal development, with the degree and type of damage depending on the age of the fetus at the time of infection.
 - Transmission during birth is caused by infection during passage through the birth canal. Examples include gonococcal and chlamydial conjunctivitis.
 - Postnatal transmission in maternal milk can transmit cytomegalovirus (CMV), human immunodeficiency virus (HIV), and hepatitis B virus (HBV).

There are several additional routes of transmission. Saliva is responsible for transmitting viruses that replicate in the salivary glands or the oropharynx, including Epstein-Barr virus (EBV). Some protozoa are spread through blood meals taken by arthropod vectors. *Zoonotic infections* are those transmitted from animals to humans, either by direct contact (including animal bites), use of animal products, or via a vector.

KEY CONCEPTS

General Principles of Microbial Pathogenesis

- Transmission of infections can occur by contact (direct and indirect), the respiratory route, the fecal-oral route, sexual transmission, vertical transmission, or insect/arthropod vectors.
- A pathogen can establish infection if it possesses virulence factors that overcome normal host defenses or if the host defenses are compromised.
- Host defenses against infection include:
 - Skin: tough keratinized barrier, low pH, fatty acids
 - Respiratory system: alveolar macrophages, mucociliary clearance by bronchial epithelium, IgA
 - GI system: acidic gastric pH, viscous mucus, pancreatic enzymes, bile, defensins, IgA, and normal microbiota
 - Urogenital tract: repeated flushing and acidic environment created by commensal microbiota
- Pathogens can proliferate locally, at the site of initial infection, or spread to other sites by direct extension (invasion) or by transport in the lymphatics, the blood, or nerves.

Microbiome

The human microbiome is defined as the vast number of microorganisms that coexist with us, residing in or on our bodies, including bacteria, fungi, viruses, and protozoa. Within the human microbiome, there are many subsets such as skin, gut, oral, lung, and vaginal microbiomes. Even within a categorized microbiome, such as the oral

microbiome, there are niche-specific microbiomes such as those on teeth, gums, saliva, tongue, buccal mucosa, throat, and tonsils. There is evidence that the specific microbiome in one body site can affect distant organ function, leading to concepts of the gut-lung, gut-skin, or gut-brain axes. Microorganisms synthesize essential vitamins and amino acids needed by the host and modulate host metabolism and immunity.

Human physiology is intricately intertwined with microorganism biological pathways. There is a wealth of associations between the microbiome and states of disease or health. However, it has been very challenging to determine if dysbiosis, or alteration in the composition of the microbiome, is the cause or consequence of an illness. Because microbial communities are extremely complex, including thousands of taxa, the specific species or combination of species responsible for a specific effect is most often not known.

Many diseases have been associated with an imbalance of the gut microbiome, including atopic dermatitis, depression, obesity, Parkinson disease, autism, and diabetes. The associations are not necessarily proposed as a primary factor in the development of the disorder, but in many cases at least as a potential modulator of the established condition. Although much of the association data are at the taxonomic level of family/order/class/phylum, in some cases, specific species have been implicated in disease: *Ruminococcus gnavus* is associated with inflammatory bowel disease flares, *Prevotella copri* is correlated with rheumatoid arthritis, and *Akkermansia muciniphila* degrades mucin and affects gut permeability. However, how the specific microbes affect disease pathogenesis is unknown in most cases.

In addition to culture-based methods, culture-free sequencing (now even at the single cell level) and mass spectrometry-based metabolomics are valuable tools for study of the microbiome but require extensive bioinformatic analyses. As many gut microbes are challenging to culture or are nonculturable, these advanced methods provide a much more comprehensive picture of the microbiome.

Therapies related to the microbiome include diet intervention and prebiotics/probiotics, and fecal microbiota transplantation (FMT), which has proven efficacious for treatment of recurrent *C. difficile* infection. In addition to FMT, fecal microbiota therapy for recurrent *C. difficile* infection includes an FDA-approved oral pill containing bacterial spores purified from donor stool. FMT has been evaluated for treatment of multiple other chronic conditions including inflammatory bowel disease, irritable bowel syndrome, metabolic syndrome, and neuropsychiatric disorders, with mixed results. Microbiome therapies and under exploration include bioengineered commensals, genetically engineered bacteriophages that deliver CRISPRs directed against microbial genes, and administration of bacterial metabolites.

Host-Pathogen Interactions

The outcome of infection is determined by the virulence of the microbe and the nature of the host immune response, which may eliminate the infection or, in some cases, exacerbate or cause tissue damage. The host has a large and complex armamentarium of defenses against pathogens, including physical barriers and components of the innate and adaptive immune systems, which were discussed in Chapter 6. Microbes undergo continuous evolution to combat host defenses. Here we discuss some specific features of microbes and the host response that are important determinants of the outcome of infections.

Immune Evasion by Microbes

Most pathogenic microbes have developed strategies to evade host defenses (Fig. 8.2).

- *Antigenic variation.* Antibodies against microbial antigens block microbial adhesion and uptake into cells, act as opsonins to facilitate

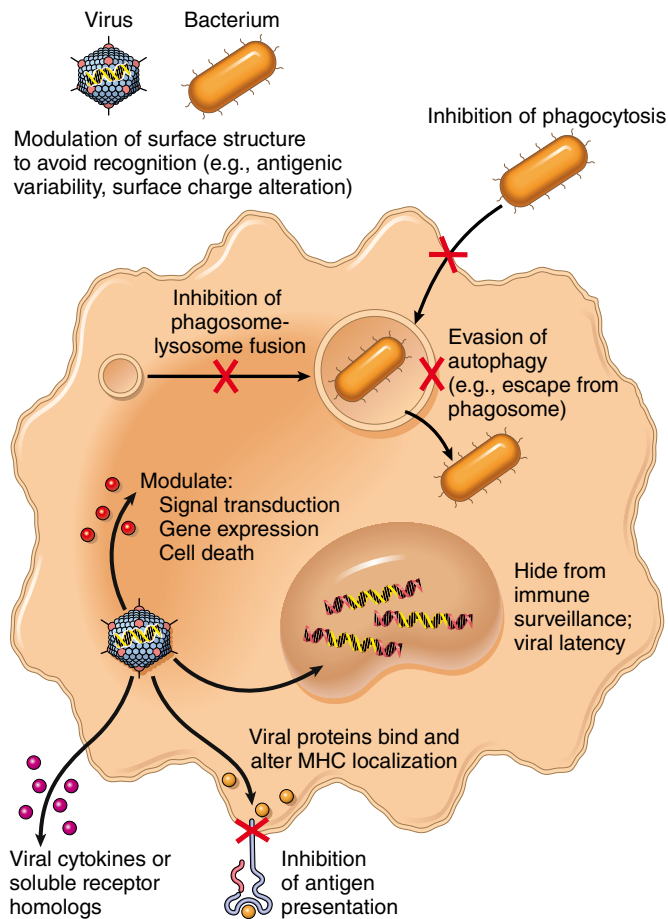


Fig. 8.2 An overview of mechanisms used by viral and bacterial pathogens to evade innate and adaptive immunity. (Modified with permission from Finlay B, McFadden G. Anti-immunology: evasion of the host immune system by bacterial and viral pathogens. *Cell*. 2006;124:767-782.)

phagocytosis, and fix complement. To escape recognition, some microbes have strategies that allow them to “change their coats” by expressing different surface antigens (*Borrelia* spp. and trypanosomes). Influenza viruses have a segmented RNA genome that allows for frequent recombination events, permitting antigenic drift (changes in antibody-binding sites of major viral envelope glycoproteins) and antigenic shift (reassortment of genomes between two viral strains to create a new strain). Other microbes generate numerous genetic variants through mutation (including SARS-CoV-2 and *Streptococcus pneumoniae*) (Table 8.2).

- **Resistance to antimicrobial peptides.** Pathogens such as *Shigella* spp., *Staphylococcus aureus*, and *Candida* spp. use several strategies to avoid killing by cationic antimicrobial peptides. These include changes in net surface charge and membrane hydrophobicity, which prevent antimicrobial peptide insertion and pore formation; secretion of proteins that inactivate or degrade the peptides; and pumps that export the peptides.
- **Resistance to killing by phagocytes.** The carbohydrate capsule on the surface of many bacteria (*S. pneumoniae*, *Neisseria meningitidis*, *Haemophilus influenzae*) prevents phagocytosis of the organisms by neutrophils. *S. aureus* expresses protein A, which binds the Fc portion of antibodies and inhibits phagocytosis by competitively reducing binding of the antibodies to phagocyte Fc receptors. Some pathogens are resistant to intracellular killing in phagocytes, including

TABLE 8.2 Mechanisms of Antigenic Variation

Type	Example	Disease
High mutation rate	Human immunodeficiency virus	AIDS
	Severe acute respiratory syndrome virus 2	COVID-19
	Influenza virus	Influenza
Genetic reassortment	Influenza virus	Influenza
	Rotavirus	Diarrhea
Genetic rearrangement (e.g., gene recombination, gene conversion, site-specific inversion)	<i>Borrelia burgdorferi</i>	Lyme disease
	<i>Neisseria gonorrhoeae</i>	Gonorrhea
	<i>Trypanosoma brucei</i>	African sleeping sickness
	<i>Plasmodium falciparum</i>	Malaria
Large diversity of serotypes	Rhinoviruses	Colds
	<i>Streptococcus pneumoniae</i>	Pneumonia and meningitis

mycobacteria (which inhibit phagosome-lysosome fusion), *Listeria monocytogenes* (which disrupts the phagosome membrane and escapes onto the cytosol), *Cryptococcus neoformans*, and certain protozoa (e.g., *Leishmania* spp., *Trypanosoma* spp., *Toxoplasma gondii*).

- **Evasion of apoptosis and manipulation of host cell metabolism.** Some viruses produce proteins that interfere with apoptosis of the host cell, buying time to replicate, enter latency, or even transform infected cells. Microbes that replicate intracellularly (viruses, some bacteria, fungi, and protozoa) also express factors that modulate autophagy, thus evading degradation.
- **Resistance to cytokine-, chemokine- and complement-mediated host defense.** Some viruses interfere with interferon (IFN) function by producing a variety of different factors: soluble homologues of type I and type II IFN receptors that function as “decoys” to inhibit the actions of secreted IFNs; inhibitors of the JAK/STAT cytokine receptor signaling pathway; and proteins that inactivate or inhibit double-stranded RNA-dependent protein kinase (protein kinase R [PKR]), through which IFNs inhibit viral replication.
- **Evasion of recognition by CD8+ cytotoxic T lymphocytes (CTLs) and CD4+ helper T cells.** T cells recognize microbial antigens presented by major histocompatibility complex (MHC) molecules, class I for CTLs and class II for CD4+ cells (Chapter 6). Several DNA viruses (e.g., HSV, CMV, EBV) bind to or alter localization of MHC class I proteins, impairing peptide presentation to CD8+ T cells. Herpesviruses also can target MHC class II molecules for degradation, impairing antigen presentation to CD4+ T-helper cells.
- **Immunoregulatory mechanisms to downregulate antimicrobial T cell responses.** Loss of T cell potency over time, termed *T-cell exhaustion*, is a feature of chronic infections by HIV, HCV, and HBV. Programmed cell death protein 1 (PD-1) is an immune checkpoint cell-surface receptor, and the PD-1 pathway, which normally functions to maintain T-cell tolerance to self-antigens, is a mediator of T-cell exhaustion during chronic viral infection.
- Another way of avoiding the immune system is to “lie low” by establishing a state of *latent infection* in which few, if any, viral genes are expressed, until later reactivation. Examples include latent infections of neurons by HSV and varicella-zoster virus, and of B lymphocytes by EBV.
- Pathogens can infect immune cells and interfere with their function (e.g., HIV, which infects and destroys CD4+ T cells).

KEY CONCEPTS

Immune Evasion by Microbes

After bypassing host tissue barriers, infectious microorganisms must also evade host innate and adaptive immunity to successfully proliferate and be transmitted to the next host. Strategies include:

- Antigenic variation
- Inactivating antibodies or complement
- Resisting phagocytosis (e.g., by producing a capsule)
- Suppressing the host adaptive immune response (e.g., by interfering with cytokines or inhibiting MHC expression and antigen presentation)
- Establishing latency, during which viruses survive in a silent state in infected cells
- Infecting and disabling or killing immune cells

Injurious Effects of Host Immunity

The host immune response to microbes can be a major cause of tissue injury. The granulomatous inflammatory reaction to *M. tuberculosis* sequesters the bacilli and prevents their spread, but it also produces tissue damage and fibrosis. Similarly, damage to hepatocytes following HBV and HCV infection is mainly because host T cells and, possibly, natural killer (NK) cells kill infected hepatocytes in an attempt to clear the virus. Antibodies produced against the streptococcal M protein can cross-react with cardiac proteins and damage the heart, leading to rheumatic heart disease. Poststreptococcal glomerulonephritis is caused by immune complexes formed between anti-streptococcal antibodies and streptococcal antigens that deposit in the renal glomeruli, producing inflammation. A cycle of inflammation and epithelial injury contributes to the pathogenesis of inflammatory bowel disease, with microbes playing a role (Chapter 17). Viruses (HBV, HCV) and bacteria (*H. pylori*) that are not known to carry or activate oncogenes are associated with cancers, presumably because these microbes trigger chronic inflammation, which provides fertile ground for the development of cancer (Chapter 7). Cytokine-mediated systemic inflammation contributes to some severe cases of COVID-19 and other viral infections.

Infections in People With Immunodeficiencies

Inherited or acquired defects in innate and adaptive immunity often impair the immune system, rendering the affected individual susceptible to infections (Chapter 6). Worldwide, the most devastating immunodeficiency is that caused by infection with HIV, the cause of AIDS. Other causes of acquired immunodeficiencies include infiltrative processes that suppress bone marrow function (such as leukemia), immunosuppressive drugs used to treat patients with autoimmune diseases and organ transplant recipients, certain drugs used to treat cancer, and hematopoietic stem cell transplantation. Decline of immune responses can result in reactivation of latent infection (e.g., herpesviruses and *M. tuberculosis*). Age-related decline in immune function may increase infections in the elderly. Nonimmune diseases or injuries also increase susceptibility to infection. Examples include *Pseudomonas aeruginosa* and *Burkholderia cepacia* in cystic fibrosis due to a defective transmembrane conductance regulator; *S. pneumoniae* in people with sickle cell disease due to loss of splenic function; and *P. aeruginosa* in burns due to barrier disruption. Finally, malnutrition can impair immune defenses.

In addition to common immunodeficiencies, rare inherited (primary) immunodeficiency diseases illuminate important aspects

of specific components of host defense as well as the unique vulnerabilities of certain pathogens.

- *Antibody deficiencies*, as seen in patients with X-linked agammaglobulinemia, lead to increased susceptibility to infections by extracellular bacteria, including *S. pneumoniae*, *H. influenzae*, and *S. aureus*, as well as a few viruses (rotavirus and enteroviruses).
- *Complement defects* involving the early components of the complement cascade lead to susceptibility to infections by encapsulated bacteria, such as *S. pneumoniae*, whereas deficiencies of the late membrane attack complex components (C5 to C9) are associated with infections with *Neisseria* spp.
- *Defects in neutrophil function*, as in chronic granulomatous disease, lead to increased susceptibility to infections with *S. aureus*, some gram-negative bacteria, and fungi.
- *Defects in Toll-like receptor (TLR) signaling pathways* have varied effects. Mutations in MyD88 or IRAK4, which are signaling proteins downstream of several TLRs, predispose to pyogenic bacterial infections (*S. pneumoniae*), and impaired TLR3 responses are associated with childhood HSV encephalitis.
- *T-cell defects* lead to susceptibility to intracellular pathogens, particularly viruses and some parasites. Inherited mutations that impair the generation of T-helper 1 (Th1) cells (such as mutations in IL-12 or IFN γ receptors, or the transcription factor STAT1) are associated with atypical mycobacterial infections, discussed later. By contrast, defects that impair the generation of Th17 cells (mutations in *STAT3*) are associated with chronic mucocutaneous candidiasis.
- *Defects in type I interferon immunity*, such as rare inborn errors of TLR3- and TLR7-dependent immunity, or the production of autoantibodies to type I IFN, are associated with more severe illness during SARS-CoV-2 infection.

Host Damage by Microbes

Infectious agents establish infection and damage tissues by a few mechanisms:

- They can contact or enter host cells and cause *cell death* directly, or cause changes in cellular metabolism and proliferation that can eventually lead to *transformation*.
- They may release *toxins* that kill cells at a distance, release *enzymes* that degrade tissue components, or damage blood vessels and cause *ischemic necrosis*.
- They can induce host *immune responses* that, though directed against the invader, cause additional tissue damage.

Mechanisms of Viral Injury

Viruses can directly damage host cells by entering them and replicating at the cell's expense. The predilection for viruses to infect certain cells and not others is called *tropism*. A major determinant of tissue tropism is the presence of molecules on host cells that viruses use as entry receptors. Viruses bind to proteins found on the surface of host cells that normally function as signaling receptors or adhesion molecules. This is presumably one way in which viruses evolve to infect, survive within cells, and spread. For example, the HIV glycoprotein gp120 binds to CD4 and to the chemokine receptors CXCR4 (mainly on T cells) and CCR5 (mainly on macrophages) (Chapter 6), and EBV binds to complement receptor 2 (also known as CR2 or CD21) on B cells. Another example is the SARS-CoV-2 spike protein that binds to the host ACE2 receptor. Other tropisms are explained by cell-lineage-specific factors. JC virus infection, which causes leukoencephalopathy (Chapter 28), is restricted to oligodendroglial cells in the

CNS because JC viral genes require glial-specific host transcription factors for their expression.

Physical barriers can contribute to tissue tropism. For example, enteroviruses replicate in the intestine in part because they can resist inactivation by acids, bile, and digestive enzymes. Rhinoviruses infect host cells within the upper respiratory tract because they replicate optimally at the lower temperatures found in sites exposed to the ambient atmosphere.

Once viruses are inside host cells, they can damage or kill the cells by several mechanisms (Fig. 8.3):

- **Direct cytopathic effects.** Some viruses kill cells by preventing synthesis of critical host macromolecules (e.g., host cell DNA, RNA, or proteins), or by producing degradative enzymes and toxic proteins. Poliovirus inactivates cap-binding protein, which is essential for translation of host cell mRNAs but leaves translation of poliovirus mRNAs unaffected. Large amounts of viral proteins are synthesized in infected cells, including unfolded or misfolded proteins that activate the endoplasmic reticulum (ER) stress response; this, too, activates pro-apoptotic pathways.
- **Antiviral immune responses.** Host lymphocytes can recognize and destroy virus-infected cells, but as mentioned earlier, CTLs can also be responsible for tissue injury.
- **Transformation of infected cells.** Oncogenic viruses can stimulate cell growth and survival by a variety of mechanisms, including expression of virus-encoded oncogenes and expression of viral proteins that inactivate key tumor suppressors (Chapter 7).

Mechanisms of Bacterial Injury

Bacterial Virulence. Bacterial damage to host tissues depends on the ability of the bacteria to adhere to host cells, to invade cells and tissues, and to deliver toxins (Fig. 8.4). Pathogenic bacteria have *virulence genes* that encode proteins responsible for these properties. An example of the importance of such genes can be found in the various strains of *S. enterica*. All *S. enterica* strains that infect humans are so closely related that they form a single species, meaning that they share many housekeeping (core) genes. Differences in a relatively small number of virulence genes determine whether an isolate of *S. enterica* causes life-threatening typhoid fever or self-limited enteritis. Virulence genes are frequently found grouped together on bacterial chromosomes or plasmids in clusters called *pathogenicity islands*.

Mobile genetic elements such as plasmids and bacteriophages can transmit functionally important genes to bacteria, including genes that influence pathogenicity and drug resistance. Genes for toxins are sometimes found in plasmids but are more often found in the genomes of lysogenic bacteriophages, including the genes that encode the toxins responsible for the pathogenesis of cholera, diphtheria, and botulism. Genes for acquired antibiotic resistance traits are more frequently found on plasmids, which can spread not only within bacterial species but also among more distantly related organisms. For example, a plasmid with genes for vancomycin resistance can spread among *Enterococcus* spp. and also to more distantly related (and virulent) *S. aureus*.

Communities of bacteria often form *biofilms* in which the organisms live within a viscous layer of extracellular polymeric substances, including polysaccharides, proteins and DNA that adhere to host tissues or devices such as intravascular catheters and artificial joints. In addition to enhancing adherence to host tissues, biofilms increase the virulence of bacteria by protecting the microbes from immune effector mechanisms and increasing their resistance to antimicrobial drugs. Biofilm formation seems to be particularly

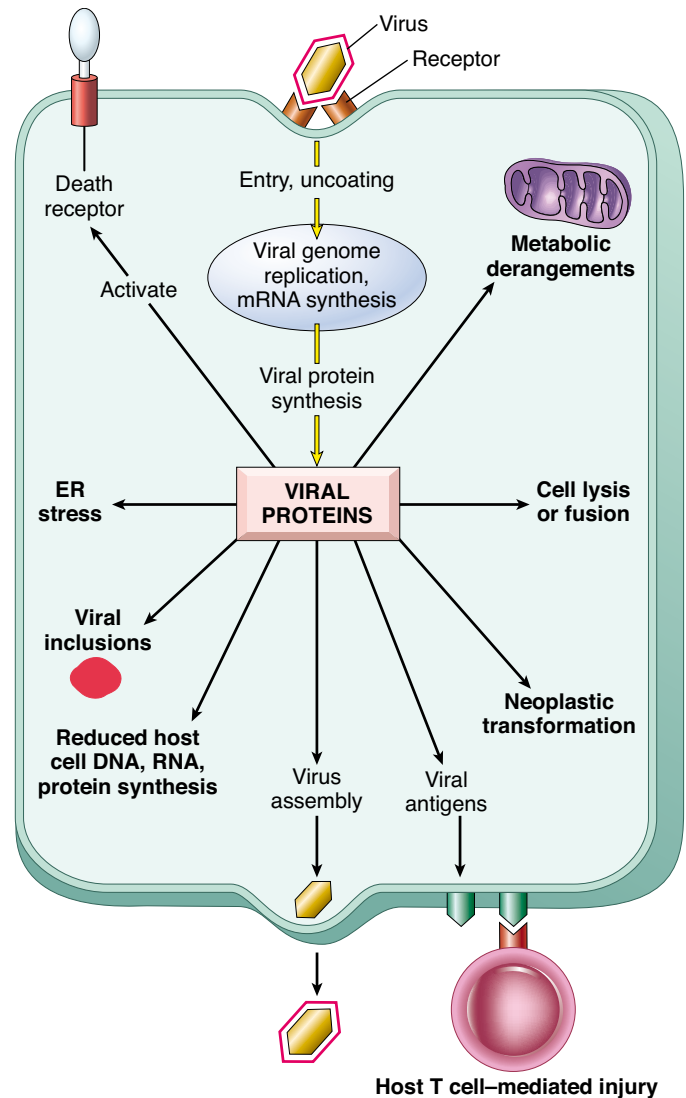


Fig. 8.3 Mechanisms by which viruses cause injury to cells. ER, Endoplasmic reticulum.

important in the persistence and relapse of bacterial endocarditis, artificial joint infections, and respiratory infections in people with cystic fibrosis.

Bacterial Adherence to Host Cells. Bacteria use various surface structures to attach to host cells and tissues. Adhesion allows bacteria to resist mechanical removal (e.g. by intestinal peristalsis) and is the first step in bacterial entry into host cells for intracellular bacteria.

- **Adhesins** are a category of bacterial surface proteins that bind the organisms to host cells or extracellular matrix. Adhesins have a broad range of host cell specificity. For example, *Streptococcus pyogenes* adheres to host tissues using the adhesins protein F and teichoic acid, which project from the bacterial cell wall and bind to fibronectin on the surface of host cells and in the extracellular matrix.
- **Pili** are filamentous structures on the surface of bacteria that can act as adhesins. Pili consist of stalks composed of conserved repeating

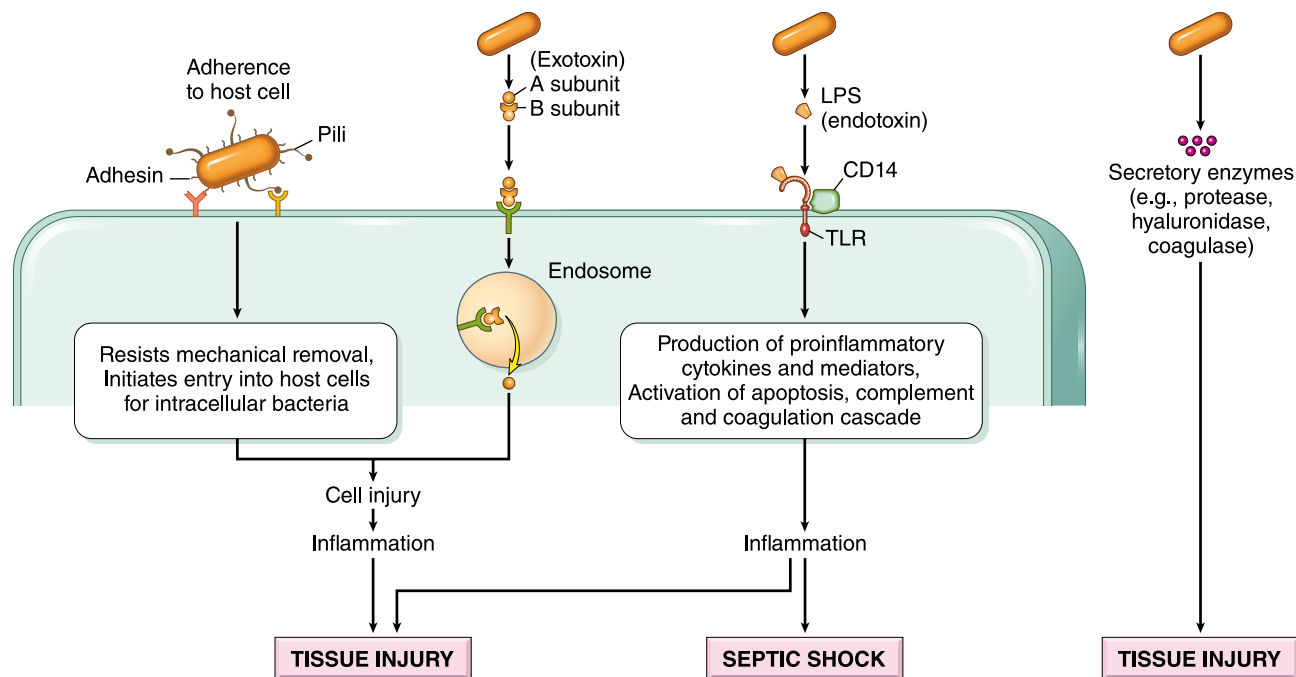


Fig. 8.4 Mechanisms by which bacteria cause injury to cells. LPS, Lipopolysaccharide; TLR, toll-like receptor.

protein subunits and variable tip fibrilli, which determine the tissue-binding specificity of the bacteria. Adhesion of *E. coli* to bladder epithelium is mediated by a tip fibrillum that binds D-mannosylated receptors, whereas adhesion to the renal epithelium depends on binding of a different tip fibrillum to galabiose-containing glycosphingolipids. Pili can be targets of the host antibody response and, in turn, some bacteria such as *N. gonorrhoeae* vary their pili to escape from the host immune system.

Intracellular Bacteria. The growth of bacteria inside cells can allow them to escape from certain effector mechanisms of the immune response (e.g., antibodies and complement), and can also facilitate the spread of the bacteria. Bacteria have evolved a variety of mechanisms for entering host cells. Some bacteria use receptors that are important in the host immune response to gain entry into macrophages. *M. tuberculosis* uses host receptors for opsonins (antibodies and C3b) as well as poorly defined nonopsonic receptors on macrophages. *S. enterica* and *Shigella* spp. use a type III secretion system to enter epithelial cells. This consists of needlelike structures projecting from the bacterial surface that bind to host cells. These proteins then form pores in the host cell membrane and inject bacterial proteins that mediate the rearrangement of the host cell cytoskeleton, which facilitates bacterial entry.

After bacteria enter the host cell, their fate (and that of the infected cell) varies greatly depending on the organism. *Shigella* spp. and *E. coli* inhibit host protein synthesis, replicate rapidly, and lyse the host cell within hours. Most bacteria are killed within macrophages when the phagosome fuses with an acidic lysosome to form a phagolysosome, where ingested microbes are destroyed, but certain bacteria elude this host defense. *M. tuberculosis* blocks fusion of the lysosome with the phagosome, allowing it to proliferate unchecked within infected macrophages. *M. tuberculosis* also spreads to draining lymph nodes and more distant sites within macrophages. Other bacteria avoid destruction in macrophages by leaving the phagosome

and entering the cytoplasm. *L. monocytogenes* produces a pore-forming protein called listeriolysin O and two phospholipases that degrade the phagosome membrane, allowing the bacterium to escape into the cytoplasm, protected from the killing mechanisms of macrophages. In the cytoplasm, *L. monocytogenes* modifies the host cell actin cytoskeleton to promote direct spreading of the organism to neighboring cells.

Bacterial Toxins. Any bacterial substance that contributes to illness can be considered a toxin. Toxins are classified as endotoxin, which is a component of gram-negative bacterial cell walls, and exotoxins, which are proteins that are secreted by many kinds of bacteria. Toxins can cause disease independent of infection, as with *Staphylococcus aureus* toxins produced by bacteria growing in food, which is then eaten. Toxins are also produced by bacteria growing within the host, as with *Vibrio cholera*, which replicates and releases a toxin within the intestine.

Bacterial endotoxin is a lipopolysaccharide (LPS) in the outer membrane of gram-negative bacteria that both stimulates host immune responses and injures the host. Lipid A, the part of LPS that anchors the molecule in the host cell membrane, has the endotoxin activity of LPS. LPS binds to CD14, either as a soluble molecule or on the cell surface, and the complex of LPS/CD14 then binds to TLR4. Other molecules in the outer structures of gram-positive bacteria can have effects similar to LPS, including lipoteichoic acid, which binds to TLR2. The response to lipid A or lipoteichoic acid is beneficial to the host in that it activates protective immunity in several ways. It induces the production of important cytokines and chemoattractants (chemokines) by immune cells and increases the expression of costimulatory molecules, which enhance T-lymphocyte activation. However, high levels of endotoxin play a pathogenic role in septic shock, disseminated intravascular coagulation (DIC), and adult respiratory distress syndrome, mainly through induction of excessive levels of cytokines such as TNF, IL-6, and IL-12.

Exotoxins are secreted bacterial proteins that cause cellular injury and disease. They can be classified into broad categories by their mechanism of action. These are briefly described next and discussed in more detail in the specific sections about each type of bacteria.

- **Enzymes.** Bacteria secrete a variety of enzymes (proteases, hyaluronidases, coagulases, fibrinolysins) that act on substrates in host tissues or on host cells. These enzymes have roles in tissue destruction and abscess formation. For example, exfoliative toxins produced by *S. aureus* cause staphylococcal scalded-skin syndrome by degrading proteins that hold keratinocytes together, causing the epidermis to detach from the deeper skin.
- **Toxins that alter intracellular signaling or regulatory pathways.** Most of these toxins have an active (A) subunit with enzymatic activity and a binding (B) subunit that binds to receptors on the cell surface and delivers the A subunit into the cell cytoplasm. The effects of these toxins are diverse and depend on the binding specificity of the B domain and the cellular pathways affected by the A domain. A-B toxins are made by many bacteria including *Bacillus anthracis*, *V. cholerae*, and some strains of *E. coli*. A-B toxins produced by *Clostridium botulinum* and *Clostridium tetani* are neurotoxins that inhibit release of neurotransmitters, resulting in paralysis. These toxins do not kill neurons; instead, the A domains interact specifically with proteins involved in secretion of neurotransmitters at the synaptic junction. Both tetanus and botulism can result in death from respiratory failure due to paralysis of the chest and diaphragm muscles.
- **Superantigens** are bacterial toxins that stimulate many T lymphocytes by binding to conserved portions of the T-cell receptor, leading to massive T-lymphocyte proliferation and cytokine release. The high levels of cytokines can lead to systemic inflammatory response syndrome.

Spectrum of Inflammatory Responses to Infection

In contrast to the vast molecular diversity of microbes, the morphologic patterns of tissue responses to microbes are limited, as are the mechanisms directing these responses (Chapter 3). Therefore, many

KEY CONCEPTS

Host Damage

- Diseases caused by microbes involve interplay between microbial virulence factors and host responses.
- Infectious agents cause death or dysfunction by directly interacting with the host cells.
- Injury may be due to local or systemic release of microbial products including endotoxin (LPS), exotoxins, or superantigens.
- Pathogens can induce immune responses that cause tissue damage. Absence of an immune response may reduce damage induced by some infections; conversely, immune compromise can allow uncontrolled expansion of microorganisms that can directly cause injury.

pathogens produce similar reaction patterns, and few features are unique or pathognomonic for a particular microorganism. Moreover, sometimes the immune status of the host determines the histologic features of the inflammatory response to microbes. Thus, pyogenic bacteria, which normally evoke vigorous leukocyte responses, may cause rapid tissue necrosis with little leukocyte exudation in a profoundly neutropenic host. Similarly, in a patient who is not immunocompromised, *M. tuberculosis* causes well-formed granulomas with few mycobacteria present, whereas in an AIDS patient the same mycobacteria multiply profusely in macrophages, which fail to coalesce into granulomas.

There are five major histologic patterns of tissue reaction in infections (Table 8.3).

Suppurative (Purulent) Inflammation

This pattern is characterized by increased vascular permeability and leukocytic infiltration, predominantly of neutrophils (see Fig. 3.14). The neutrophils are attracted to the site of infection by release of chemoattractants from the “pyogenic” (pus-forming) bacteria that evoke this response, mostly extracellular gram-positive cocci and gram-negative rods. Masses of dying and dead neutrophils and liquefactive necrosis of the tissue form pus. The sizes of purulent lesions range from

TABLE 8.3 Spectrum of Inflammatory Responses to Infection

Type of Response	Pathogenesis	Examples
Suppurative (purulent) inflammation	Increased vascular permeability Leukocyte infiltration (neutrophils) Chemoattractants from bacteria Formation of pus	Pneumonia (<i>Staphylococcus aureus</i>) Abscesses (<i>Staphylococcus</i> spp., anaerobic and other bacteria)
Mononuclear and granulomatous inflammation	Mononuclear cell infiltrates (monocytes, macrophages, plasma cells, lymphocytes) Cell-mediated immune response to chronic (persistent) pathogens Formation of granulomas	Syphilis Tuberculosis
Cytopathic-cytoproliferative reactions	Viral transformation of cells Necrosis or proliferation (including multinucleation) Linked to neoplasia	Cervical cancer (human papillomavirus) Chicken pox, shingles Herpes Infectious mononucleosis (Epstein-Barr virus)
Tissue necrosis	Toxin- or lysis-mediated destruction Lack of inflammatory cells Rapidly progressive processes	Gangrene (<i>Clostridium perfringens</i>) Fulminant acute hepatitis (hepatitis B virus)
Chronic inflammation/scarring	Repetitive injury leads to fibrosis Loss of normal parenchyma	Chronic hepatitis with cirrhosis (hepatitis B and C viruses)
No reaction	Severe immune compromise	<i>Mycobacterium avium</i> in untreated AIDS (T-cell deficiency) Mucormycosis in bone marrow transplant patients (neutropenia)

tiny microabscesses formed in multiple organs during bacterial sepsis secondary to a colonized heart valve, to diffuse involvement of entire lobes of the lung in pneumonia. How destructive the lesions are depends on their location and the organism involved. For example, *S. pneumoniae* usually spare alveolar walls and cause lobar pneumonia that resolves completely, whereas *S. aureus* and *Klebsiella pneumoniae* destroy alveolar walls and form abscesses that heal with scar formation. Bacterial pharyngitis (*S. pyogenes*) can resolve without sequelae, whereas untreated acute bacterial inflammation of a joint can destroy the joint in a few days.

Mononuclear and Granulomatous Inflammation

Diffuse, predominantly mononuclear, interstitial infiltrates are a common feature of all chronic inflammatory processes, but when they develop acutely they often are a response to viruses, intracellular bacteria, or intracellular parasites. In addition, spirochetes and helminths provoke chronic inflammatory responses. Which mononuclear cell predominates within the inflammatory lesion depends on the host immune response to the organism. For example, plasma cells are abundant in the primary and secondary lesions of syphilis (Fig. 8.5), whereas lymphocytes predominate in HBV infection or viral infections of the brain. The presence of these lymphocytes reflects cell-mediated immune responses against the pathogen or pathogen-infected cells. At the other extreme, macrophages may become filled with organisms, as occurs in *Mycobacterium avium* complex infections in AIDS patients, who cannot mount an effective immune response to the organisms.

Granulomatous inflammation is a distinctive form of mononuclear inflammation usually evoked by infectious agents that resist eradication and can stimulate strong T cell-mediated immunity (e.g., *M. tuberculosis*, *Histoplasma capsulatum*, schistosome eggs). Granulomatous inflammation is characterized by accumulation and aggregation of activated macrophages called “epithelioid” cells (because they morphologically resemble epithelial cells), some of which may fuse to form giant cells. Granulomas caused by tubercle bacilli may contain a central area of necrosis (see Chapter 3 and discussion of tuberculosis later in this chapter).

Cytopathic-Cytoproliferative Reaction

These reactions are usually produced by viruses. The lesions are characterized by cell necrosis or cellular proliferation, usually with sparse inflammatory cells. Some viruses replicate within cells and make viral aggregates that are visible as inclusion bodies (e.g., herpesviruses or

adenovirus) or induce cells to fuse and form multinucleated cells called *polykaryons* (e.g., measles virus or herpesviruses). Focal cell damage in the skin may cause epithelial cells to become detached, forming blisters (Fig. 8.6). Some viruses can cause epithelial cells to proliferate (e.g., venereal warts caused by HPV or the umbilicated papules of molluscum contagiosum caused by poxviruses). Finally, some viruses such as Epstein-Barr virus and oncogenic strains of HPV can contribute to the development of malignant neoplasms (Chapter 7).

Tissue Necrosis

Clostridium perfringens and other organisms such as *Corynebacterium diphtheriae* that secrete powerful exotoxins cause such rapid and severe necrosis that tissue damage is the dominant feature. The parasite *Entamoeba histolytica* causes colonic ulcers and liver abscesses characterized by extensive tissue destruction with liquefactive necrosis and little inflammatory infiltrate. Some viruses can cause widespread and severe necrosis of host cells associated with inflammation, as exemplified by total destruction of the temporal lobes of the brain by HSV or the liver by HBV.

Chronic Inflammation and Scarring

Many infections elicit chronic inflammation, which can lead either to healing or to extensive scarring. For example, chronic HBV infection may cause cirrhosis of the liver, in which dense fibrous septae surround nodules of regenerating hepatocytes, leading to loss of normal liver architecture and consequent changes in blood flow. Sometimes the exuberant scarring response is the major cause of dysfunction (e.g., “pipestem” fibrosis of the liver or fibrosis of the bladder wall caused by schistosome eggs [Fig. 8.7] or constrictive fibrous pericarditis in tuberculosis).

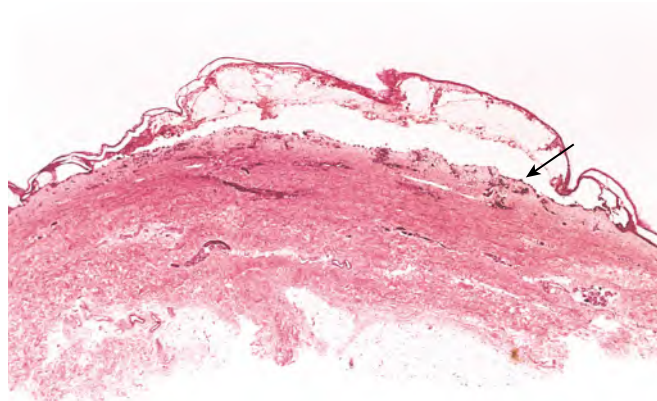


Fig. 8.6 Herpesvirus blister in mucosa (arrow). See Fig. 8.12 for viral inclusions.

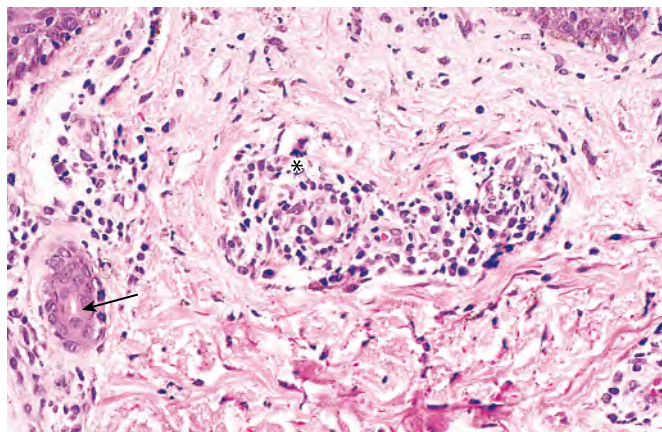


Fig. 8.5 Secondary syphilis in the dermis with perivascular lymphoplasmacytic infiltrate (asterisk) and endothelial proliferation (arrow).

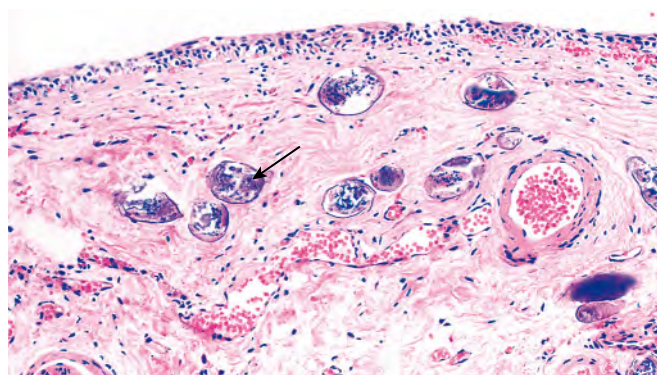


Fig. 8.7 *Schistosoma haematobium* infection of the bladder with numerous calcified eggs (arrow) and extensive scarring.

These patterns of tissue reaction are useful guidelines for analyzing microscopic features of infectious processes, but they rarely appear in pure form because different types of host reactions often occur at the same time. For example, the lung of an AIDS patient may be infected with CMV, which causes cytological changes in alveolar lining cells, and at the same time by *Pneumocystis* spp., which causes interstitial inflammation. Similar patterns of inflammation can also be seen in tissue responses to physical or chemical agents and in inflammatory diseases of unknown cause (Chapter 3).

Sexually Transmitted Infections

A variety of organisms can be transmitted through sexual contact (Table 8.4). Groups at greater risk for sexually transmitted infections (STIs) include adolescents and young adults, men who have sex with men, and people who use illegal drugs parenterally. Although the increased risk among these groups is partially due to sexual practices, limited access to health care is often a contributing factor. The presence of an STI in a child, unless acquired during birth, strongly suggests sexual abuse.

Some pathogens, such as *Chlamydia trachomatis* and *N. gonorrhoeae*, are almost always spread by sexual intercourse, whereas others, such as *Shigella* spp. and *E. histolytica*, are typically spread by the fecal-oral route, but are also occasionally spread by oral-anal sex. To reduce the spread of STIs, these infections are often reported to public health authorities so that people who have had sexual contact with the person may be tested and treated.

Although the various pathogens that cause STIs differ in many ways, some general features should be noted.

- STIs may become established locally and then spread from the urethra, vagina, cervix, rectum, or oral pharynx. Organisms that cause

STIs depend on direct contact for person-to-person spread because these pathogens do not survive in the environment. Transmission of STIs often occurs from asymptomatic people who do not realize that they have an infection.

- Infection with one STI-associated organism increases the risk for additional STIs. This is mainly because the risk factors are the same for all STIs. In addition, the epithelial injury caused by *N. gonorrhoeae* or *C. trachomatis* can increase the chance of co-infection with the other, as well as risk of HIV infection with concomitant exposure.
- The microbes that cause STIs can be spread from a pregnant woman to the fetus and cause severe damage to the fetus or child. Perinatally acquired *C. trachomatis* causes conjunctivitis, and neonatal HSV infection is much more likely to cause visceral and CNS disease than is infection acquired later in life. Syphilis frequently causes miscarriage or manifestations discussed above. Untreated HIV infection may be fatal to children infected with the virus prenatally or perinatally. Diagnosis of STIs in pregnant females is critical, because intrauterine or neonatal transmission of STIs can often be prevented by treatment of the mother or newborn. Antiretroviral treatment of pregnant females with HIV infection and their newborn infants can reduce the transmission of HIV to offspring from 25% to less than 1%.

Syphilis is discussed later in this chapter, and other STIs are described in Chapters 21 and 22.

This concludes our discussion of the general principles of the pathogenesis and pathology of infectious disease. We now turn to specific infections caused by viruses, bacteria, fungi, and parasites, and focus on their pathogenic mechanisms and pathologic effects rather than details

TABLE 8.4 Classification of Important Sexually Transmitted Infections

Disease or Syndrome and Population Principally Affected			
Pathogens	Males	Females	Both
Viruses			
Herpes simplex virus			Primary and recurrent herpes, neonatal herpes
Hepatitis B virus			Hepatitis
Human papillomavirus	Cancer of penis	Cervical dysplasia and cancer, vulvar cancer	Condyloma acuminatum
Human immunodeficiency virus			Acquired immunodeficiency syndrome
Chlamydia			
<i>Chlamydia trachomatis</i>	Urethritis, epididymitis, proctitis	Urethral syndrome, cervicitis, Bartholinitis, salpingitis, and sequelae	Lymphogranuloma venereum
Mycoplasmas			
<i>Ureaplasma urealyticum</i>	Urethritis		
Bacteria			
<i>Neisseria gonorrhoeae</i>	Epididymitis, prostatitis, urethral stricture	Cervicitis, endometritis, Bartholinitis, salpingitis, and sequelae (infertility, ectopic pregnancy, recurrent salpingitis)	Urethritis, proctitis, pharyngitis, disseminated gonococcal infection
<i>Treponema pallidum</i>			Syphilis
<i>Haemophilus ducreyi</i>			Chancroid
<i>Klebsiella granulomatis</i>			Granuloma inguinale (donovanosis)
Protozoa			
<i>Trichomonas vaginalis</i>	Urethritis, balanitis	Vaginitis	

of clinical features, which are available in other textbooks. Infections that typically involve a specific organ are discussed in other chapters.

VIRAL INFECTIONS

Viruses are the cause of many clinically important acute and chronic infections, which may affect virtually every organ system (Table 8.5).

Acute (Transient) Infections

The viruses that cause transient infections are structurally heterogeneous, but all elicit effective immune responses that eliminate the pathogens, limiting the durations of these infections. However, specific viruses exhibit widely differing degrees of genetic diversity, a variable that has an important impact on the susceptibility of the host to reinfection by viruses of the same type. The mumps virus, for example, has only one genetic subtype and infects people only once, whereas other viruses, such as influenza viruses, can repeatedly infect the same individual because new genetic variants arise periodically in nature. Immunity to some viruses, particularly respiratory and gastrointestinal viruses, wanes with time, and this too allows the same virus to infect the host repeatedly.

COVID-19

COVID-19 (Coronavirus disease of 2019) is a major modern pandemic. It is discussed in some detail because of its enormous impact worldwide on health and economies and because it illustrates important features of an acute cytopathic viral infection.

History and Epidemiology. COVID-19 was first noted in December of 2019 in Wuhan, China, as a cluster of cases of pneumonia unresponsive to treatment. Many of the initial cases occurred in people who had been at the Huanan Wholesale Seafood Market, which is near the Wuhan Institute of Virology. Four years after the start of the pandemic, the two major hypotheses about the origin of the virus are intensely debated: natural zoonotic spillover versus laboratory leak. The causal agent was quickly identified and the genome sequence of the virus causing COVID-19, severe acute respiratory syndrome (SARS) coronavirus-2 (SARS-CoV-2), was publicly available within a month of the initial cases. By March 2020, there were over 100,000 known cases in 114 countries, leading the World Health Organization (WHO) to declare COVID-19 a pandemic. The emergence of variants of SARS-CoV-2 in the summer of 2021 (Delta variant) and the winter of 2021–2022 (Omicron variant) led to large surges in the numbers of cases and deaths. With the widespread availability of vaccines for COVID-19 in high-resource nations in early 2021 and large numbers of infections in all nations, much of the world population now has some level of immunity to SARS-CoV-2. The virus remains endemic in humans and animals and is expected to remain among the seasonal respiratory viruses. The morbidity and mortality of the COVID-19 pandemic were very high. As of October 2023, 770,000,000 confirmed cases and 6.9 million deaths have been reported to the WHO, but this is likely to be well below the actual values due to cases and deaths that were not reported.

During 2020–2022, in the United States, the rates of infection and death from COVID-19 varied widely across individuals of differing geographic origins and ethnicity. The age-adjusted rates of infection were higher in African-American, Hispanic, Native American or Alaska native, and native Hawaiian people. Several risk factors contributed to these striking differences between groups, including the likelihood of work that cannot be done

TABLE 8.5 Selected Human Viruses and Viral Diseases

Organ System	Species	Disease
Respiratory	Adenovirus	Upper and lower respiratory tract infections, conjunctivitis, diarrhea
	Severe acute respiratory syndrome virus (SARS-CoV-2)	COVID-19
	Rhinovirus	Upper respiratory tract infection
	Influenza viruses A, B	Influenza
	Respiratory syncytial virus	Bronchiolitis, pneumonia
Digestive	Mumps virus	Mumps, pancreatitis, orchitis
	Rotavirus	Childhood gastroenteritis
	Norovirus	Gastroenteritis
	Hepatitis A virus	Acute viral hepatitis
	Hepatitis B virus	Acute or chronic hepatitis
	Hepatitis D virus	With hepatitis B virus, acute or chronic hepatitis
	Hepatitis C virus	Acute or chronic hepatitis
Systemic with skin eruptions	Hepatitis E virus	Acute viral hepatitis
	Measles virus	Measles (rubeola)
	Rubella virus	German measles (rubella)
	Varicella-zoster virus	Chickenpox, shingles
	Herpes simplex virus 1	Oral herpes ("cold sore")
Systemic with hematologic disorders	Herpes simplex virus 2	Genital herpes
	Cytomegalovirus	Cytomegalic inclusion disease
	Epstein-Barr virus	Infectious mononucleosis
Arboviral and hemorrhagic fevers	Human immunodeficiency viruses 1 and 2	Acquired immunodeficiency syndrome
	Dengue viruses 1 to 4	Dengue hemorrhagic fever
Skin/genital warts	Yellow fever virus	Yellow fever
	Human papillomavirus	Condyloma; cervical carcinoma
Central nervous system	Poliovirus	Poliomyelitis
	JC virus	Progressive multifocal leukoencephalopathy (opportunistic)

remotely leading to clustering of people close to one another, more residents in households, and use of public transportation. Higher death rates are associated with lack of access to both COVID-19 vaccines and medical care for those infected.

Transmission and Clinical Manifestations. Several specific risk factors for and patterns of transmission have been observed.

- SARS-CoV-2 spreads primarily from person to person by both droplets and aerosols. Droplet transmission is probably the main route of transmission, via wet respiratory particles between people within about 2 meters of each other. Aerosol transmission can take place over longer distances and is more likely to occur in enclosed spaces with inadequate ventilation.
- Transmission is most likely to occur during the first 7 to 10 days of infection, when levels of the virus in the respiratory tract are highest.
- Most transmission occurs where people are gathered indoors for prolonged periods of time, such as households, housing shelters, jails, dormitories, and health care and long-term care facilities. Overall transmission within households is approximately 20%, but Omicron variants are transmitted at about twice that rate.
- People with asymptomatic infections and those in the incubation period can transmit the virus, although infection is more likely to occur during symptomatic infection.
- Infection of humans from animal sources, such as mink, deer, and Syrian hamsters, occurs but is uncommon.

Following infection, the incubation period ranges from 2 to 14 days, with a median of 4 to 5 days to onset of symptoms. The greatest risk factor for severe COVID-19 is age; the risk of death in people 50 to 64 years of age is 25-fold higher than in those 18 to 29 years of age and rises further with increasing age. Other risk factors for severe disease include race and ethnicity, which are associated with access to care, underlying lung diseases (including asthma), cancer, types 1 and 2 diabetes mellitus, heart disease, immunocompromise (HIV infection, medications, primary immunodeficiencies, prior organ transplantation), obesity, smoking, pregnancy, dementia, mental health (depression and schizophrenia), and chronic kidney or liver disease. The case fatality rate of COVID-19 in early 2020, at the start of the pandemic, is estimated to have been as high as 10% to 20%, but it is now less than 0.3% due to improved treatment, including antiviral drugs, and widespread immunity due to previous infections and vaccination.

A range of symptoms can persist beyond acute COVID-19. This is referred to as “long COVID” or “postacute sequelae of SARS-CoV-2 infection.” Persistence of symptoms which occur at least 1 month after acute COVID-19 and last at least 2 months is generally considered long COVID, but there is no consensus about what symptoms define the syndrome. Because the definition of long COVID varies between studies, the prevalence is difficult to determine. One recent study estimated the overall incidence to be approximately 6% among people who have had acute COVID-19. Risk factors for long COVID include female sex, severe acute COVID-19, lack of vaccination, and age over 20 years. Fatigue, weakness, dyspnea, chest pain, and cough are among the more common physical symptoms. Posttraumatic stress disorder, problems with memory or concentration, anxiety and depression are among the common psychological symptoms. Most people recover within a year. Among those with persistent symptoms, a relapsing-remitting pattern is probably most common with long COVID, but it can also follow a progressive pattern.

Pathogenesis

SARS-CoV-2 is a coronavirus that is endemic among humans and a cause of severe infections. Before the emergence of SARS-CoV-2, the coronaviruses endemic in humans infected only the upper

respiratory tract in nearly all cases, causing the common cold. The zoonotic coronaviruses, severe acute respiratory syndrome (SARS) virus and Middle East respiratory syndrome virus (MERS), caused more severe infections, but did not spread efficiently among people. The single-stranded positive-sense RNA genome of SARS-CoV-2 is approximately 30 kilobases, among the largest of RNA viruses. Important structural proteins of the virus include the spike and membrane proteins, both located in the lipid envelope of the virus, as well as the nucleocapsid protein, which is bound to the viral RNA.

The S1 subunit of the trimeric spike protein of SARS-CoV-2 binds to angiotensin-converting enzyme 2 (ACE2). ACE2 is expressed on the surface of epithelial cells, including those of the respiratory tract. The portion of S1 that binds to ACE2 is the *receptor binding domain* (RBD). Once bound to ACE2, the spike protein is cleaved by a host cell protease, TMPRSS2. This activates the S2 subunit of the spike protein, which drives the fusion of the host and viral membranes, after which the viral RNA enters the host cell and viral replication begins. The spike protein, and specifically the RBD, is the main target of antibodies that neutralize SARS-CoV-2, which prevent the virus from infecting host cells.

After the viral RNA enters the host cell, it is translated by host ribosomes into viral proteins. These proteins include the viral RNA-dependent RNA polymerase, which synthesizes additional positive-strand viral RNA genomes via a double-stranded RNA intermediate. Virus-specific RNA structures and viral double-stranded RNA stimulate production of type 1 IFN in the host, which inhibits replication of the virus within the infected and neighboring cells; however, SARS-CoV-2 has evolved to evade these responses. A remarkable testament to the importance of immune evasion is that 10 of the 29 proteins made by the virus are dedicated to dampening the production of or response to IFNs.

To reduce recognition of RNA by innate immune receptors such as RIG-I, the virus produces proteins that remove a phosphate and then cap and methylate the RNA, making it resemble host RNA. SARS-CoV-2 has two strategies for reducing recognition of double-strand RNA. In addition, the virus produces an RNA-endonuclease that minimizes accumulation of negative-strand RNA and double-strand RNA. Also, viral proteins use membranes from the host endoplasmic reticulum to enclose the viral RNA within a double-membrane-bound vesicle, where the double-strand RNA intermediates cannot be detected by RIG-I-like receptors. Finally, several viral proteins inhibit the signaling cascade that leads to production of IFNs by binding to or modifying proteins required for this cascade. These mechanisms allow the virus to escape detection and elimination. The importance of type I IFNs in antiviral immunity is further highlighted by the observation that individuals who make autoantibodies against their own type I IFN or have genetic defects affecting IFN production or signaling are at increased risk for developing severe COVID-19 following infection with SARS-CoV-2.

Clinical Features

The manifestations of COVID-19 are variable (Fig. 8.8). The infection is asymptomatic in one-third or more of those afflicted. The most common manifestations are moderate respiratory disease with fever, cough, fatigue, diarrhea, and myalgia. The pulmonary disease is described in Chapter 15.

Severe COVID-19 is characterized by a hyperinflammatory response with increased levels of cytokines, including TNF and IL-6, and dysregulation of the antiviral immune response manifested by abnormalities in myeloid cell populations and lymphopenia with

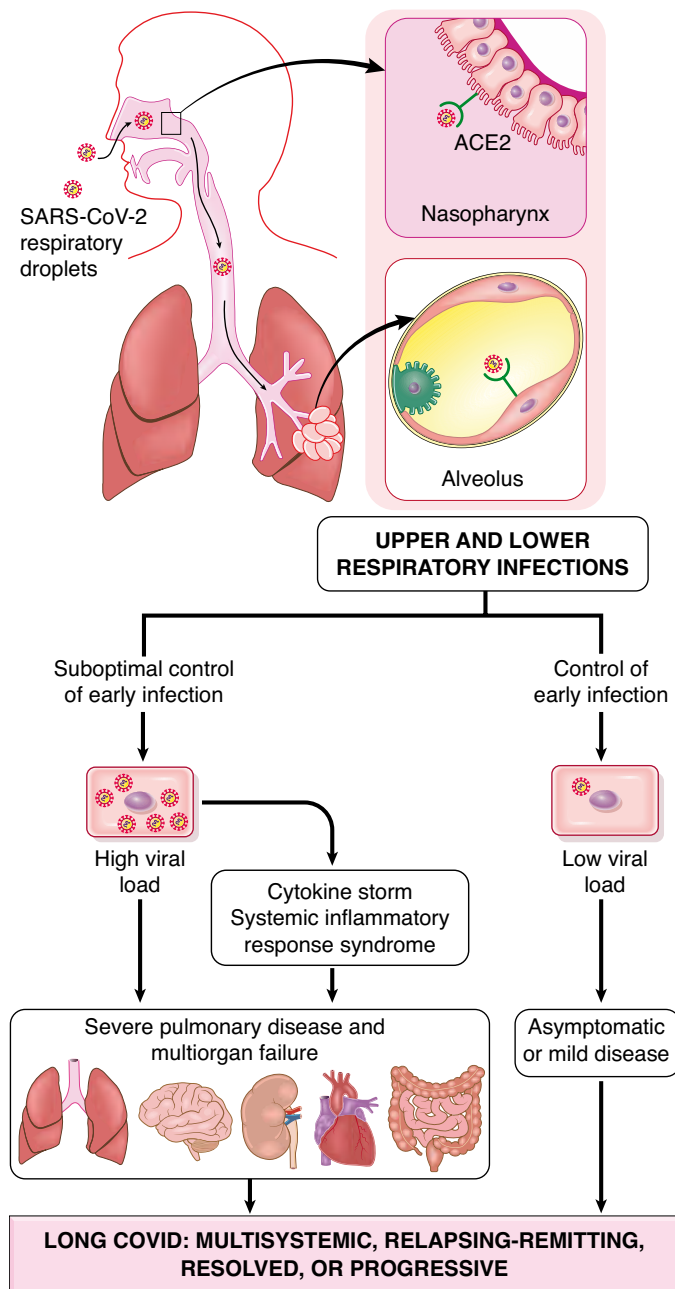


Fig. 8.8 Pathogenesis of COVID-19. ACE2, Angiotensin-converting enzyme 2; SARS-CoV-2, severe acute respiratory syndrome coronavirus 2.

reduced numbers of T cells and NK cells. It results in onset of dyspnea and hypoxia with progression to acute respiratory distress syndrome, which can lead to death. This hyperinflammatory response has several unusual features.

- The patterns of multiorgan manifestations of severe COVID-19 vary with age. Children, mainly over the age of 5 years, can have *multisystem inflammatory syndrome (MIS-C)*, which usually begins 2 to 6 weeks after acute COVID-19. MIS-C involves several organ systems, including the gastrointestinal system (pain, vomiting, diarrhea), the skin (rash), the mucosa (swollen lips, reddened tongue), and CNS (confusion, lethargy). Cardiac manifestations are common and include myocardial dysfunction (depressed contractility), coronary artery dilation, mitral regurgitation and

pericardial effusion. MIS-C can present with or progress to shock, which can be life-threatening. Fortunately, most children recover from MIS-C; however, it is not yet known whether there are long-term sequelae.

- Among adults, some develop characteristics of *hemophagocytic lymphohistiocytosis (HLH)* (Chapter 13), with cytopenias, hemophagocytosis, and elevated acute phase reactants. HLH is known to be associated with many other viral infections, including other respiratory viruses. *Systemic vasculitis* can occur in various body sites, including the skin, CNS, lungs, and gastrointestinal tract. Additional signs of inflammation with severe COVID-19 include myositis and arthritis.
- Coagulopathy*, with increased coagulation and reduced fibrinolysis, occurs with severe COVID-19. Coagulation may be triggered by exposure of tissue factor following endothelial damage by viral infection or inflammation. In addition, it is common for patients with severe COVID-19 to develop antiphospholipid antibodies and the anti-phospholipid antibody syndrome (Chapter 3), which is also associated with thrombosis. Fibrinolysis is reduced, possibly due to elevated levels of the plasminogen activator inhibitor 1 (PAI-1), which inhibits two proteins, tissue plasminogen activator and urokinase, that activate fibrinolysis by converting plasminogen to plasmin. The inhibition of fibrinolysis exacerbates the coagulopathy by further tipping the balance toward coagulation.

The most frequently used tests for infection rely on identification of the SARS-CoV-2 nucleocapsid protein in nasal swabs by antibody binding assays, though PCR-based diagnostic approaches are also used.

Variants of SARS-CoV-2. Accumulation of mutations, insertions, and deletions in the viral RNA, as well as recombination events between SARS-CoV-2 infecting a single host cell, leads to the generation of variants of the virus. The WHO has defined *variants of concern (VOCs)* as causing more severe disease or impacting public health interventions (e.g., because they are more transmissible), or for which available vaccines are relatively ineffective. The WHO assigns a Greek letter to each variant of concern.

- Some SARS-CoV-2 VOCs are transmitted more efficiently. For example, a change from aspartic acid to glycine in the spike protein was associated with increased replication and transmission of the virus.
- Other SARS-CoV-2 VOCs can escape from the immune response. The Omicron variant emerged in late 2021 and caused a massive increase in COVID-19 cases worldwide in early 2022. This variant had many changes in the spike protein, which allowed it to avoid recognition by host antibodies.

MORPHOLGY

Deaths from COVID-19 are associated with acute respiratory distress syndrome, with **diffuse alveolar damage** seen on histology (Chapter 15). The lungs are grossly congested and edematous. There is congestion of the capillaries. The alveoli are filled with edema and lined with hyaline membranes composed of fibrinous exudate, leading to severe compromise of gas exchange, and there is hyperplasia of type II pneumocytes. The alveolar ducts are dilated, and some alveoli are collapsed. Thrombi are found in the arterioles and alveolar capillaries in most cases. Electron microscopy demonstrates intracellular viral particles, primarily within type I and type II pneumocytes.

Measles

Measles is an acute viral infection that affects multiple organs and causes a wide range of disease from mild, self-limited infections to severe systemic manifestations. A major initiative to increase vaccination for measles, led by several governments and international bodies, including the WHO, is estimated to have prevented 56 million deaths between 2000 and 2021; however, there were still almost 128,000 deaths from measles in 2021. The COVID-19 pandemic interrupted the measles virus campaign, and in 2022 only 83% of children got one dose of measles vaccine by 1 year of age, the lowest rate since 2008. Because of poor nutrition and lack of access to medical care, children in low-resource countries are 10 to 1000 times more likely to die of measles than are children in high-resource countries. Measles can produce severe disease in people with defects in cellular immunity (e.g., people infected with HIV or people with a hematologic malignancy). In high-resource nations, epidemics of measles occur when the virus is introduced by individual(s) traveling from an area of endemic disease, and then spreads, primarily to unvaccinated individuals. Such outbreaks typically occur several times each year in the United States and, after falling during 2020 and 2021, cases rose in 2022. The diagnosis may be made clinically, by serology, or by detection of viral RNA by polymerase chain reaction (PCR) in respiratory secretions or urine.

Pathogenesis

Measles virus is a single-stranded RNA virus of the Paramyxoviridae family, which includes mumps, respiratory syncytial virus (RSV), parainfluenza virus (a cause of croup), and human metapneumovirus. There is only one serotype of measles virus. Measles virus is very efficiently transmitted by the airborne route via aerosolized respiratory secretions. Three cell-surface receptors have been identified for measles hemagglutinin protein. Signaling lymphocytic activation molecule family member 1 (SLAMF1) is expressed on activated lymphocytes, dendritic cells, and monocytes, and it serves as the initial receptor for viral infection. Nectin-4 is found on the basal surface of epithelial cells and is thought to be important for entry and replication of the virus within the respiratory tract before spread of the virus in respiratory secretions.

The virus initially multiplies within the respiratory epithelium and then spreads to local lymphoid tissues. Replication of the virus in lymphatic tissue is followed by viremia and dissemination to many tissues, including the conjunctiva, skin, respiratory tract, urinary tract, small blood vessels, lymphatic system, and CNS. Most children develop T cell-mediated immunity to measles virus that helps control the infection and produces the measles rash. Hence, the rash is less frequent in people with deficient cell-mediated immunity. In addition, in malnourished children with poor medical care, measles virus may cause croup, pneumonia, diarrhea and protein-losing enteropathy, keratitis leading to scarring and blindness, encephalitis, and hemorrhagic rashes. Subacute sclerosing panencephalitis (Chapter 28) and measles inclusion body encephalitis (in immunocompromised individuals) are rare late complications of measles. In subacute sclerosing panencephalitis, the virus accumulates a number of mutations such that fully infectious viral particles are not formed, and the fusion (F) protein promotes efficient fusion of host cells allowing spread of the defective virus without receptor-mediated endocytosis.

Antibody-mediated immunity to measles virus protects against reinfection. Measles can also cause transient but profound immunosuppression, resulting in secondary bacterial and viral infections, which are responsible for much of measles-related morbidity and mortality. Delayed-type hypersensitivity responses are reduced following measles infection, indicating a reduction in lymphocyte responses.

MORPHOLOGY

The blotchy, reddish-brown rash of measles virus infection on the face, trunk, and proximal extremities is produced by dilated skin vessels, edema, and a mononuclear perivascular infiltrate. Ulcerated mucosal lesions in the oral cavity near the opening of the Stensen ducts (the pathognomonic **Koplik spots**) are marked by necrosis, neutrophilic exudate, and neovascularization. The lymphoid organs typically have marked follicular hyperplasia, large germinal centers, and randomly distributed multinucleate giant cells, called **Warthin-Finkeldey cells**, which have eosinophilic nuclear and cytoplasmic inclusion bodies (Fig. 8.9). These are pathognomonic of measles and are also found in the lung and sputum. The milder forms of measles pneumonia show the same peribronchial and interstitial mononuclear cell infiltration that is seen in other nonlethal viral infections.

Mumps

Mumps is an acute systemic viral infection usually associated with pain and swelling of the salivary glands. Like measles virus, mumps virus is a member of the Paramyxoviridae family. Mumps virus has two surface glycoproteins, one with hemagglutinin and neuraminidase activities and the other with cell fusion activity. Mumps viruses enter the upper respiratory tract through inhalation of or contact with respiratory droplets, spread to draining lymph nodes where they replicate in lymphocytes (preferentially in activated T cells), and then spread through the blood to the salivary and other glands. Mumps virus infects salivary gland ductal epithelial cells, resulting in desquamation of involved cells, edema, and inflammation that leads to salivary gland pain and swelling. Mumps virus can also spread to other sites, including the CNS, testis, ovary, and pancreas. Orchitis is the most common complication, occurring in one-third of unvaccinated and 6% of vaccinated men. In the United States, outbreaks of mumps have occurred in populations with close contact (e.g., university or high-school settings) in recent years. The number of cases in the United States in recent years peaked at over 6000 in 2017 and fell into the hundreds in 2019 to 2022. It is important to note that this is still more than 99% fewer cases than occurred annually in the United States before use of the mumps vaccine. The diagnosis is usually made clinically, but serology or detection of viral RNA in saliva or, less often, urine or cerebrospinal fluid (CSF), can be used for definitive diagnosis.

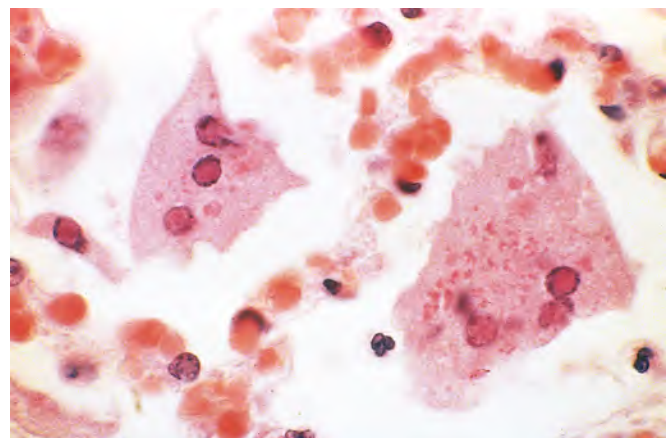


Fig. 8.9 Measles giant cells in the lung. Note the glassy eosinophilic intranuclear inclusions.

MORPHOLOGY

Mumps parotitis is bilateral in 70% of cases. The affected glands are enlarged, have a doughy consistency, and are moist, glistening, and reddish-brown in cross-section. On microscopic examination, the gland interstitium is edematous and diffusely infiltrated by macrophages, lymphocytes, and plasma cells, which compress acini and ducts. Neutrophils and necrotic debris may fill the duct lumen and cause focal damage to the lining epithelium.

In **mumps orchitis**, testicular swelling may be marked, caused by edema, mononuclear cell infiltration, and focal hemorrhages. Because the testis is tightly contained within the tunica albuginea, parenchymal swelling may compromise the blood supply and cause areas of infarction. The testicular damage can lead to scarring, atrophy, and, if severe, sterility.

Infection and damage of acinar cells in the pancreas may release digestive enzymes, causing parenchymal and fat necrosis and neutrophil-rich inflammation.

Mumps encephalitis is associated with perivenous demyelination and perivascular mononuclear cuffing.

Poliomyelitis

Poliovirus causes an acute systemic viral infection marked by a wide range of manifestations, from mild, self-limited infections to paralysis of limb muscles and respiratory muscles. Poliovirus is a spherical, unencapsulated RNA virus of the *Enterovirus* genus. Other enteroviruses cause childhood diarrhea as well as rashes (coxsackievirus A), conjunctivitis (enterovirus 70), viral meningitis (coxsackieviruses and echovirus), and myopericarditis (coxsackievirus B). There are three serotypes of poliovirus, but only type 1 is a cause of endemic disease. The inactivated (injected) poliovirus vaccine protects against all three serotypes, and the attenuated (oral) poliovirus vaccine is available in various combinations of one, two, or all three serotypes, although only formulations containing one or two serotypes are currently in use. Use of these vaccines has nearly eradicated polio because poliovirus infects only humans, shows limited genetic variation, and is effectively neutralized by antibodies generated by immunization. As mentioned, type 1 poliovirus is the only wild (nonvaccine) poliovirus that continues to cause disease, and transmission is currently limited to Pakistan and Afghanistan. Poliovirus from attenuated virus in the oral vaccine can rarely acquire mutations and become pathogenic, causing disease in infants, those with impaired antibody responses, and close contacts of oral vaccine recipients. Cases of poliomyelitis from vaccine-derived poliovirus types 1 and 2 occur mainly in sub-Saharan African nations. In some countries, infection from the attenuated oral vaccine strain is becoming more frequent than natural infection, requiring a change to an injected vaccine using killed virus, which is logistically more challenging.

Poliovirus, like other enteroviruses, is transmitted by the fecal-oral route or, less often, by pharyngeal secretions. The virus infects human cells by binding to CD155, a molecule expressed on a variety of cell types, including epithelial cells, lymphocytes, and neurons. The virus is ingested and replicates in the mucosa of the pharynx and gut, including tonsils and Peyer patches in the ileum. Poliovirus then spreads through lymphatics to lymph nodes and eventually the blood, producing transient viremia and fever. Although most poliovirus infections are asymptomatic, in about 1 in 100 infected persons, poliovirus invades the CNS and replicates in motor neurons of the spinal cord (spinal poliomyelitis) or brainstem (bulbar poliomyelitis). Antiviral antibodies control the disease in most cases; it is not known why they fail to contain the virus in some individuals. Viral spread to the nervous system may be through the blood or by retrograde transport of the virus along axons of motor neurons. The diagnosis can be made by viral culture or detection of viral RNA in throat secretions or, more

often, stool, or by serology. The neurologic features and neuropathology of poliovirus infection are described in Chapter 28.

West Nile Virus

West Nile Virus causes an acute systemic infection that has two very different presentations: a mild, self-limited infection, and neuroinvasive disease associated with long-term neurologic sequelae. West Nile virus is an arthropod-borne virus (arbovirus) of the flavivirus group, which also includes viruses that cause dengue fever and yellow fever. West Nile virus has a broad geographic distribution, including Africa, the Middle East, Europe, Southeast Asia, and Australia. It was first detected in the United States in 1999 during an outbreak in New York City and has since spread across the United States, causing approximately 1000 to 3000 reported cases per year since 2013. West Nile virus is transmitted by mosquitoes to birds and to mammals. Infected birds develop prolonged viremia and are the major reservoir for the virus. Humans are incidental hosts. Most affected patients acquire the infection from a mosquito bite; less commonly, human-to-human transmission occurs by blood transfusion, organ transplantation, breastfeeding, or vertically by transplacental spread or during breast feeding.

After inoculation by a mosquito, West Nile virus replicates in skin dendritic cells, which then migrate to lymph nodes. Here, the virus replicates further, enters the bloodstream, and, in some individuals, crosses the blood-brain barrier and infects neurons in the CNS.

West Nile virus infection is usually asymptomatic, but in 20% of infected individuals it causes fever, headache, myalgia, fatigue, anorexia, vomiting and diarrhea. A maculopapular rash is seen in approximately one-half of symptomatic cases. CNS complications (meningitis, encephalitis, meningoencephalitis) occur in about 1 in 150 infections. Meningoencephalitis has a mortality rate of about 10% and results in long-term cognitive and neurologic impairment in many survivors. Immunosuppressed people and older adults are at the greatest risk for severe disease. Rare complications include hepatitis, myocarditis, and pancreatitis. The diagnosis is usually made by detection of virus-specific IgM in serum or cerebrospinal fluid, but viral culture and PCR-based tests are also used.

MORPHOLOGY

Perivascular and leptomeningeal chronic inflammation, microglial nodules (Chapter 28), and neuronophagia predominantly involving the temporal lobes and brainstem have been observed in patients who died of West Nile virus infection.

Viral Hemorrhagic Fever

Viral hemorrhagic fever is a severe life-threatening multisystem syndrome in which there is vascular damage, leading to widespread hemorrhage and shock. Viral hemorrhagic fever is caused by enveloped RNA viruses belonging to four different groups: *Arenaviridae*, *Filoviridae*, *Bunyavirales*, and *Flaviviridae*. These viruses can produce a spectrum of illnesses, ranging from a mild acute disease characterized by fever, headache, myalgia, rash, neutropenia, and thrombocytopenia to severe, life-threatening disease in which there is sudden hemodynamic deterioration and shock. These viruses pass through an animal or insect host during their life cycles, and therefore their ranges are restricted to areas in which their hosts reside. Humans are incidental hosts who are infected when they encounter infected hosts (typically rodents or bats) or insect vectors (mosquitoes and ticks). Some viruses that cause hemorrhagic fever also can spread from

person-to-person by contact with symptomatic people or cadavers, including Lassa, Crimean-Congo, Ebola, and Marburg viruses.

The pathogenesis of the infection and its complications vary among the different viruses, but there are some common features. Damage to blood vessels is often prominent. It may be caused by direct infection of and damage to endothelial cells, or infection of macrophages and dendritic cells leading to production of inflammatory cytokines.

There was a major outbreak of Ebola virus in Sierra Leone, Liberia, and Guinea in 2014 to 2016, with over 28,000 cases and 11,000 deaths reported to the WHO. Outbreaks in the Democratic Republic of Congo have occurred several times since 2017. The outbreaks were characterized by explosive spread of the virus by person-to-person transmission through exposure to mucosal secretions. Ebola carries a very high mortality rate, about 40% during the 2019 outbreak. Effective vaccines for Ebola have been developed and used during outbreaks in Guinea and the Democratic Republic of Congo. Ebola virus (and other, related filoviruses) has specific mechanisms by which it can evade the immune response, allowing the virus to rapidly replicate to reach high levels. Two viral proteins, VP24 and VP35, inhibit the action of type I IFN. VP24 blocks the production of type I IFN by inhibiting the nuclear translocation of the transcription factor STAT1. VP35 inhibits type I IFN signaling and also binds to double-stranded viral RNA in infected host cells, preventing detection of the RNA by cytoplasmic RIG-I receptors.

MORPHOLOGY

With viral hemorrhagic fever, there may be hemorrhagic manifestations, including petechiae, caused by a combination of thrombocytopenia or platelet dysfunction, endothelial injury, cytokine-induced DIC, and deficiency of clotting factors because of hepatic injury. Hemorrhages may be prominent in some infections (e.g., Congo-Crimean fever) but are rarely life-threatening. Necrosis of tissues secondary to the vascular lesions and hemorrhages may be seen and varies from mild and focal to massive, but the attendant inflammatory response is usually minimal. In Ebola virus infection, there is widespread hemorrhage and viral replication in many organs including the liver with hepatocellular necrosis and scant inflammation, and the spleen with lymphocyte apoptosis and viral replication in dendritic cells, fibroblasts, and monocytes.

Zika Virus

Zika virus is a Flavivirus that was discovered in 1947 and was subsequently found to be widespread in Africa, Asia, and the Middle East. In 2015, an outbreak of Zika virus in Brazil caused up to 1.3 million suspected cases. Subsequently, a marked increase in the occurrence of microcephaly in infants born in Brazil was noted, and Zika virus has now been clearly linked to this tragic outcome. This Zika virus outbreak spread to at least 33 countries and territories in the Americas, with one of the largest outbreaks occurring in Colombia, which had over 50,000 suspected cases. In the United States, most of the approximately 5600 cases occurred in travelers returning from affected areas; however over 200 cases were acquired by local transmission. Since 2017, cases have declined worldwide and there have been no confirmed cases in the United States since 2019. Zika virus is mainly transmitted by *Aedes* mosquitoes, and also prenatally, by blood transfusion, and by sexual contact.

The manifestations of Zika virus infection in adults are usually mild and nonspecific, with fever, myalgia, arthralgia, conjunctivitis, and maculopapular rash lasting a few days to 1 week. The infection is associated with a small number of cases of neurologic complications in adults, primarily Guillain-Barré syndrome. Perinatal transmission of the virus can result in fetal death or microcephaly with moderate to severe brain defects in the fetus and newborn child. Zika virus replicates initially in the

placenta followed by infection of the fetus, and then viral replication occurs in neural precursor cells, leading to reduced brain size and thinning of the cortex. The mechanism of tropism for neural progenitor cells is not known.

MORPHOLOGY

Cerebral abnormalities associated with in utero infection by Zika virus include calcifications (Fig. 8.10A), atrophy and hypoplasia, and ventricular enlargement. Microcephaly occurs in less than 5% of infants born to infected females, with one-half of cases being proportionate microcephaly in children who are small for gestational age, and the other half being disproportionate microcephaly, meaning that the head size is small relative to the size of the child. Ocular abnormalities, including pigment mottling, chorioretinal atrophy, and optic nerve abnormalities, are also common.

Findings in newborn children include microcephaly, ventriculomegaly, congenital joint contractures (arthrogryposis), and pulmonary hypoplasia. Severe neuronal degeneration (Fig. 8.10B) and associated thinning of the brain parenchyma may also be seen in association with microcalcifications and microglial nodules.

Dengue

Dengue virus is a flavivirus transmitted by *Aedes* mosquitoes. Transmission may also occur less frequently via prenatal or perinatal routes,

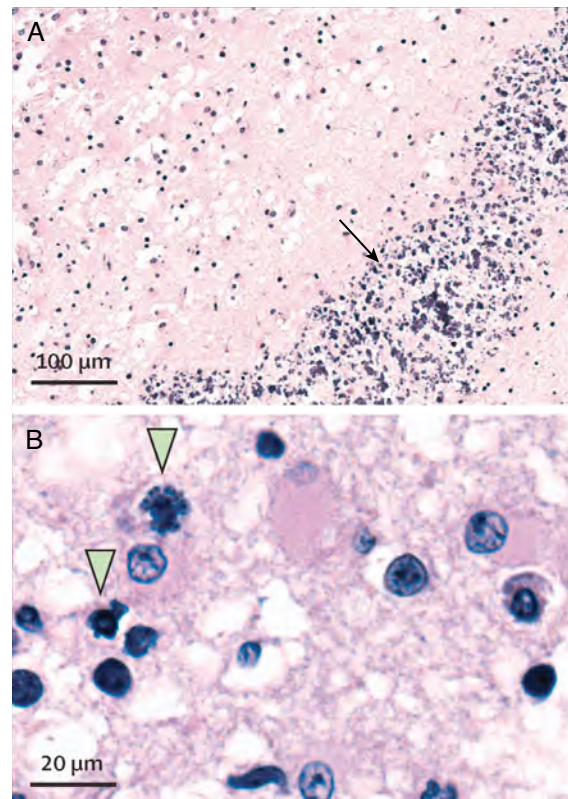


Fig. 8.10 Morphology of brain from a 2-month-old child with congenital Zika virus infection. (A) Subcortical band of degenerating cells with prominent calcifications (arrow). (B) Cortex with degenerating neurons (arrowheads). (From Martines RB, Bhatnagar J, de Oliveira Ramos AM, et al. Pathology of congenital Zika syndrome in Brazil: a case series. *Lancet*. 2016;388(10047):898-904.)

and by blood transfusion or organ transplantation. It is estimated to infect as many as 400 million people each year, 100 million of whom are symptomatic and 40,000 of whom die from severe dengue. Dengue is widespread in South and Central America, Mexico, Africa, and Asia. Local vector-borne spread has caused limited infections in the continental United States, the Caribbean region, and European nations including France, Italy, and Spain. Of interest, intentional release of *Aedes* mosquitoes infected with *Wolbachia* bacteria, which decrease viral replication and transmission, has reduced cases of dengue by 95% to 97% in densely populated areas in Columbia, a novel approach to viral control that may be expanded. Clinical manifestations of dengue vary from fever with headache, macular rash, and severe myalgias (breakbone fever) to severe dengue (*dengue hemorrhagic fever*) with bleeding, liver failure, reduced consciousness, organ failure, and plasma leakage leading to shock and respiratory distress. Severe dengue is marked by widespread hemorrhages throughout the body, hepatitis, and diffuse alveolar damage with hyaline membrane formation (Chapter 15).

The immune response against dengue virus is thought to determine, in part, the severity of the infection. There are four serotypes of dengue virus. Infection with each serotype stimulates protective immunity against that serotype, but also stimulates a cross-reactive antibody response that is weak and does not protect against other serotypes of the virus. Severe dengue usually occurs in people who have had a previous infection with a different serotype than the one associated with their severe illness. It is more likely to occur with infection of dengue-2 than other serotypes. Antibody-dependent enhancement, in which the cross-reactive antibodies enhance uptake of virus into macrophages via Fc receptors, is thought to increase infectivity of the virus and contribute to severe dengue. Severe dengue also occurs in infants who have maternal antibodies against dengue virus, consistent with this model.

Poxviruses

Mpox (previously referred to as monkeypox) is a zoonotic viral disease endemic in parts of Africa. In 2022, the WHO declared a Public Health Emergency of International Concern due to a global mpox outbreak that led to more than 85,000 cases and over 100 deaths in over 100 countries, propagated primarily by human-to-human transmission, especially via sexual contact in men who have sex with men in high-resource countries. The virus is transmitted through contact with bodily fluids, skin or mucosal membrane lesions, respiratory droplets, infected animals, and contaminated objects. It can also be transmitted vertically from the pregnant mother transplacentally or during birth. Unlike variola virus, which has no animal reservoir, mpox virus has an animal reservoir that includes squirrels, Gambian rats, dormice, and other animals, and it can be transmitted to monkeys, which are accidental hosts.

This zoonosis was a neglected tropical disease in sub-Saharan Africa for many years prior to the 2022 outbreak. Clade I, with a 10% reported mortality rate, was primarily seen in central Africa and Clade II was primarily in West Africa. The 2022 outbreak was due to clade IIb. By early 2023, the global outbreak had waned due to a multi-pronged control strategy, but endemic areas still have significant numbers of cases. Smallpox vaccination reduces the risk of infection and Mpox cases increased as smallpox vaccination ended worldwide. Thousands of cases are reported in the Democratic Republic of Congo each year. The JYNNEOS vaccine is a smallpox/mpox vaccine that is a live, attenuated, nonreplicating orthopoxvirus from a modified vaccinia strain. Mpox vaccination is estimated to be 66% to 85% effective in preventing infection after two doses.

Mpox disease is similar to, but less severe than, smallpox, which was caused by the closely related variola virus that was eradicated

worldwide in 1980. The genome of mpox is 96% similar to that of the variola virus. It is an enveloped double-stranded DNA virus of the Orthopoxvirus genus with about 200 genes. All poxviruses replicate their nucleic acid in the host cell cytoplasm. Some essential proteins of monkeypox include (1) viral entry protein, (2) proteins that release monkeypox from host cells, and (3) proteins that modulate host cell signaling pathways and host immunity. Mpox does not require a specific host receptor for entry. A few examples, among many, of mpox factors that modulate the host immune response include (1) viral chemokine binding protein, which regulates leukocyte trafficking to tissues; (2) viral cytokine response-modifying protein B, a decoy receptor for TNF; and (3) phosphatase H1, which dephosphorylates STAT1, blocking IFN-induced antiviral responses.

MORPHOLOGY

Mpox, which can last 2 to 4 weeks, presents as a painful or itchy rash on hands, feet, chest, mouth, anus, or genitals. The lesions evolve over 2 to 4 weeks from macules, to papules, to vesicles, to pustules, to ulcerative lesions. The center of the lesion dips before crusting over, then scabs fall off (Fig. 8.11). The lesions can look similar to other rashes, so PCR diagnosis from a lesion specimen is critical. The histopathological findings during the pustular stage include epidermal necrosis, with areas of nonviable keratinocytes and “shadow cell” appearance characterized as enlarged cells with eosinophilic cytoplasm and poorly defined nuclei. The hair follicle and acrosyringial epithelium may be affected. There are pale ballooning keratinocytes, with intracellular edema, **Guarnieri bodies** (a cytoplasmic eosinophilic inclusion), a ground glass appearance of the keratinocyte nuclei with a “basophilic halo,” multinucleation of keratinocytes, dense mixed inflammatory cell infiltrate, and neutrophil exocytosis.

Clinical Features

In addition to the skin lesions, symptoms may include fever, chills, lymphadenopathy, fatigue, muscle aches, headache, and upper respiratory symptoms. Although most cases in the 2022 global outbreak were mild, complications can include bacterial infection of lesions leading to abscess, pneumonia, corneal infection with loss of vision, difficulty swallowing, vomiting, diarrhea with dehydration, sepsis, encephalitis, myocarditis, proctitis, balanitis, urethritis, or death.

Latent Infections (Herpesvirus Infections)

Latency is defined as the persistence of viral genomes in cells without production of infectious virus. Dissemination of the infection and tissue injury stem from reactivation of the latent virus, which leads to viral replication and release of infectious virions. The viruses that most frequently establish latent infections in humans are herpesviruses. These are large, encapsulated viruses with double-stranded DNA genomes that encode approximately 70 proteins. Herpesviruses cause acute infection followed by latent infection in which the viruses persist in a noninfectious form with periodic reactivation and shedding of infectious virus.

There are eight types of human herpesviruses, belonging to three subgroups that are defined by the type of cell most frequently infected and the site of latency.

- The α -group viruses, including HSV-1, HSV-2, and VZV, infect epithelial cells and produce latent infection in postmitotic neurons.
- The β -group viruses include CMV and human herpesviruses 6 and 7 (HHV-6 and HHV-7), which infect and produce latent infection in a variety of cell types. HHV-6 and HHV-7 cause exanthem subitum, also known as *roseola infantum*, and sixth disease, a benign rash of infants, and have been associated with encephalitis, pneumonitis, hepatitis, and myelitis on reactivation.

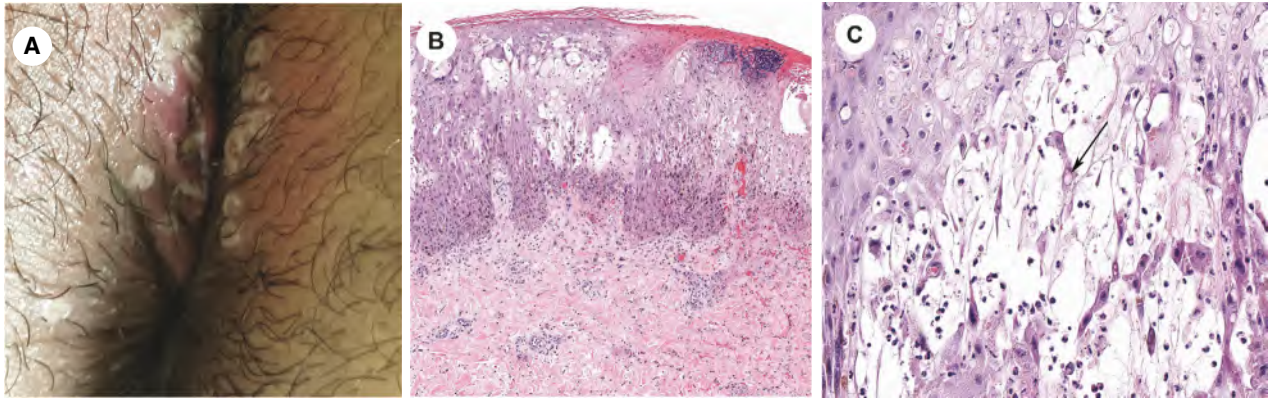


Fig. 8.11 Morphology and histopathology of mpox lesions. (A) Morphology of one stage of mpox skin pseudopustular lesions. (B) Lesion histopathology with spongiosis, ballooning degeneration, and multinucleation of keratinocytes, focal ulceration, dermal edema, and mixed inflammatory infiltrate. (C) Epidermis with ballooning degeneration, multinucleation, and intracytoplasmic eosinophilic inclusion bodies (Guarnieri bodies indicated by arrow). (A, From Prasad S, Galvan Casas C, Strahan AG, et al. A dermatological assessment of 101 mpox (monkeypox) cases from 13 countries during the 2022 outbreak: skin lesion morphology, clinical course, and scarring. *J Am Acad Dermatol.* 2023;88:1066-1073; B and C, from McMahon DE, Schuetz AN, Kovarik CL. Emerging infectious diseases of the skin: a review of clinical and histologic findings. *Hum Pathol.* 2023;140:196-213.)

- The γ -group viruses, including EBV and Kaposi sarcoma herpesvirus (KSHV/HHV-8), produce latent infection mainly in lymphoid cells.
- *Herpesvirus simiae* (monkey B virus) is an Old World monkey virus that resembles HSV-1 and can cause fatal neurologic disease in animal handlers, usually resulting from an animal bite.

Herpes Simplex Virus

Herpes simplex virus (HSV)-1 and HSV-2 are closely related genetically and cause a similar set of primary and recurrent infections. They can be distinguished serologically by antibodies to variant forms of glycoprotein G specific for each virus. Herpes simplex viruses are common throughout the world, with an estimated 4 billion infections globally. Both viruses replicate in the skin and the mucous membranes at their site of entry (usually oropharynx or genitals), where they produce infectious virions and cause vesicular lesions of the epidermis. The viruses spread to sensory neurons that innervate these primary sites of viral replication and release. Viral nucleocapsids are transported along axons to the neuronal cell bodies, where the viruses establish latent infection. In immunocompetent hosts, primary HSV infection resolves in a few weeks, although the virus remains latent in nerve cells. During latency, the viral DNA remains within the nucleus of the neuron, and only latency-associated viral RNA transcripts (LATs) are synthesized. No viral proteins appear to be produced during latency. LATs may contribute to latency by conferring resistance to apoptosis, silencing lytic gene expression through heterochromatin formation, and serving as precursors for micro-RNAs that downregulate expression of critical HSV lytic genes. Reactivation of HSV-1 and HSV-2 may occur repeatedly with or without symptoms, and results in the spread of virus from the neurons to the skin or to mucous membranes. It is proposed that during reactivation, release from epigenetic silencing occurs through a methylation/phosphorylation process that is linked to the cellular stress response pathways in neurons. Reactivation can occur in the presence of host immunity, because HSVs have developed ways to avoid immune recognition. Examples include evasion of antiviral CTLs by inhibiting the MHC class I recognition pathway; neutralizing cross-reactive antibodies by expressing glycoprotein gE, which binds to the IgG Fc domain and interferes with C1q binding

and antibody-dependent cellular cytotoxicity; and blocking the alternative complement pathway by expressing glycoprotein gC, which binds to and accelerates the decay of complement component C3 and its activation products. HSVs can infect multiple cell types, including dendritic cells that are important for the antiviral immune response.

In addition to causing cutaneous and genital lesions, HSV-1 is the major infectious cause of corneal blindness in the United States. Corneal epithelial disease is thought to be due to direct viral damage, and corneal stromal disease appears to be immune-mediated. HSV-1 is also the major cause of fatal sporadic encephalitis in the United States. When the infection spreads to the brain, it usually involves the temporal lobes and orbital gyri of the frontal lobes. Inherited mutations in TLR3 or components of its signaling pathway increase the risk of HSV encephalitis. In addition, neonates and individuals with compromised cellular immunity (e.g., secondary to HIV infection or chemotherapy) may experience disseminated herpesvirus infections. Due to ulceration and dampening of the immune response, HSV-2 infection increases the risk of HIV transmission four-fold.

MORPHOLOGY

HSV-infected cells contain large, pink to purple **intranuclear inclusions (Cowdry type A)** that consist of viral replication proteins and virions at various stages of assembly that push the host cell chromatin out to the edges of the nucleus (Fig. 8.12). Due to cell fusion, HSV also produces inclusion-bearing multinucleated syncytia.

HSV-1 and HSV-2 cause lesions ranging from self-limited cold sores and gingivostomatitis to life-threatening disseminated visceral infections and encephalitis. **Fever blisters** or **cold sores** favor the facial skin around mucosal orifices (lips, nose), where their distribution is frequently bilateral and independent of skin dermatomes. Intraepithelial vesicles (blisters), which are formed by intracellular edema and ballooning degeneration of epidermal cells, frequently burst and crust over, but some may result in superficial ulcerations.

Gingivostomatitis, which is usually encountered in children, is caused primarily by HSV-1. It is a vesicular eruption extending from the tongue to the retropharynx and causing cervical lymphadenopathy. Swollen, painful, erythematous HSV lesions of the fingers or palm (**herpetic whitlow**) occur in infants and, occasionally, in health care workers due to occupational exposure.

Genital herpes is more often caused by HSV-2 than by HSV-1. It is characterized by vesicles on the genital mucous membranes as well as on the external genitalia that are rapidly converted into superficial ulcerations that are rimmed by an inflammatory infiltrate (Chapter 22). Herpesvirus (usually HSV-2) can be transmitted to neonates during passage through the birth canal of infected mothers. Although HSV-2 infection in the neonate may be mild, more commonly it is fulminating and marked by generalized lymphadenopathy, splenomegaly, and necrotic foci throughout the lungs, liver, adrenals, and CNS.

Two forms of corneal lesions are caused by HSV (Chapter 29). **Herpes epithelial keratitis** shows typical virus-induced cytolysis of the superficial epithelium. In contrast, **herpes stromal keratitis** is characterized by infiltrates of mononuclear cells around keratinocytes and endothelial cells, leading to neovascularization, scarring, opacification of the cornea, and eventual blindness. Here, the damage is caused by an immunologic reaction to the HSV infection, rather than the cytopathic effects of the virus itself.

Herpes simplex encephalitis is described in Chapter 28.

Disseminated skin and visceral herpes infections are usually encountered in hospitalized patients with underlying cancer or immunosuppression. **Herpes esophagitis** is frequently complicated by superinfection with bacteria or fungi. **Herpes bronchopneumonia**, sometimes stemming from intubation of a patient with active oral lesions, is often necrotizing, and **herpes hepatitis** may cause liver failure.

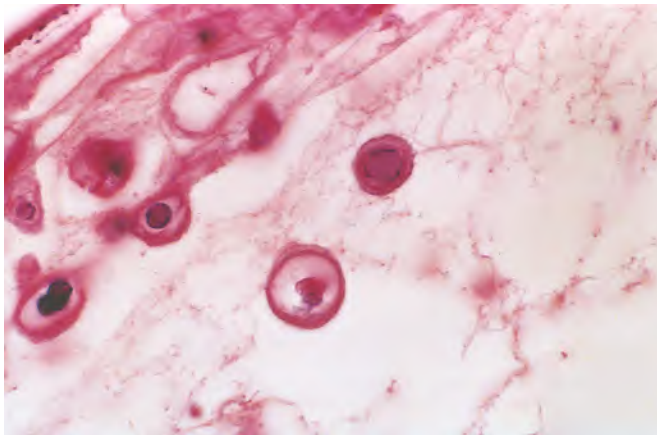


Fig. 8.12 Glassy intranuclear viral inclusion bodies in a herpesvirus blister.

Varicella-Zoster Virus

Acute infection with varicella-zoster virus (VZV) causes chickenpox, and reactivation of latent VZV causes shingles (also called **herpes zoster**). Chickenpox is mild in children but more severe in adults and in immunocompromised people. Shingles is a source of morbidity in older and immunosuppressed persons. Like HSV, VZV infects mucous membranes, skin, and neurons and causes a self-limited primary infection in immunocompetent individuals. Like HSV, VZV evades immune responses and establishes a latent infection in sensory ganglia. VZV, unlike HSV, is transmitted in epidemic fashion by respiratory aerosols, disseminates hematogenously, and causes widespread vesicular skin lesions. Latent VZV infection is seen in neurons and/or satellite cells around neurons in the dorsal root ganglia. Reactivation and clinical recurrences causing shingles are uncommon but may occur many years after the primary infection. Localized recurrence of VZV is most frequent and painful in dermatomes innervated by the trigeminal ganglia, where the virus is most

likely to be latent. Shingles rarely recurs in immunocompetent individuals (in only 1% to 4% of infected individuals), but immunosuppressed or older persons can have multiple recurrences. For this reason, vaccination with a live, attenuated VZV vaccine is recommended in children and a recombinant vaccine is recommended in adults older than 50 years and younger adults who are at increased risk of shingles due to immunodeficiency or immunosuppression. VZV infection is diagnosed primarily by PCR.

MORPHOLOGY

The **chickenpox** rash occurs approximately 2 weeks after respiratory infection. Lesions appear in multiple waves centrifugally from the torso to the head and extremities. Each lesion progresses rapidly from a macule to a vesicle, which resembles a dewdrop on a rose petal. On histologic examination, chickenpox lesions show intraepithelial vesicles (Fig. 8.13) with intranuclear inclusions in epithelial cells at the base of the vesicles. After a few days, most chickenpox vesicles rupture, crust over, and heal by regeneration, leaving no scars. However, bacterial superinfection of vesicles that are ruptured by trauma may lead to destruction of the basal epidermal layer and residual scarring.

Shingles occurs when VZV that has long remained latent in the dorsal root ganglia after a previous chickenpox infection is reactivated and infects sensory nerves that carry it to one or more dermatomes. There, the virus infects keratinocytes and causes vesicular lesions, which, unlike chickenpox, are often associated with intense itching, burning, or sharp pain because of concomitant radiculoneuritis. This pain is especially severe when the trigeminal nerves are involved; rarely, the geniculate nucleus is involved, causing facial paralysis (**Ramsay Hunt syndrome**). The sensory ganglia contain a dense, predominantly mononuclear infiltrate, with herpetic intranuclear inclusions within neurons and their supporting cells (Fig. 8.14). VZV can also cause interstitial pneumonia, encephalitis, transverse myelitis, vasculopathy, and necrotizing visceral lesions, particularly in immunosuppressed people. Similar to HSV, patients with mutations in TLR3 have an increased risk of VZV encephalitis.

Cytomegalovirus

Cytomegalovirus (CMV), a β -group herpesvirus, can produce a variety of disease manifestations, depending on the age of the host, and, more importantly, on the host's immune status. CMV latently infects monocytes and their bone marrow progenitors and can be reactivated

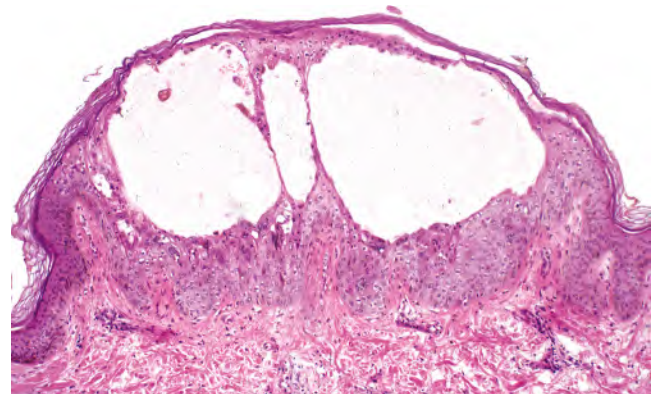


Fig. 8.13 Skin lesion of chickenpox (varicella-zoster virus) with intraepithelial vesicle.

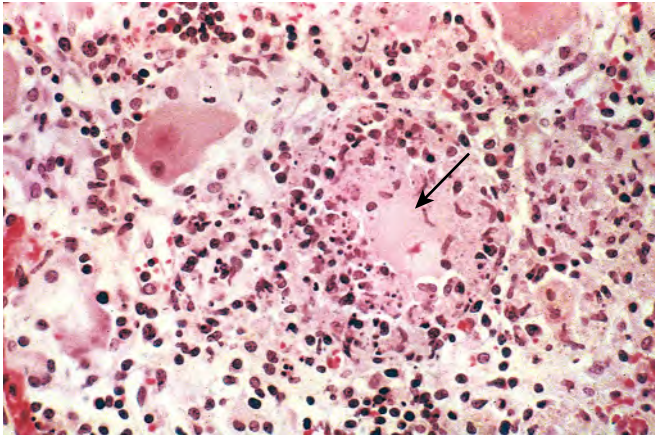


Fig. 8.14 Dorsal root ganglion with varicella-zoster virus infection. Note the ganglion cell necrosis (arrow) and associated inflammation. (Courtesy Dr. James Morris, Radcliffe Infirmary, Oxford, United Kingdom.)

when cellular immunity is depressed. CMV causes an asymptomatic or mononucleosis-like infection in healthy individuals but can cause devastating systemic infections in neonates and in immunocompromised people, in whom the virus may infect many different cell types and tissues. As its name implies, CMV-infected cells exhibit gigantism of both the entire cell and its nucleus, which typically contains a large inclusion surrounded by a clear halo ("owl's eye").

Transmission of CMV can occur by several routes, depending on the age group affected. These include:

- *Transplacental transmission*, from a newly acquired or primary infection in a mother who does not have protective antibodies (congenital CMV)
- *Neonatal transmission*, through cervical or vaginal secretions at birth, or later through breast milk from a mother who has active infection (perinatal CMV)
- *Transmission through saliva* may occur during preschool years, especially in day care centers. Toddlers so infected readily transmit the virus to their parents
- *Transmission by the genital route* is the dominant mode after about 15 years of age. Spread may also occur via respiratory secretions and the fecal-oral route
- *Iatrogenic transmission* can occur at any age through organ transplant or blood transfusion

Acute CMV infection induces transient but severe immunosuppression. CMV can infect dendritic cells and impair antigen processing and the ability of dendritic cells to stimulate T lymphocytes. Like other herpesviruses, CMV can evade immune defenses by down-modulating MHC class I and II molecules and by producing homologues of TNF receptor superfamily members, IL-10, and MHC class I molecules. Interestingly, CMV can evade NK cells by producing ligands that block activating receptors and MHC class I-like proteins that engage inhibitory receptors. Adoptive transfer of CMV-specific T cells has been successfully used to prevent CMV disease after hematopoietic stem cell transplantation, and the CD8⁺ T cell response has been thought to be the most important effector response. CD4⁺ T cells, $\gamma\delta$ T cells, and NK cells may also play a role in immune control of the infection. CMV encodes several Fc γ -binding glycoproteins that effectively operate as adversaries of host Fc γ receptors, inhibiting IgG-mediated immunity. Thus, CMV both hides from and actively suppresses immune responses.

MORPHOLOGY

Infected cells are strikingly enlarged, often to a diameter of 40 μ m, and show cellular and nuclear pleomorphism. Prominent **intranuclear basophilic inclusions** spanning one-half of the nuclear diameter are usually set off from the nuclear membrane by a clear halo (Fig. 8.15). Within the cytoplasm of infected cells, smaller basophilic inclusions can also be seen. Parenchymal epithelial cells are infected in glandular organs, as well as neurons in the brain, alveolar macrophages, epithelial cells, and endothelial cells in the lungs, and tubular epithelial and glomerular endothelial cells in the kidneys. CMV causes focal necrosis with minimal inflammation in virtually any organ.

Congenital Infections. Infection acquired in utero may take several forms and in many cases is asymptomatic. About 1 in 200 babies is born with congenital CMV, 1 in 5 of whom will have some long-term health consequence. When the virus is acquired from a mother with primary infection (who does not have protective antibodies), classic cytomegalic inclusion disease can develop. Affected infants may experience intrauterine growth restriction and present with jaundice, hepatosplenomegaly, anemia, bleeding due to thrombocytopenia, and encephalitis. In fatal cases, the brain is often smaller than normal (microcephaly) and may show foci of calcification. Diagnosis of neonatal CMV is usually made by PCR of viral DNA in urine or saliva.

Infants who survive usually have permanent deficits, including intellectual disability, hearing loss, and other neurologic impairments. The congenital infection is not always devastating, however, and may take the form of interstitial pneumonitis, hepatitis, or a hematologic disorder. Most infants with this milder form of cytomegalic inclusion disease recover, although a few develop intellectual disability later. Uncommonly, a totally asymptomatic infection may be followed months to years later by neurologic sequelae, including delayed-onset intellectual disability and deafness.

Perinatal Infections. Infection acquired during passage through the birth canal or from breast milk is usually asymptomatic due to protective maternal anti-CMV antibodies, which are transmitted to the fetus. Despite the lack of symptoms, many of these infants continue to excrete CMV in their urine or saliva for months to years. Subtle effects on hearing and intelligence later in life have been reported in some studies. Much less commonly, infected infants develop interstitial pneumonitis, failure to thrive, rash, or hepatitis.

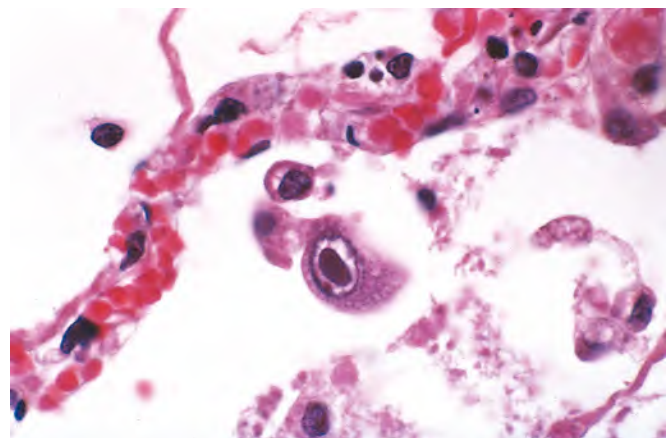


Fig. 8.15 Cytomegalovirus: distinct nuclear and ill-defined cytoplasmic inclusions in the lung.

CMV Mononucleosis. In healthy young children and adults, the disease is nearly always asymptomatic. The CMV seroprevalence worldwide is about 80%. The most common clinical manifestation of CMV infection in immunocompetent hosts beyond the neonatal period is an infectious mononucleosis–like illness, with fever, atypical lymphocytosis, lymphadenopathy, and hepatitis, marked by hepatomegaly and abnormal liver function tests. The diagnosis is made by serology. Most people recover without any sequelae, but the virus may continue to be excreted in body fluids for months to years. Irrespective of the presence or absence of symptoms, infected individuals remain seropositive for life, and the virus is never cleared, persisting in latently infected leukocytes.

CMV in Immunosuppressed Individuals. Immunocompromised individuals (e.g., transplant recipients and HIV-infected individuals) are susceptible to severe CMV infection, which may be either a new infection or reactivation of latent CMV. In the past, CMV was the most common opportunistic viral pathogen in AIDS. Although CMV is still a significant cofactor in HIV disease, especially in Africa, the frequency of serious CMV infection in HIV-positive people has been greatly reduced by antiretroviral treatment. Recipients of solid-organ transplants (heart, liver, kidney) also may contract CMV from the donor organ.

In all these settings, serious, even life-threatening, disseminated CMV infections in immunosuppressed people primarily affect the lungs (pneumonitis) and gastrointestinal tract (colitis). With pulmonary infection, an interstitial mononuclear infiltrate with foci of necrosis develops, accompanied by the typical enlarged cells with inclusions. The pneumonitis can progress to full-blown acute respiratory distress syndrome. Intestinal necrosis and ulceration can develop and be extensive, leading to the formation of pseudomembranes and debilitating diarrhea. CMV has been associated with vasculopathy and neurocognitive impairment. Diagnosis of CMV infection is made by demonstration of characteristic morphologic alterations in tissue sections, viral culture, rising antiviral antibody titer, and PCR-based detection of CMV DNA. Quantitative PCR-based assays have revolutionized the approach to monitoring CMV infection in people after transplantation.

Chronic Productive Infections

In some infections, the immune system is unable to eliminate the virus, and continued viral replication leads to persistent viremia. The high mutation rate of viruses such as HIV and HBV may contribute to their escape from control by the immune system. HIV and HBV infection are described in Chapters 6 and 18, respectively.

Transforming Viral Infections

Some viruses can transform infected cells into benign or malignant tumor cells. Oncogenic viruses can stimulate cell growth and survival by a variety of mechanisms. Several viruses have been implicated in the causation of human cancer, including EBV, HPV, HBV, and HTLV-1 (discussed in Chapters 7, 13, 18, and 22).

Epstein-Barr Virus

Epstein-Barr virus (EBV) causes infectious mononucleosis, a benign, self-limited lymphoproliferative disorder, and is associated with the pathogenesis of several human tumors, most commonly certain lymphomas and nasopharyngeal carcinoma (Chapters 7 and 13).

Infectious mononucleosis is characterized by fever, sore throat, generalized lymphadenopathy, splenomegaly, and the appearance in the blood of atypical activated T lymphocytes (mononucleosis cells). Some people develop hepatitis, meningoencephalitis, and pneumonitis.

Infectious mononucleosis occurs principally in late adolescents or young adults among upper socioeconomic classes in highly resourced nations. In the rest of the world, primary infection with EBV occurs in childhood and is usually asymptomatic.

Pathogenesis

EBV is transmitted by close human contact, frequently through the saliva during kissing. Infection of B cells may take one of two forms. In a minority of B cells, infection is lytic, leading to viral replication and eventual cell lysis accompanied by release of virions, which may infect other B cells. In most B cells, however, EBV establishes latent infection, during which the virus persists as an extrachromosomal episome.

Most B cells that are latently infected with EBV are activated by expression of EBV proteins such as LMP1 and EBNA2, which drive the proliferation and dissemination of the infected cells. This uncontrolled, expanding polyclonal population of EBV-infected B cells secretes antibodies with many specificities, including antibodies that recognize sheep or horse red cells. These so-called *heterophile antibodies* are detected in diagnostic tests for mononucleosis. EBV-infected B cells may also produce autoantibodies, for example against platelets, leading to transient immune-mediated thrombocytopenia in a small subset of patients with mononucleosis.

The symptoms of infectious mononucleosis appear on initiation of the host immune response. The *atypical lymphocytes* seen in the blood, characteristic of this disease, are mainly EBV-specific CD8⁺ CTLs, but also include NK cells. The reactive proliferation of T cells is largely centered in lymphoid tissues, which accounts for the lymphadenopathy and splenomegaly. Early in the course of the infection, IgM antibodies are formed against viral capsid antigens; later, IgG antibodies are formed that persist for life. In otherwise healthy persons, the fully developed humoral and cellular responses to EBV act as brakes on viral shedding, resulting in the elimination of B cells expressing the full complement of EBV latency-associated genes. However, a small fraction of infected B cells downregulate the expression of immunogenic EBV polypeptides and persist indefinitely. In hosts with acquired defects in cellular immunity, reactivation of expression of the full panoply of EBV-encoded proteins in latently infected B cells may lead to lymphoma (Chapters 7 and 13). Studies have shown a sharply increased risk of multiple sclerosis in individuals who have been infected with EBV, but the mechanism underlying this association remains to be elucidated.

MORPHOLOGY

The major alterations in infectious mononucleosis involve the blood, lymph nodes, spleen, liver, CNS, and, occasionally, other organs. The **peripheral blood** shows absolute lymphocytosis; more than 60% of white blood cells are lymphocytes. Between 5% and 80% of these are large, **atypical lymphocytes**, 12 to 16 μm in diameter, characterized by an abundant cytoplasm containing multiple clear vacuolations; an oval, indented, or folded nucleus; and scattered cytoplasmic azurophilic granules (Fig. 8.16). These atypical lymphocytes, most of which express CD8, are sufficiently distinctive to strongly suggest the diagnosis.

The **lymph nodes** are typically discrete and enlarged throughout the body, particularly in the posterior cervical, axillary, and inguinal regions. On histologic examination, the most striking feature is the expansion of paracortical areas due to activation of T cells (immunoblasts). A minor population of EBV-infected B cells expressing Epstein-Barr nuclear antigen 2 (EBNA2), latent membrane protein 1 (LMP1), and other latency-specific genes can also be detected in the paracortex using antibodies specific for the gene products. EBV-infected B cells resembling Reed-Sternberg cells (the malignant cells of Hodgkin lymphoma, Chapter 13) may be found. B-cell areas (follicles) may also show mild hyperplasia. The T-cell proliferation is sometimes so exuberant that it is difficult to distin-

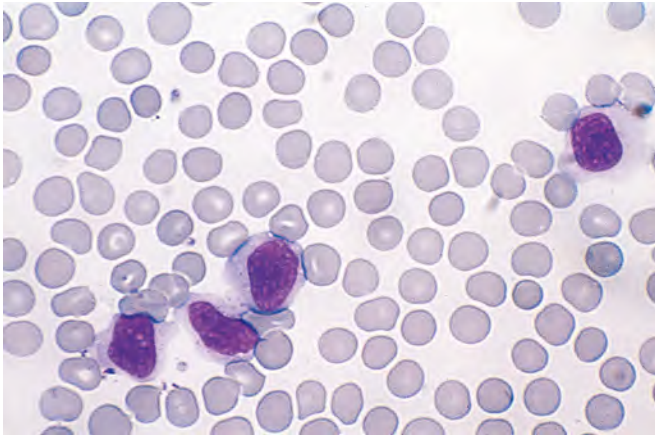


Fig. 8.16 Atypical lymphocytes in infectious mononucleosis.

guish the nodal morphology from that seen in malignant lymphomas. Similar changes commonly occur in the tonsils and lymphoid tissue of the oropharynx.

The **spleen** is enlarged in most cases, weighing between 300 and 500 g. It is usually soft and fleshy, with a hyperemic cut surface. The histologic changes are analogous to those of the lymph nodes, showing an expansion of white pulp follicles and red pulp sinusoids due to the presence of numerous activated T cells. These spleens are especially vulnerable to rupture, possibly in part because the rapid increase in size produces a tense, fragile splenic capsule.

The **liver** is usually involved to some degree, although hepatomegaly is at most moderate. On histologic examination, atypical lymphocytes are seen in the portal areas and sinusoids, and scattered, isolated cells or foci of parenchymal necrosis may be present. This histologic picture is similar to that of other forms of viral hepatitis.

Clinical Features

EBV infection in young children classically presents with fever, sore throat, and lymphadenitis. However, malaise, fatigue, and lymphadenopathy are the common presentation in young adults with infectious mononucleosis and can raise the specter of leukemia or lymphoma, hepatitis resembling one of the hepatotropic viral syndromes, or a febrile rash resembling rubella. The diagnosis depends on the following findings (in increasing order of specificity): (1) lymphocytosis with the characteristic atypical lymphocytes in the peripheral blood, (2) a positive heterophile antibody reaction (Monospot test), and (3) a rising titer of antibodies specific for EBV antigens (viral capsid antigens, early antigens, or EBNA). In most patients, infectious mononucleosis resolves within 4 to 6 weeks, but sometimes the fatigue lasts longer. Hepatosplenomegaly, jaundice, and some degree of hepatitis are common. Rarer complications include immune cytopenias, hemophagocytic lymphohistiocytosis (HLH, Chapter 13), and diverse abnormalities of the nervous system, kidneys, bone marrow, lungs, eyes, and heart. Splenic rupture is rare, but can occur even with minor trauma, leading to hemorrhage that may be fatal.

Serious consequences of EBV infection occur in individuals with X-linked lymphoproliferative disease (also known as *Duncan disease*), a disorder caused by mutations in the *SH2D1A* gene, which encodes a signaling protein that participates in T-cell and NK-cell activation and antibody production. This rare inherited immunodeficiency is characterized by an ineffective immune response to EBV. Patients are usually normal until they are acutely infected with EBV, often during adolescence. In more than half the cases, EBV causes an acute overwhelming infection that may be fatal. Others succumb to EBV-positive B-cell lymphoma or infections related to hypogammaglobulinemia.

BACTERIAL INFECTIONS

Different classes of bacteria are responsible for diverse infections (Table 8.6). Selected examples of the most common or clinically significant infections are discussed next.

Gram-Positive Bacterial Infections

Common gram-positive cocci include *Staphylococcus* spp., *Streptococcus* spp., and *Enterococcus* spp., each of which causes many types of infections. Diphtheria, listeriosis, anthrax, and nocardiosis are less common infections caused by gram-positive rods.

Staphylococcal Infections

S. aureus causes a myriad of skin lesions (boils, carbuncles, impetigo, and scalded-skin syndrome) as well as abscesses, sepsis, osteomyelitis, pneumonia, endocarditis, food poisoning, and toxic shock syndrome (Fig. 8.17). There are more than 1 million infections in the United States alone, with 20,000 attributable deaths per year. An estimated 20% of individuals are persistent nasal carriers of *S. aureus*. Patients with an *S. aureus* bacteremia have a 27% mortality rate within 3 months of infection. Since 2005, there has been some success in combating staphylococcal infections in hospitals due to extensive screening policies. *S. aureus* is a pyogenic gram-positive coccus that forms clusters resembling bunches of grapes. Coagulase-negative staphylococci, such as *S. epidermidis*, characteristically cause opportunistic infections in catheterized patients, patients with prosthetic cardiac valves, and intravenous drug users. *S. saprophyticus* is a common cause of urinary tract infection in young females.

Pathogenesis

S. aureus produces a multitude of virulence factors, which include surface proteins involved in adherence and evasion of the host immune response, secreted enzymes that degrade host structures, secreted toxins that damage host cells, and proteins that cause antibiotic resistance (see Fig. 8.17).

***S. aureus* Surface Proteins.** *S. aureus* expresses surface receptors for fibrinogen (called clumping factor), fibronectin, and vitronectin and uses these molecules to bind to host endothelial cells. *S. aureus* has a polysaccharide capsule, which enables attachment to artificial materials such as prosthetic valves as a biofilm, resulting in significant prosthetic valve and catheter-associated infection and a resistance to host cell phagocytosis. *S. aureus* also expresses protein A, which binds the Fc portion of immunoglobulins, allowing the organism to escape antibody-mediated killing.

***S. aureus* Toxins.** *S. aureus* produces multiple *membrane-damaging (hemolytic) toxins*. These include α -toxin, a protein that intercalates into the plasma membrane of host cells, forming pores that allow toxic levels of calcium to leak into cells; β -toxin, a sphingomyelinase; and δ -toxin, which is a detergent-like peptide. *S. aureus* γ -toxin and leukocidin lyse red cells and phagocytes, respectively.

The *exfoliative A and B toxins* produced by *S. aureus* are serine proteases that cleave the desmosomal protein desmoglein 1, which normally holds epidermal cells tightly together. This causes keratinocytes to detach from one another and from the underlying basement membrane, resulting in a loss of barrier function that often leads to secondary skin infections. Exfoliation may occur locally at the site of infection (bullous impetigo) or may result in widespread loss of the superficial epidermis (staphylococcal scalded-skin syndrome).

Superantigens produced by *S. aureus* cause food poisoning and toxic shock syndrome (TSS). TSS came to public attention in the later

TABLE 8.6 Selected Human Bacterial Pathogens and Associated Diseases

Organ System	Species	Frequent Disease Presentations
Respiratory	<i>Streptococcus pyogenes</i>	Pharyngitis
	<i>Corynebacterium diphtheriae</i>	Diphtheria
	<i>Bordetella pertussis</i>	Pertussis
	<i>Streptococcus pneumoniae</i>	Lobar pneumonia
	<i>Mycobacterium tuberculosis</i>	Tuberculosis
	<i>Legionella pneumophila</i>	Legionnaires' disease
Gastrointestinal	<i>Helicobacter pylori</i>	Peptic ulcers
	<i>Vibrio cholerae</i> , enterotoxigenic <i>Escherichia coli</i>	Noninflammatory gastroenteritis
	<i>Shigella</i> spp., <i>Salmonella</i> spp., <i>Campylobacter jejuni</i> , enterohemorrhagic <i>Escherichia coli</i>	Inflammatory gastroenteritis
	<i>Salmonella typhi</i>	Enteric (typhoid) fever
	<i>Clostridioides difficile</i>	Pseudomembranous colitis
Nervous system	<i>Neisseria meningitidis</i> , <i>Streptococcus pneumoniae</i> , <i>Haemophilus influenzae</i> , <i>Listeria monocytogenes</i>	Acute meningitis
	<i>Clostridium tetani</i> , <i>Clostridium botulinum</i>	Paralytic intoxications, tetanus, and botulism
Urogenital	<i>Escherichia coli</i> , <i>Pseudomonas aeruginosa</i> , <i>Enterococcus</i> spp.	Urinary tract infections
	<i>Neisseria gonorrhoeae</i>	Gonorrhea
	<i>Chlamydia trachomatis</i>	Chlamydia
	<i>Treponema pallidum</i>	Syphilis
Skin and adjacent soft tissue	<i>Staphylococcus aureus</i>	Abscess, cellulitis
	<i>Streptococcus pyogenes</i>	Impetigo, erysipelas, necrotizing fasciitis
	<i>Clostridium perfringens</i>	Gas gangrene
	<i>Bacillus anthracis</i>	Cutaneous anthrax
	<i>Pseudomonas aeruginosa</i>	Burn infections
	<i>Mycobacterium leprae</i>	Leprosy
Disseminated infections	<i>Yersinia pestis</i>	Plague
	<i>Borrelia burgdorferi</i>	Lyme disease
	<i>Brucella</i> spp.	Brucellosis (undulant fever)
Disseminated neonatal infection	<i>Streptococcus agalactiae</i> , <i>Listeria monocytogenes</i>	Neonatal bacteremia, meningitis
	<i>Treponema pallidum</i>	Congenital syphilis

1970s because of its association with the use of hyperabsorbent tampons, which became colonized with *S. aureus* during use. TSS is now rare, but still has a mortality rate of 8% and it is now clear that TSS can be caused by growth of *S. aureus* at many sites, most commonly the vagina and infected surgical sites. This syndrome is characterized by hypotension (shock), renal failure, coagulopathy, liver disease, respiratory distress, a generalized erythematous rash, and soft-tissue necrosis at the site of infection. If not promptly treated, it can be fatal. TSS can also be caused by *S. pyogenes*. Bacterial superantigens cause polyclonal T-cell proliferation by binding to conserved portions of MHC molecules and to relatively conserved portions of T-cell receptor β chains. In this manner, superantigens may stimulate up to 20% of T lymphocytes, leading to release of cytokines such as TNF and IL-1, in such large amounts that they may trigger the systemic inflammatory response syndrome (Chapter 4). Superantigens produced by *S. aureus* also cause vomiting when ingested in food, presumably by affecting the CNS or the enteric nervous system.

Antibiotic resistance is a significant problem in treatment of *S. aureus* infections. Methicillin-resistant *S. aureus* (MRSA) bacteria are resistant to all penicillin and nearly all cephalosporin antibiotics due to an altered penicillin-binding protein encoded by the *mecA* gene, a part of a larger mobile genetic element. Previously, MRSA was mainly found in health care facilities, but community-acquired MRSA infections have been recognized since the 1980s and are common. It is estimated that 25% to 50% of staphylococcal infections in hospitals are due to MRSA and patients with a MRSA infection are 64% more likely to die than those with a more susceptible strain.

MORPHOLOGY

Whether the lesion is located in the skin, lungs, bones, or heart valves, *S. aureus* causes pyogenic inflammation that is distinctive for its local destruction of host tissue.

Excluding impetigo, which is restricted to the superficial epidermis, staphylococcal skin infections are centered around the hair follicles where they begin. A **furuncle**, or **boil**, is a focal suppurative inflammation of the skin and subcutaneous tissue. They may be solitary or multiple, or recur in successive crops. Furuncles are most frequent in moist, hairy areas, such as the face, axillae, groin, legs, and submammary folds. Beginning in a single hair follicle, a boil develops into a growing and deepening abscess that eventually “comes to a head” by thinning and rupturing the overlying skin. A **carbuncle** is a deeper suppurative infection that spreads laterally beneath the deep subcutaneous fascia and then burrows superficially to erupt in multiple adjacent skin sinuses. Carbuncles typically appear beneath the skin of the upper back and posterior neck, where fascial planes favor their spread. **Hidradenitis** is chronic suppurative infection of apocrine glands, most often in the axilla. Infections of the nail bed (**paronychia**) or on the palmar side of the fingertips (**felons**) are exquisitely painful. They may follow trauma or embedded splinters and, if deep enough, destroy the bone of the terminal phalanx or detach the fingernail.

S. aureus lung infections (Fig. 8.18) have a polymorphonuclear infiltrate similar to that of *S. pneumoniae* infections, but they cause much more tissue destruction. Lung infections usually arise from a hematogenous source, such as an infected thrombus, or in the setting of a predisposing condition such as influenza.

Staphylococcal scalded-skin syndrome, also called **Ritter disease**, most frequently occurs in children with *S. aureus* infection of the nasopharynx or skin. There is a sunburn-like rash that spreads over the entire body and evolves into fragile bullae that lead to partial or total skin loss. The desquamation of the epidermis in staphylococcal scalded-skin syndrome occurs at the level of the granulosal layer, distinguishing it from toxic epidermal necrolysis, or Lyell disease, which is secondary to drug hypersensitivity and causes desquamation at the level of the epidermal-dermal junction (Chapter 25).

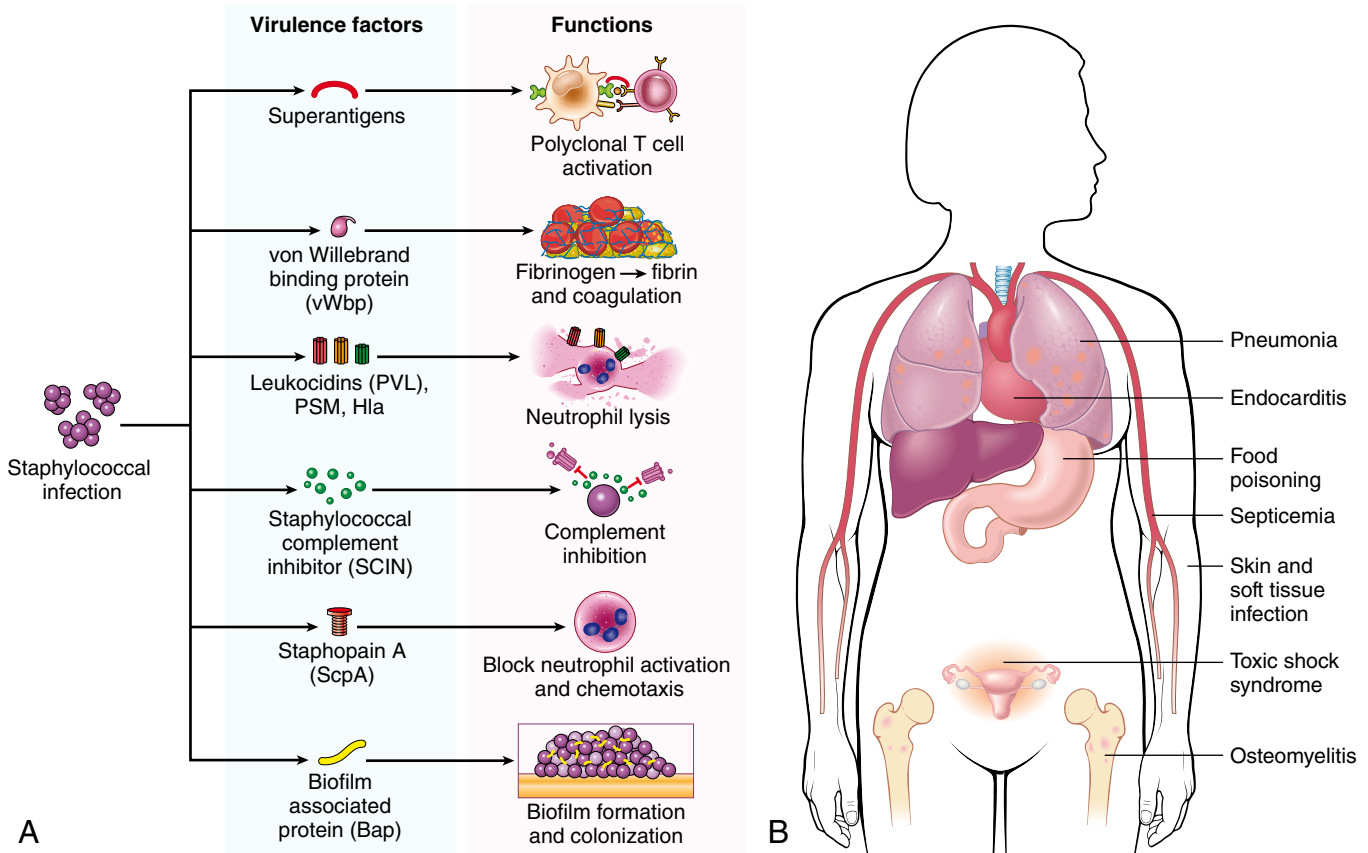


Fig. 8.17 *Staphylococcus aureus* infection. (A) Virulence factors of *S. aureus* and their functions. (B) Clinical manifestations of *S. aureus* infection. *Hla*, Alpha-hemolysin; PSM, phenol-soluble modulins; PVL, Pantone-Valentine leucocidin.

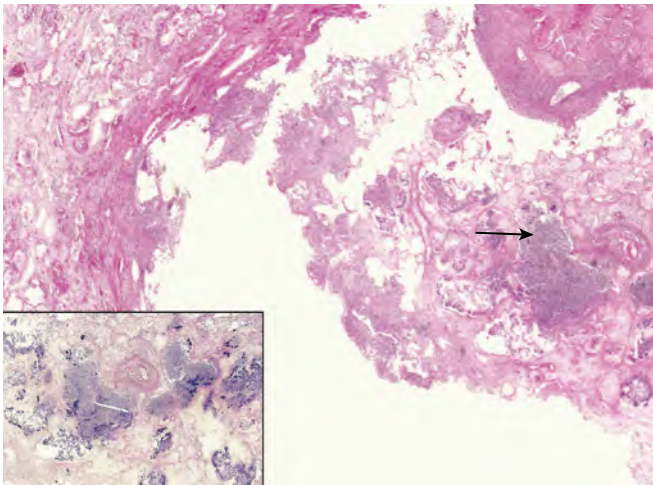


Fig. 8.18 Staphylococcal abscess of the lung with extensive neutrophilic infiltrates (arrow) and destruction of the alveoli. Inset shows the same area on Gram stain highlighting clusters of bacteria.

Streptococcal and Enterococcal Infections

Streptococci cause suppurative infections of the skin, oropharynx, lungs, and heart valves. They are also responsible for several postinfectious syndromes, including rheumatic fever (Chapter 12), poststreptococcal glomerulonephritis (Chapter 20), and erythema

nodosum (Chapter 25). These bacteria are gram-positive cocci that grow in pairs or chains. β -hemolytic streptococci are typed according to their surface carbohydrate antigens. *S. pyogenes* (group A) causes pharyngitis, scarlet fever, erysipelas, impetigo, rheumatic fever, TSS, and glomerulonephritis. *Streptococcus agalactiae* (group B) colonizes the female genital tract and causes chorioamnionitis in pregnancy. It is the leading cause of neonatal sepsis and meningitis in many developed and developing countries. *S. pneumoniae*, the most important α -hemolytic streptococcus, is a common cause of community-acquired pneumonia in older adults and meningitis in children and adults (Chapter 15). There are several effective vaccines against *S. pneumoniae*, including several conjugate vaccines and a polysaccharide vaccine, recommended for young children, older adults, and those with specific risk factors. The *viridians* group streptococci include α -hemolytic and nonhemolytic streptococci found in normal oral microbiota that are a common cause of endocarditis. *Streptococcus mutans* is the major cause of dental caries. Streptococcal infections are diagnosed by culture, and, in those with pharyngitis, by the rapid streptococcal antigen test.

Enterococci are commensal gut bacteria and are the most common cause of nosocomial (hospital-acquired) urinary tract infections. Enterococci are responsible for 10% of endocarditis cases worldwide. The death rates from enterococcal bacteremia are 13% to 39%. *E. faecalis* and *E. faecium* are the most common species; they also grow in pairs and chains and are hence difficult to distinguish from streptococci by morphology alone. They cause meningitis, intraabdominal infections, and wound infections.

Pathogenesis

The different species of streptococci produce many virulence factors and toxins. *S. pyogenes*, *S. agalactiae*, and *S. pneumoniae* have capsules that resist phagocytosis. *S. pyogenes* also expresses M protein, a surface protein that prevents bacteria from being phagocytosed, and a complement C5a peptidase that degrades this chemotactic peptide. *S. pyogenes* secretes a phage-encoded pyrogenic exotoxin that causes fever and rash in scarlet fever. Rheumatic fever is probably caused by antistreptococcal M protein antibodies and T cells that cross-react with cardiac proteins. Virulent *S. pyogenes* have been called flesh-eating bacteria because they cause a rapidly progressive necrotizing fasciitis. Although the antiphagocytic capsule is the most important virulence factor of *S. pneumoniae*, this organism also produces pneumolysin, a toxin that inserts into host cell membranes and lyses cells, greatly increasing tissue damage. *S. mutans* produces caries by metabolizing sucrose to lactic acid (which causes demineralization of tooth enamel) and by secreting high-molecular-weight glucans that promote aggregation of bacteria and plaque formation.

Enterococci do not have toxins like *Staphylococcus* and *Streptococcus*, but they do have an antiphagocytic capsule responsible for biofilm formation and produce enzymes that injure host tissues. They persist in the environment and are the second most common cause of hospital-acquired infection. Enterococci have intrinsic resistance to multiple antibiotics and transfer additional resistance on mobile genetic elements. One important mechanism of vancomycin resistance in enterococci is mediated by the *van A* gene, which encodes a factor that catalyzes the replacement of D-Ala-D-alanine in the peptidoglycan of the cell with D-Ala-D-lactate, which reduces the binding affinity of vancomycin. The incidence of enterococcal infection is higher in immunosuppressed patient populations, such as those undergoing organ or stem cell transplantation, who receive frequent antimicrobial agents and can have altered commensal microbiota.

MORPHOLOGY

Streptococcal infections are characterized by diffuse interstitial neutrophilic infiltrates with minimal destruction of host tissues. The skin lesions caused by streptococci (furuncles, carbuncles, and impetigo) resemble those of staphylococci.

Erysipelas is caused by exotoxins produced by *S. pyogenes*. It is characterized by rapidly spreading erythematous cutaneous swelling that may begin on the face or, less frequently, on the body or an extremity. The rash has a sharp, well-demarcated, serpiginous border and may form a “butterfly” distribution on the face (Fig. 8.19). On histologic examination, there is a diffuse, edematous, neutrophilic inflammatory reaction in the dermis and epidermis extending into the subcutaneous tissues. Microabscesses may be formed, but tissue necrosis is usually minor.

Streptococcal pharyngitis, which is the major antecedent of poststreptococcal glomerulonephritis (Chapter 20), is marked by edema, epiglottic swelling, and punctate abscesses of the tonsillar crypts, sometimes accompanied by cervical lymphadenopathy. Swelling associated with severe pharyngeal infection may encroach on the airways, especially if there is peritonsillar or retropharyngeal abscess formation.

Scarlet fever, associated with pharyngitis caused by *S. pyogenes*, is most common between 3 and 15 years of age. It is manifested by a punctate erythematous rash that is most prominent over the trunk and inner aspects of the arms and legs. The face is also involved, but usually a small area about the mouth remains relatively unaffected, producing circumoral pallor. The skin usually becomes hyperkeratotic and scaly during defervescence.

S. pneumoniae is an important cause of lobar pneumonia (Chapter 15).



Fig. 8.19 Streptococcal erysipelas.

Diphtheria

Diphtheria is caused by *Corynebacterium diphtheriae*, a slender gram-positive rod with clubbed ends that spreads from person-to-person in respiratory droplets or skin exudate. Diphtheria was once a major cause of childhood mortality, but cases are now rare in the United States. However, the World Health Organization reported more than 5000 cases globally in 2022, with large outbreaks still occurring in areas with displaced populations and infrastructure failures. Respiratory diphtheria causes pharyngeal or, less often, nasal or laryngeal infection. Damage to the heart, nerves, and other organs may be present. Cutaneous diphtheria causes chronic ulcers with a dirty gray membrane but does not cause systemic damage. *C. diphtheriae* produces a phage-encoded A-B toxin that blocks host cell protein synthesis. The A fragment does this by catalyzing the covalent transfer of adenosine diphosphate (ADP)-ribose to elongation factor-2 (EF-2). This inhibits EF-2 function, which is required for the translation of mRNA into protein. A single molecule of diphtheria toxin can kill a cell by ADP-ribosylating, and thereby inactivating, more than a million EF-2 molecules. Immunization with diphtheria toxoid (formalin-fixed toxin) stimulates production of toxin-neutralizing antibodies that protect people from the lethal effects of the toxin.

MORPHOLOGY

Inhaled *C. diphtheriae* carried in respiratory droplets proliferate at the site of attachment on the mucosa of the nasopharynx, oropharynx, larynx, or trachea. The bacteria also form satellite lesions in the esophagus or lower airways. Release of exotoxin causes necrosis of the epithelium, accompanied by an outpouring of a dense fibrinosuppurative exudate. The coagulation of this exudate on the ulcerated necrotic surface creates a tough, dirty, gray to black superficial membrane, sometimes called **pseudo-membrane** because it is not formed by viable tissue (Fig. 8.20). There is an intense neutrophilic infiltrate in the underlying tissues with marked vascular congestion, interstitial edema, and fibrin exudation. When the membrane sloughs off its inflamed and vascularized bed, bleeding and asphyxiation may occur. With control of the infection, the membrane is coughed up or removed by enzymatic digestion, and the inflammatory reaction subsides.

Although the bacterial invasion remains localized, with entry of exotoxin into the blood and its systemic distribution, there may be fatty change in the myocardium with isolated myofiber necrosis, polyneuritis with degeneration of the myelin sheaths and axis cylinders, and (less commonly) fatty change and focal necrosis of parenchymal cells in the liver, kidneys, and adrenals.

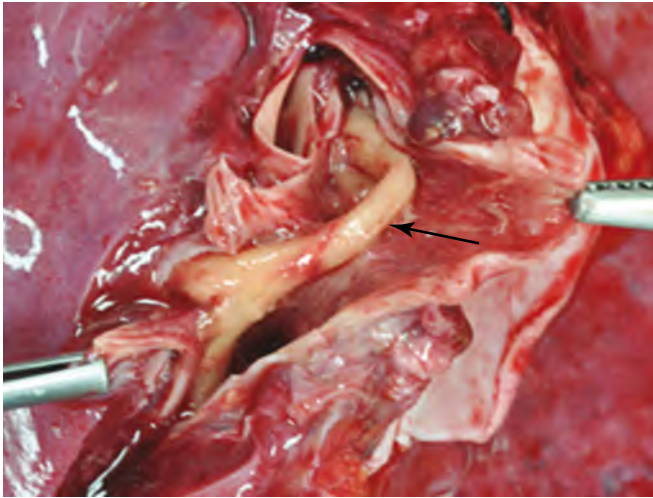


Fig. 8.20 Pseudo-membrane of diphtheria (arrow) lying within a transverse bronchus. (Courtesy Dr. Robin A. Cooke, Department of Anatomical Pathology, Princess Alexandra Hospital, Brisbane, Australia.)

Listeriosis

Listeria monocytogenes is a gram-positive bacillus that causes gastroenteritis in most individuals who ingest it in sufficient quantity, and severe food-borne infections in vulnerable hosts. Outbreaks of *L. monocytogenes* infection have been linked to many foods, but most are associated with contaminated dairy products, prepared meats, or processed fruits and vegetables. In the United States, all clinical isolates of *L. monocytogenes* have been typed for epidemiologic surveillance since 1998, resulting in a five-fold increase in the number of outbreaks detected. Pregnant females, neonates, older adults, and immunosuppressed persons are particularly susceptible to severe *L. monocytogenes* infection. In pregnant females, *L. monocytogenes* causes an amnionitis that may result in abortion, stillbirth, or neonatal sepsis. In neonates and immunosuppressed adults, it can cause disseminated disease (granulomatosis infantiseptica of the newborn) and exudative meningitis.

L. monocytogenes is a facultative intracellular pathogen. The bacteria bind to receptors on host epithelial cells and macrophages and are phagocytosed. The bacteria escape from the phagolysosome using a pore-forming protein, listeriolysin O, and two phospholipases. Rapid replication of the bacteria in the host cytosol depends on uptake of intermediates of host cell glucose; this uptake is mediated by a hexose phosphate permease, Hpt, which is upregulated in the environment of the host cell cytosol. Listeriolysin O also affects membrane-bound organelles, causing rounding and shrinking of mitochondria, inhibition of the endoplasmic reticulum, and damage to the lysosomal membrane. In the host cell cytoplasm, Act A, a bacterial surface protein, binds to the Arp2/3 complex, an actin nucleating complex, and induces actin polymerization. This generates force to propel the bacteria into membrane-bound protrusions of the infected cell that are phagocytosed by adjacent cells and macrophages, after which the bacteria escape from the membrane-bound vacuoles and begin another round of replication. Resting macrophages fail to kill the intracellular bacteria and may facilitate their spread in the host, whereas macrophages that are activated by IFN- γ can. Accordingly, an effective host response to *L. monocytogenes* depends on IFN- γ produced by NK cells early in the course of the infection and Th1 and CD8⁺ T cells in chronic infection. Patients with defects in cell-mediated immunity, such as those with reduced levels of CD4⁺ lymphocytes, are at increased risk for listeriosis.

MORPHOLOGY

In acute infections, *L. monocytogenes* evokes an exudative pattern of inflammation with numerous neutrophils. The meningitis due to *L. monocytogenes* is macroscopically and microscopically indistinguishable from meningitis due to other pyogenic bacteria (Chapter 28). **The finding of gram-positive bacilli in the CSF is virtually diagnostic.** More varied lesions may be encountered in neonates and immunosuppressed adults. Focal abscesses alternating with grayish or yellow nodules representing necrotic amorphous tissue debris can occur in any organ, including the lung, liver, spleen, and lymph nodes. In infections of longer duration, macrophages appear in large numbers, but granulomas are rare. Infants born with *L. monocytogenes* sepsis often have a papular red rash over the extremities, and listerial abscesses can be seen in the placenta. A smear of the meconium will disclose the gram-positive bacilli.

Anthrax

Anthrax is characterized by necrotizing inflammatory lesions in the skin, gastrointestinal tract, or other organs. It is caused by *Bacillus anthracis*, a large, spore-forming gram-positive rod-shaped bacterium found in environmental sources. Livestock become infected by spores in their environment or feed. Humans usually become infected by eating or handling meat or products from infected animals (e.g., wool or hides). There are a small number of cases of anthrax each year, most of which occur in low-resource nations. Anthrax spores can be made into a fine powder, creating a potent biologic weapon that is a potential bioterrorism threat. In 1979, accidental release of *B. anthracis* spores at a military research institute in Russia killed 66 people. In 2001, 22 people in the United States were infected with *B. anthracis*, mostly through spores delivered in the mail in an act of domestic bioterrorism.

There are three major forms of anthrax:

- *Cutaneous anthrax*, which makes up 95% of naturally occurring infections, begins as a painless, pruritic papule that develops into a vesicle within 2 days. As the vesicle enlarges, striking edema may occur around it, with development of regional lymphadenopathy. After the vesicle ruptures, the remaining ulcer becomes covered with a characteristic black eschar, which dries and falls off as the person recovers. Bacteremia is rare.
- *Inhalational anthrax* occurs when airborne spores are inhaled. The spores are carried by phagocytes to lymph nodes where they germinate, producing bacilli that release toxins that cause hemorrhagic mediastinitis. After a prodromal illness of 1 to 6 days characterized by fever, cough, and chest or abdominal pain, there is abrupt onset of increased fever, hypoxia, and sweating. Frequently, meningitis develops from bacteremia. Shock and death frequently ensue within 1 to 2 days.
- *Gastrointestinal anthrax* is usually contracted by eating undercooked meat contaminated with *B. anthracis*. Initially, the person has nausea, abdominal pain, and vomiting, followed by severe bloody diarrhea and, sometimes, bacteremia. Mortality is approximately 40%.

Pathogenesis

B. anthracis produces potent toxins and an antiphagocytic polyglutamate capsule. The mechanisms of action of anthrax toxins are well understood (Fig. 8.21). There are two A subunits and one B subunit. The two A subunits, *edema factor* (EF) and *lethal factor* (LF), are named for their effects in experimental animals. The B subunit is called the *protective antigen* (PA) because antibodies against it protect against the effects of the toxins. PA is not toxic, but it serves to deliver the toxic EF and LF into cells. The bacterium releases each subunit as a separate

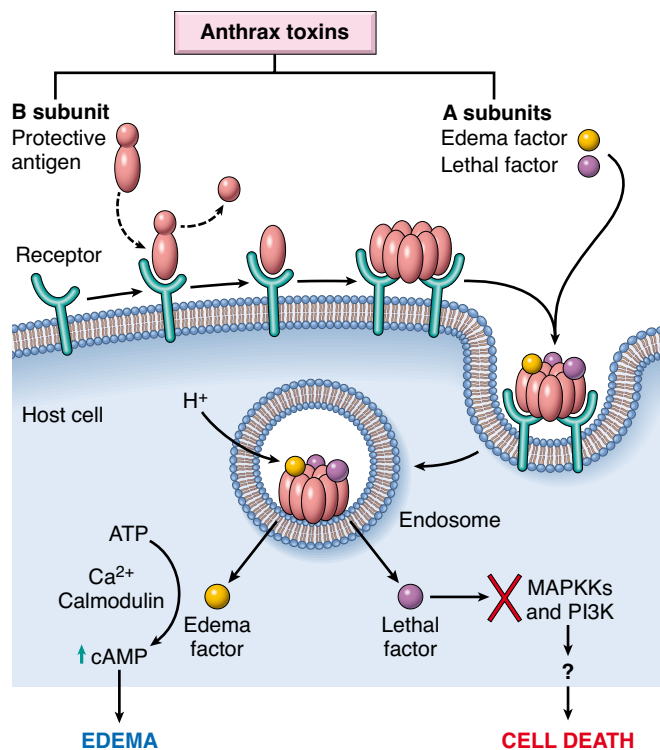


Fig. 8.21 Mechanism of action of anthrax toxins. Note that each cluster of B subunits binds either the edema factor or the lethal factor, but not both (as shown for simplicity). *ATP*, Adenosine triphosphate; *cAMP*, cyclic adenosine monophosphate; *MAPKKs*, mitogen-activated protein kinase kinases; *PI3K*, phosphoinositide 3-kinase. (Modified from Mourez M, Lacy DB, Cunningham K, et al. 2001: A year of major advances in anthrax toxin research. *Trends Microbiol.* 2002;10:287-293.)

protein. PA binds to one of two cell-surface receptors, called anthrax toxin receptor and capillary morphogenesis protein 2. A host protease removes a fragment of the PA, and the remaining fragment self-associates to form a heptamer. One to three molecules of the EF or LF bind to a PA heptamer, and this complex is endocytosed into the host cell. The low pH of the endosome causes a conformational change in the PA heptamer, which then forms a channel in the endosome membrane through which the EF or LF moves into the cytoplasm. In the cytoplasm, EF binds to calcium and calmodulin to form an adenylate cyclase. The active enzyme converts adenosine triphosphate (ATP) to intracellular cyclic adenosine monophosphate (cAMP), altering cell function. LF has a different mechanism of action. LF is a protease that cleaves both mitogen-activated protein kinase kinases (MAPKKs) and regulatory subunits of phosphoinositide-3 kinase (PI3K) disrupting the RAS and PI3K signaling pathways, respectively, both of which are important regulators of cell growth and differentiation (Chapter 1). The mechanism of cell death caused by dysregulation of these pathways is not understood.

MORPHOLOGY

Anthrax lesions at any site are typified by necrosis and exudative inflammation rich in neutrophils and macrophages. The presence of large, boxcar-shaped gram-positive extracellular bacteria in chains, seen histologically using the Brown and Brenn stain or grown in culture, suggests the diagnosis.

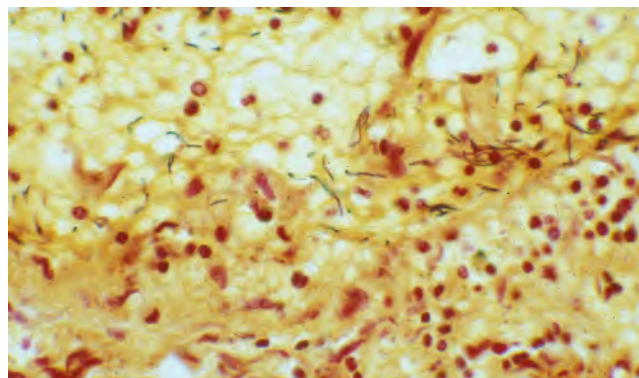


Fig. 8.22 *Bacillus anthracis* in the subcapsular sinus of a hilar lymph node of a patient who died of inhalational anthrax. (Courtesy Dr. Lev Grinberg, Department of Pathology, Hospital 40, Ekaterinburg, Russia, and Dr. David Walker, UTMB Center for Biodefense and Emerging Infectious Diseases, Galveston, TX.)

Inhalational anthrax causes numerous foci of hemorrhage in the mediastinum and hemorrhagic lymphadenitis of hilar and peribronchial lymph nodes. The lungs typically show a perihilar interstitial pneumonia with infiltration of macrophages and neutrophils associated with vasculitis. Hemorrhagic lung lesions associated with vasculitis are present in about one-half of cases. Mediastinal lymph nodes are expanded by edema and by macrophages containing phagocytosed apoptotic lymphocytes. *B. anthracis* is most likely to be seen in the alveolar capillaries and venules and, to a lesser degree, within the alveolar space and draining hilar lymph nodes (Fig. 8.22). In fatal cases, however, the organism may be found in multiple organs (spleen, liver, intestines, kidneys, adrenal glands, and meninges).

Nocardial Infections

***Nocardia* spp. are aerobic gram-positive bacteria found in soil that cause opportunistic infections.** The organism grows in distinctive branched chains. In culture, *Nocardia* form thin aerial filaments resembling hyphae. Despite this morphologic similarity to molds, *Nocardia* are bacteria.

Nocardia spp. cause respiratory infections, most often in patients with defects in immunity due to prolonged steroid use, HIV infection, or diabetes. Respiratory infection with *Nocardia* spp. manifests as an indolent illness with fever, weight loss, and cough, which may be mistaken for tuberculosis or malignancy. In some cases, *Nocardia* spp. infections disseminate from the lungs to the CNS. Infections of the CNS are also indolent and cause neurologic deficits that vary depending on the site of the lesions. *N. brasiliensis* causes skin infections following injuries that introduce contaminated soil. Manifestations include cellulitis, lymphocutaneous disease, and formation of nodules that may progress to form chronic draining fistulae.

MORPHOLOGY

Nocardia spp. appear in tissue as slender gram-positive organisms arranged in branching filaments (Fig. 8.23). Irregular staining gives the filaments a beaded appearance. *Nocardia* spp. stain with modified acid-fast stains (Fite-Faraco stain), unlike *Actinomyces* spp., which may appear similar on Gram stain of tissue. *Nocardia* spp. elicit a suppurative response with central liquefaction and surrounding granulation and fibrosis.

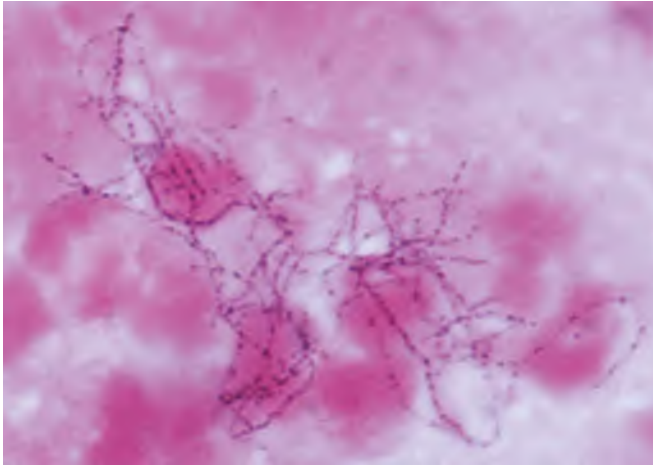


Fig. 8.23 *Nocardia asteroides* in a Gram-stained sputum sample. Note the beaded, branched gram-positive organisms. (Courtesy Dr. Ellen Jo Baron, Stanford University Medical Center, CA.)

Gram-Negative Bacterial Infections

There are a large number of pathogenic gram-negative bacteria. Many of these bacteria are becoming increasingly resistant to antibiotics, including carbapenem-resistant Enterobacterales (enteric gram-negative bacilli) and cephalosporin-resistant *N. gonorrhoeae*. Only a few gram-negative bacteria are discussed in this section. Several important gram-negative pathogens are discussed in the appropriate chapters of organ systems, including bacterial causes of gastrointestinal infections (Chapter 17) and urinary tract infections (Chapter 22). Anaerobic gram-negative organisms are considered later in this chapter. Gram-negative bacterial infections are usually diagnosed by culture.

Neisserial Infections

Neisseria spp. are gram-negative diplococci that are flattened on the adjoining sides, giving the pair the shape of a coffee bean. These aerobic bacteria have stringent nutritional requirements and grow best on enriched media such as lysed sheep's blood (chocolate) agar. The two clinically significant *Neisseria* spp. are *N. meningitidis* and *N. gonorrhoeae*.

***N. meningitidis* is a significant cause of bacterial meningitis, particularly among adolescents, young adults, and the elderly.** The organism is a common colonizer of the oropharynx and is spread by the respiratory route. An immune response leads to elimination of the colonizing organism in most people, and this response is protective against subsequent infection with the same serotype of bacteria. There are several capsular serotypes of *N. meningitidis*, five of which cause most cases of meningitis. Invasive disease mainly occurs when people encounter new strains to which they are not immune, as may happen to young children or to young adults living in crowded quarters such as military barracks or college dormitories. *N. meningitidis* is endemic in the United States, but epidemics occur every 5 to 12 years in sub-Saharan Africa during the dry season and cause thousands of deaths. Highly effective conjugate vaccines for *N. meningitidis* composed of capsular polysaccharides conjugated to antigenic proteins are available for four serogroups of *N. meningitidis* (A, C, W, and Y), and recombinant protein vaccines are available for serogroup B.

Even in the absence of preexisting immunity, only a small fraction of people infected with *N. meningitidis* develop meningitis. The bacteria must invade respiratory epithelial cells and enter the blood. In the blood, the bacterial capsule inhibits opsonization and destruction

of the bacteria by complement proteins. The importance of complement as a first-line defense against *N. meningitidis* is shown by the 1000- to 2000-fold increased rates of serious infection among people who have inherited defects in the complement proteins (C5 to C9) that form the membrane attack complex, or patients with hemolytic uremic syndrome or paroxysmal nocturnal hemoglobinuria (Chapter 14) who are treated with an antibody inhibitor of membrane attack complex formation. If *N. meningitidis* escapes the host response, the consequences can be severe. Although antibiotic treatment greatly reduces the mortality of *N. meningitidis* infection, about 10% of infected patients still die. The pathology of pyogenic meningitis is discussed in Chapter 28.

***N. gonorrhoeae* is an important cause of STI**, with more than 700,000 cases reported each year in the United States. It is second only to *C. trachomatis* among bacterial STIs. Infection in men causes urethritis. In females, *N. gonorrhoeae* infection is often asymptomatic and so may go unnoticed. Untreated gonorrhea may lead to pelvic inflammatory disease, which can cause infertility or ectopic pregnancy (Chapter 22). Infection is diagnosed by nucleic-acid amplification tests or, less often, culture. Increased resistance to oral cephalosporins has led to the recommendation that gonorrhea be treated with intramuscular ceftriaxone and oral azithromycin. Ceftriaxone-resistant *N. gonorrhoeae* is rare but has been detected in the United Kingdom, Canada, and Japan. Unsuccessful treatment of antibiotic resistant *N. gonorrhoeae* is almost always associated with infection of the throat, a site where antibiotic penetration is poor.

Although *N. gonorrhoeae* infection usually manifests locally in the genital or cervical mucosa, pharynx, or anorectum, disseminated infections may occur. Like *N. meningitidis*, *N. gonorrhoeae* is much more likely to become disseminated in people who lack the complement proteins that form the membrane attack complex. Disseminated infection in adults and adolescents usually causes septic arthritis accompanied by a rash with hemorrhagic papules and pustules. Neonatal *N. gonorrhoeae* infection causes conjunctivitis that may lead to blindness and, rarely, sepsis. The eye infection, which is preventable by instillation of silver nitrate or antibiotics in the newborn's eyes, remains an important cause of blindness in some low-resource nations.

Pathogenesis

Neisseria spp. adhere to and invade nonciliated epithelial cells at the site of entry (nasopharynx, urethra, or cervix). Adherence of *N. gonorrhoeae* to epithelial cells is initially mediated by long pili, which bind to CD46, a protein expressed on all human nucleated cells. Opa proteins (so named because they make bacterial colonies opaque) located in the outer membrane of the bacteria increase binding of *Neisseria* spp. to epithelial cells and promote entry of bacteria into cells. Once bound to epithelial cells, *N. gonorrhoeae* enters, kills, and passes beyond these cells, leading to superficial invasion.

***Neisseria* spp. use antigenic variation as a strategy to escape the immune response.** The existence of multiple capsular serotypes of *N. meningitidis* results in meningitis in some people on exposure to a new strain, as discussed earlier. In addition, *Neisseria* spp. can generate new antigens by genetic mechanisms, which permit a single bacterial clone to change its expressed antigens and escape immune defenses. Such mechanisms involve both pili and Opa proteins:

- **Recombination of genes encoding pili proteins.** The pili are composed of polypeptides encoded by the pilin gene, which consists of a promoter and coding sequences for 10 to 15 pili protein variants. At any point in time, only one of these coding sequences is adjacent to the promoter, allowing it to be expressed. Homologous recombination periodically shuttles one of the other pilin coding sequences

next to the promoter, resulting in expression of a different pilin variant.

- **Expression of different Opa proteins.** Each OPA gene has several repeats of a five-nucleotide sequence, which are frequently deleted or duplicated. These changes shift the reading frame of the gene so that it encodes new sequences. Stop codons are also introduced by the additions and deletions. All these can lead to the expression of a myriad of Opa proteins at any time.

Pertussis

Pertussis, or whooping cough, caused by the gram-negative coccobacillus *Bordetella pertussis*, is an acute, highly communicable illness transmitted via aerosolized droplets that is characterized by paroxysms of violent coughing followed by a loud inspiratory “whoop” as the patient gasps for air. Infants younger than 1 year of age are at highest risk of death. Children with pertussis can have coughing spells for up to 10 weeks. Worldwide, it is estimated that there are 24 million cases per year and more than 160,000 deaths in children younger than 5 years of age. For diagnosis, PCR is more sensitive than culture, but the specificity of the assay is not absolute, so some level of culture confirmation is critical in an outbreak setting. Widespread use of a vaccine resulted in a dramatic decrease in cases; however, concern about side effects of the whole-cell vaccine increased after the 1980s and led to a decrease in use. An acellular vaccine became available in the 1990s and is used more commonly in high-resource countries, but protection wanes over time and although it protects against disease, it does not prevent colonization and transmission as effectively. Consequently, there has been an increased incidence in recent years, even in areas where vaccination rates appear satisfactory. A rapid expansion of strains lacking pertactin, a common vaccine antigen, has been observed, and new vaccines are in clinical trials. Outbreaks observed every 3 to 5 years primarily affect older children, adolescents, and adults. Increased resistance to macrolide antimicrobials, the primary treatment, is worrisome. There are 16 species of *Bordetella*, but human infection with the other species is uncommon.

Pathogenesis

B. pertussis colonizes the brush border of the bronchial epithelium and invades macrophages. It contains fimbriae and filamentous hemagglutinin that binds to carbohydrates on the surface of respiratory epithelial cells, as well as to the integrin Mac-1(CR3) on macrophages. Virulence factors of *B. pertussis* include pertussis toxin, adenylate cyclase toxin, dermonecrotic toxin, tracheal cytotoxin, lipooligosaccharides, and the autotransporter pertactin. Pertussis toxin is a typical A-B toxin that is composed of five subunits. The A unit, like cholera toxin, ADP-ribosylates and inactivates guanine nucleotide-binding proteins, so these G proteins can no longer transduce signals, interrupting the effect of chemokines that use G protein-coupled receptors. The B component contains four subunits that bind to extracellular molecules and allow the A subunit to enter cells. The B subunit can also bind to cell-surface molecules such as TLR4, and through these it can initiate signaling events in cells. Collectively, pertussis toxin subunits impair host defenses by inhibiting neutrophil and macrophage recruitment and activation and paralyzing cilia, among other effects.

B. pertussis also produces adenylate cyclase toxin that enters host cells and converts ATP to supraphysiologic levels of cAMP. The rise in cAMP inhibits phagocytosis, the oxidative burst, and nitric oxide-mediated killing in neutrophils and macrophages, and the formation of neutrophil extracellular traps.

MORPHOLOGY

Bordetella spp. cause a laryngotracheobronchitis that in severe cases features bronchial mucosal erosion, hyperemia, and copious mucopurulent exudate. On histological examination, bacilli can be seen entangled with the cilia of respiratory epithelium (Fig. 8.24). Unless superinfected, the lung alveoli remain open and intact. In parallel with a striking peripheral lymphocytosis (up to 90%), there is hypercellularity and enlargement of the mucosal lymphoid follicles and peribronchial lymph nodes.

Pseudomonal Infections

***Pseudomonas aeruginosa* is an opportunistic aerobic gram-negative bacillus that is a frequent, deadly pathogen in people with cystic fibrosis, severe burns, or neutropenia.** This bacterium can be very resistant to antibiotics, making these infections difficult to treat. The source of infection is environmental or, less often, person-to-person spread. People with cystic fibrosis become infected with *P. aeruginosa* in the respiratory tract beginning around age 5, with prevalence of chronic infection increasing to approximately 70% in adults. It often infects extensive skin burns, which can lead to sepsis. *P. aeruginosa* is a common cause of hospital-acquired infections, such as respiratory tract infections associated with intubation and urinary tract infections associated with urinary catheters. It also causes corneal keratitis in wearers of contact lenses, endocarditis and osteomyelitis in intravenous drug abusers, external otitis (swimmer's ear) in healthy individuals, and severe external otitis in people with diabetes.

Pathogenesis

P. aeruginosa produces several toxins that contribute to local tissue damage. Early during infection of the lungs of people with cystic fibrosis, the organism secretes an A-B exotoxin called exotoxin A that, like diphtheria toxin, inhibits protein synthesis by ADP-ribosylating the ribosomal protein EF-2, leading to the death of host cells. At this early stage, the organism uses a type III secretion system to transport effector proteins into host cells; these reduce the ability of host cells to make antibacterial reactive oxygen species and induce apoptosis of

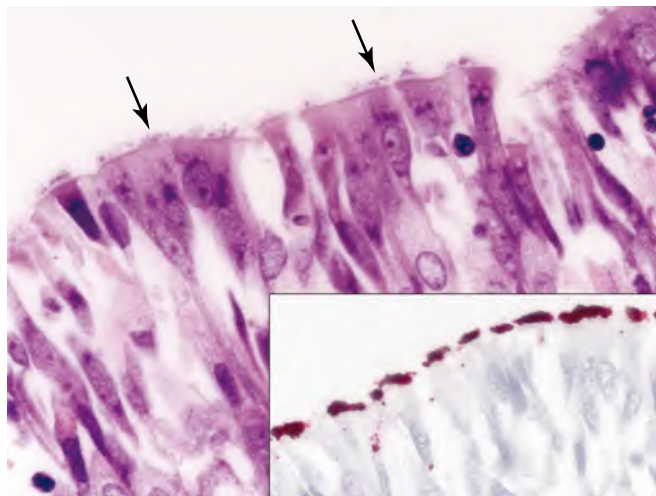


Fig. 8.24 Whooping cough showing a haze of bacilli (arrows) entangled with the cilia of bronchial epithelial cells. The inset highlights the bacilli (brown) after staining with a monoclonal antibody reactive to the lipooligosaccharide A of *Bordetella pertussis*. (Courtesy Dr. Christopher Padock, Centers for Disease Control and Prevention, Atlanta.)

the cells. Later in chronic infection in the lungs of people with cystic fibrosis, the bacteria become organized into biofilms composed, in part, of alginate and exopolysaccharides that the bacteria secrete. Within the biofilm, the bacteria are protected from antibodies, complement, phagocytes, and antibiotics. In addition, the release of exotoxin A is reduced, as is the expression of the type III secretion system, flagella and pili, and the organism evolves to become less virulent, although it continues to damage the host by stimulating inflammation and by releasing enzymes (proteases and elastases) that damage tissue. During chronic infection, the organism develops antibiotic resistance both through biofilm production and genetic changes, making treatment difficult.

MORPHOLOGY

Pseudomonas spp. cause **necrotizing pneumonia** that is distributed through the terminal airways in a fleur-de-lis pattern, with striking pale necrotic centers and red, hemorrhagic peripheral areas. On microscopic examination, masses of organisms are seen, often concentrated in the walls of blood vessels, where host cells undergo coagulative necrosis (Fig. 8.25). This picture of gram-negative **bacterial vasculitis** accompanied by thrombosis and hemorrhage, although not pathognomonic, is highly suggestive of *P. aeruginosa* infection.

Bronchial obstruction caused by mucus plugging and subsequent *P. aeruginosa* infection are frequent complications of cystic fibrosis. Despite antibiotic treatment and the host immune response, chronic *P. aeruginosa* infection may result in bronchiectasis and pulmonary fibrosis (Chapter 15).

In skin burns, *P. aeruginosa* proliferates widely, penetrating deeply into the veins and spreading hematogenously. Well-demarcated necrotic and hemorrhagic oval skin lesions, called **ecthyma gangrenosum**, often appear. Disseminated intravascular coagulation (DIC) is a frequent complication of *P. aeruginosa* bacteremia.

Plague

Plague is an invasive, frequently fatal, infection caused by *Yersinia pestis*, a gram-negative facultative intracellular bacterium. There are 200 species of mammals that can be infected by *Y. pestis*. Transmission to humans occurs via flea bites (especially from rodents), animal bites, contact with contaminated carcasses or clothing, contamination of open wounds by infected body fluids, or consumption of contaminated meat. Infection can spread from one human to another by aerosols. Plague caused three great pandemics that killed many millions of

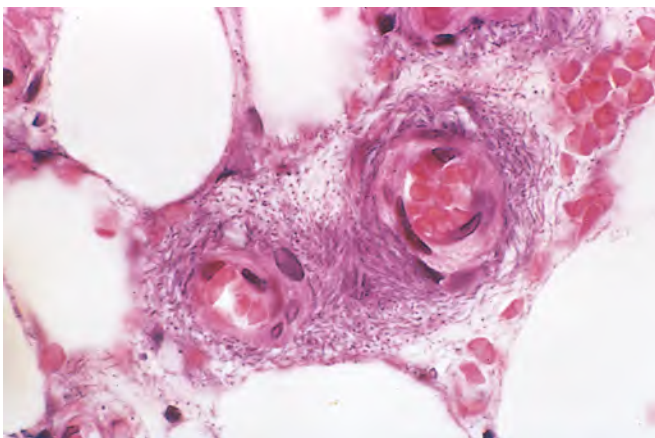


Fig. 8.25 *Pseudomonas* vasculitis in which masses of organisms form a perivascular purple-blue haze.

people; the devastation wreaked by the epidemic in Europe in the 1300s gave the infection the dramatic name *Black Death*. Plague can be found in animals on all continents except Oceania and is particularly prevalent in the Democratic Republic of Congo, Madagascar, and Peru. *Y. pestis* can exist in soil as amoeba and convert to a cell wall-deficient L-form of bacteria under stress, which helps them persist in the environment. Wild rodents in the rural western United States are infected with *Y. pestis* and are the source of a handful of human cases each year. *Y. enterocolitica* and *Y. pseudotuberculosis* are genetically similar to *Y. pestis*; these bacteria cause fecal-oral transmitted ileitis and mesenteric lymphadenitis. Antibiotics are effective, so early diagnosis is critical.

Y. pestis has two growth phases in the flea and the bacterial gene regulation shifts on sensing relocation from the insect gut to the human host. The bacteria spread from the site of inoculation to lymphoid tissues, where they proliferate, including inside macrophages, and inhibit the host from mounting an effective response. *Y. pestis* has a plasmid-borne complex of genes, the Yop virulon, which encodes a type III secretion system, a hollow syringe-like structure that projects from the bacterial surface, binds to host cells, and injects bacterial proteins called *Yops* (*Yersinia* outercoat proteins) into the cell. YopE, YopH, and YopT block phagocytosis by inactivating molecules that regulate actin polymerization. YopJ inhibits the signaling pathways that are activated by LPS, blocking the production of inflammatory cytokines. *Y. pestis* has additional virulence factors such as a siderophore that allows acquisition of iron from the blood. A human genetic variant believed to have been protective during the Black Death is associated with increased susceptibility to autoimmune diseases, suggesting that pandemics can influence long-term disease susceptibility in a population.

MORPHOLOGY

Y. pestis causes lymph node enlargement, pneumonia, or sepsis with a striking neutrophilia. The distinctive histologic features include (1) massive proliferation of the organisms, (2) early appearance of protein- and polysaccharide-rich effusions with few inflammatory cells, (3) necrosis of tissues and blood vessels associated with hemorrhage, thrombosis, and marked tissue swelling, and (4) neutrophilic infiltrates that accumulate adjacent to necrotic areas as healing begins.

In **bubonic plague**, the infected flea bite is usually on the legs, where it forms a small pustule or ulcer. The draining lymph nodes enlarge dramatically within a few days and become soft, pulpy, and plum colored, and may infarct or rupture through the skin. These lymph nodes were called **buboes**, from the Greek word for “groin,” giving rise to the name for this form of plague. In **pneumonic plague**, there is a severe, confluent, hemorrhagic, and necrotizing bronchopneumonia, often with fibrinous pleuritis. In **septicemic plague**, lymph nodes throughout the body as well as organs rich in mononuclear phagocytes develop foci of necrosis. Fulminant bacteremia also induces DIC with widespread hemorrhages and thrombi.

Chancroid (Soft Chancre)

Chancroid is an acute, ulcerative STI caused by *Haemophilus ducreyi*. The disease is most common in sub-Saharan Africa and parts of Southeast Asia, South America, and Central America. The prevalence of chancroid has fallen in Thailand and several African nations. Chancroid is an important cofactor in the transmission of HIV infection. Chancroid is uncommon in the United States, with fewer than 20 cases per year reported to the CDC in the past several years. The organism is difficult to detect in Gram stains from the lesions and it grows poorly in culture, requiring uncommon and highly enriched media. Moreover, PCR-based tests are not widely available. Thus, chancroid may be underdiagnosed.

MORPHOLOGY

Four to seven days after inoculation, a tender erythematous papule involving the external genitalia develops. In males, the primary lesion is usually on the penis; in females, most lesions occur in the vagina or the periurethral area. Over several days, the surface of the primary lesion erodes to produce an irregular, painful ulcer. **In contrast to the primary chancre of syphilis, the ulcer of chancroid is not indurated, and multiple lesions may be present.** The base of the ulcer is covered by shaggy, yellow-gray exudate. The regional lymph nodes become enlarged and tender in about 50% of cases within 1 to 2 weeks after primary infection. If the infection is not treated, the enlarged nodes (buboes) may erode the overlying skin to produce chronic, draining ulcers.

Microscopically, the ulcer of chancroid contains a superficial zone of neutrophilic debris and fibrin, and an underlying zone of granulation tissue containing areas of necrosis and thrombosed vessels. A dense, lymphoplasmacytic inflammatory infiltrate is present beneath the layer of granulation tissue. Coccobacilli are sometimes demonstrable in Gram or silver stains, but they are often obscured by other bacteria that colonize the ulcer base.

Granuloma Inguinale

Granuloma inguinale, or donovanosis, is a chronic inflammatory STI resulting in genital ulcers caused by *Klebsiella granulomatis* (formerly called *Calymmatobacterium donovani*), a minute, encapsulated coccobacillus. Granuloma inguinale is an STI that is rare in the United States and Western Europe, but more common in India, South Africa, and South America. Untreated cases are characterized by the development of extensive scarring, often associated with lymphatic obstruction and lymphedema of the external genitalia (called “genital elephantiasis,” akin to “elephantiasis” in lymphatic filariasis). Culture of the organism is difficult, and PCR assays are not widely available, so the diagnosis is made by microscopic examination of smears or biopsy samples of the ulcer.

MORPHOLOGY

Granuloma inguinale begins as a raised papular lesion on the moist stratified squamous epithelium of the genitalia or, rarely, the oral mucosa or pharynx. The lesion eventually ulcerates and develops abundant granulation tissue, which manifests grossly as a protuberant, soft, painless mass. As the lesion enlarges, its borders become raised and indurated. Disfiguring scars may develop in untreated cases and are sometimes associated with urethral, vulvar, or anal **strictures**. Regional lymph nodes typically are spared or show only nonspecific reactive changes, in contrast to chancroid.

Microscopic examination of active lesions reveals marked epithelial hyperplasia at the borders of the ulcer, sometimes mimicking carcinoma (pseudoeplitheliomatous hyperplasia). A mixture of neutrophils and mononuclear inflammatory cells is present at the base of the ulcer and beneath the surrounding epithelium. The organisms are demonstrable in Giemsa-stained smears of the exudate as minute, encapsulated coccobacilli (**Donovan bodies**) in macrophages. Silver stains (e.g., the Warthin-Starry stain) may also demonstrate the organism.

Mycobacterial Infections

Bacteria in the genus *Mycobacterium* are slender, aerobic rods that grow in straight or branching chains. Mycobacteria have a unique waxy cell wall composed of unusual glycolipids and lipids including mycolic acid, which makes them acid-fast, meaning they will retain stains, even on treatment with a mixture of acid and alcohol. They are weakly gram-positive.

Tuberculosis

Tuberculosis is a chronic pulmonary and systemic disease caused most often by *M. tuberculosis*, and the leading infectious cause of death worldwide. The source of transmission is humans with active

tuberculosis who release mycobacteria into the sputum. Oropharyngeal and intestinal tuberculosis contracted by drinking milk contaminated with *Mycobacterium bovis* is rare in countries where milk is routinely pasteurized, but it is still seen in countries that have tuberculous dairy cows and unpasteurized milk.

Epidemiology. According to the WHO, about 10.6 million people developed tuberculosis worldwide in 2022, leading to an estimated 1.3 million deaths, 167,000 of which occurred in people with HIV infection. In the United States there were 8300 cases of tuberculosis reported, 73% of which occurred in foreign-born people.

Tuberculosis flourishes wherever there is poverty, crowding, and chronic debilitating illness. In the United States, tuberculosis is mainly a disease of older adults, immigrants from high-burden countries, those living in crowded settings (prisons, homeless shelters, long-term care), and people with HIV infection. In addition to HIV infection, certain disease states also increase the risk: diabetes, Hodgkin lymphoma, chronic lung disease (particularly silicosis), chronic renal failure, malnutrition, alcohol use disorder, and immunosuppression.

It is important that infection with *M. tuberculosis* be differentiated from active disease. Infection refers to the presence of bacteria in the body, which may be symptomatic (active disease) or not (latent infection). Most infections are acquired by person-to-person transmission of airborne organisms from an active case to a susceptible host. Primary tuberculosis is usually asymptomatic in healthy individuals, although it may cause transient fever and pleural effusion. Generally, the only evidence of infection, if any remains, is a tiny, fibrocalcific pulmonary nodule (Ghon focus) at the site of the infection. Viable organisms may remain dormant in such lesions for decades. If immune defenses are lowered, the infection may be reactivated, producing communicable and potentially life-threatening disease.

Pathogenesis

The outcome of infection in a previously unexposed, immunocompetent person depends on the development of antimycobacterial T cell-mediated immunity. These T cells control the host response to the bacteria but result in development of pathologic lesions, such as caseating granulomas and cavitation.

Infection by *M. tuberculosis* proceeds in steps, from initial infection of macrophages to a subsequent Th1 response that both contains the infection and causes tissue damage (Fig. 8.26). The steps in infection are:

- **Entry into macrophages.** *M. tuberculosis* enters macrophages by phagocytosis mediated by several receptors expressed on the phagocyte, including mannose-binding lectin and the type 3 complement receptor (CR3). Alveolar macrophages are likely the first cells to be infected, and they migrate into the lung interstitium, where the bacteria can spread to additional macrophages and dendritic cells.
- **Replication in macrophages.** *M. tuberculosis* inhibits maturation of the phagosome and blocks formation of the phagolysosome, allowing the bacterium to replicate unchecked within the vesicle, protected from the microbicidal mechanisms of lysosomes. The bacterium blocks phagolysosome formation by recruiting a host protein called coronin to the membrane of the phagosome. Coronin activates the phosphatase calcineurin, leading to inhibition of phagosome-lysosome fusion. Thus, during the earliest stage of primary tuberculosis (<3 weeks) in the nonsensitized individual, bacteria proliferate in the pulmonary alveolar macrophages and air spaces, resulting in bacteremia and seeding of multiple sites. Despite the bacteremia, most people at this stage are asymptomatic or have a mild flulike illness.
- **Innate immunity.** Multiple pathogen-associated molecular patterns (PAMPs, Chapter 6) made by *M. tuberculosis* are recognized by innate immune receptors. Mycobacterial lipoarabinomannan binds TLR2, and unmethylated CpG nucleotides bind TLR9. These

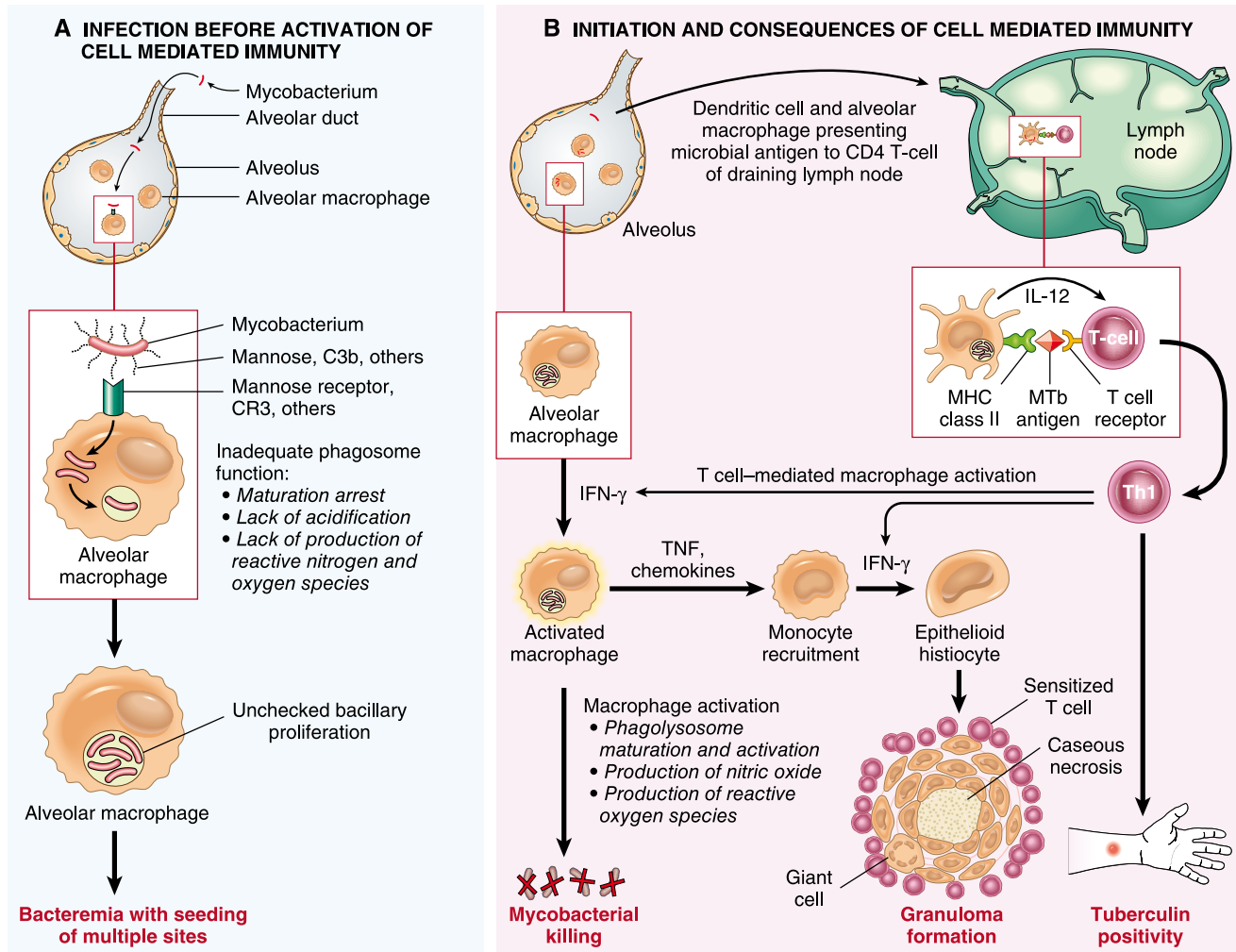


Fig. 8.26 Sequence of events in primary pulmonary tuberculosis, commencing with inhalation of virulent *Mycobacterium tuberculosis* organisms and culminating in the development of cell-mediated immunity and delayed hypersensitivity to the organism. (A) Events occurring during early infection, before activation of T cell-mediated immunity. (B) The initiation and consequences of T cell-mediated immunity. The development of resistance to the organism is accompanied by the appearance of a positive tuberculin test. CR3, Complement receptor 3; IFN-γ, interferon-gamma; MHC, major histocompatibility complex; MTb, *M. tuberculosis*; TNF, tumor necrosis factor.

interactions initiate and enhance the innate and adaptive immune responses to *M. tuberculosis*, as described later. *M. tuberculosis* also infects neutrophils, resists killing by these cells, and, after replicating, induces death of the infected cells.

- **Th1 response.** About 3 weeks after infection, a Th1 response is mounted that activates macrophages, enabling them to become bactericidal. The response is initiated by dendritic cells that migrate from the lungs to draining lymph nodes and display mycobacterial antigens to T cells. Differentiation of Th1 cells depends on IL-12 and IL-18, which are produced by antigen-presenting cells that have encountered the mycobacteria. Stimulation of TLR2 by mycobacterial ligands promotes production of IL-12 by dendritic cells.
- **Th1-mediated macrophage activation and killing of bacteria.** Th1 cells, both in lymph nodes and in the lung, produce IFN-γ. IFN-γ is the critical mediator that activates macrophages and enables them to contain the *M. tuberculosis* infection by several mechanisms. First, IFN-γ stimulates production of microbicidal nitric oxide and reactive oxygen species in the phagolysosomes of infected macrophages, which destroy the bacteria in the lethal acidic environment of the

vesicles. Second, IFN-γ mobilizes antimicrobial peptides (defensins), such as cathelicidin, against the bacteria. Third, IFN-γ stimulates autophagy, a process that sequesters and then destroys damaged organelles and intracellular bacteria such as *M. tuberculosis*.

- **Granulomatous inflammation and tissue damage.** In addition to stimulating macrophages to kill mycobacteria, the Th1 response orchestrates the formation of granulomas and caseous necrosis. Macrophages activated by IFN-γ differentiate into the “epithelioid histiocytes” that aggregate to form granulomas; some epithelioid cells may fuse to form giant cells. In many people this response halts the infection before significant tissue destruction or illness occur. In other people the infection progresses due to advanced age or immunosuppression, and the ongoing immune response results in caseous necrosis, sometimes without the formation of granulomas. Activated macrophages also secrete TNF and chemokines, which promote recruitment of more monocytes. The importance of TNF is underscored by the fact that patients with rheumatoid arthritis who are treated with TNF antagonists have an increased risk of tuberculosis reactivation.

- **Host susceptibility to disease.** AIDS is a major risk factor for progression to active disease, due to the loss of CD⁺ Th1 cell function. Other forms of immunosuppression, including glucocorticoids, TNF inhibitors, and treatment for transplants (solid organ and hematopoietic stem cell) also carry increased risk, as do renal failure and malnutrition. Rare inherited mutations that interfere with the Th1 response, such as loss of the IL-12 receptor β 1 protein, result in increased susceptibility to severe tuberculosis and even symptomatic infection with normally avirulent (so-called “atypical”) mycobacteria, such as the *Mycobacterium avium* complex (MAC), discussed later, or with the attenuated bacillus Calmette-Guérin (BCG) vaccine strain. This group of genetic disorders is called *Mendelian susceptibility to mycobacterial disease*.

In summary, **immunity to *M. tuberculosis* is primarily mediated by Th1 cells, which stimulate macrophages to kill the bacteria. This immune response, although largely effective, comes at the cost of accompanying tissue destruction.** Reactivation of the infection or re-exposure to the bacilli in a previously sensitized host results in rapid mobilization of a defensive reaction associated with tissue necrosis. Just as T-cell immunity and resistance are correlated, so, too, the loss of T-cell immunity (indicated by tuberculin negativity in a previously tuberculin-positive individual) is an ominous sign that resistance to the organism has faded.

Clinical Features

Clinical tuberculosis is separated into two important types that differ in pathophysiology: primary tuberculosis, which occurs with the first

infection, and postprimary (secondary) tuberculosis, which occurs in an individual who has been previously infected by *M. tuberculosis* (Fig. 8.27).

Primary tuberculosis is the form of disease that develops in a previously unexposed and therefore unsensitized person. Clinically significant disease develops in about 5% of newly infected people. With primary tuberculosis, the source of the organism is exogenous. In most people, the primary infection is contained, but in others, primary tuberculosis is progressive. The diagnosis of progressive primary tuberculosis in adults can be difficult. In contrast to secondary tuberculosis (apical disease with cavitation; see later), progressive primary tuberculosis more often resembles acute bacterial pneumonia with consolidation of the lobe, hilar lymphadenopathy, and pleural effusion. Lymphatic and hematogenous dissemination following primary infection may result in the development of tuberculous meningitis and miliary tuberculosis (discussed later).

Secondary tuberculosis, also known as postprimary tuberculosis, is the pattern of disease that arises in a previously sensitized host. It may follow shortly after primary tuberculosis, but more commonly it appears months to years after the initial infection, usually when host resistance is weakened. It stems from reactivation of a latent infection or from exogenous reinfection in the face of waning host immunity or when a large inoculum of virulent bacilli overwhelms the host immune system. Reactivation is more common in low-prevalence areas, and reinfection plays an important role in regions of high prevalence.

Secondary pulmonary tuberculosis classically involves the apex of the upper lobes of one or both lungs. Because of the preexistence of

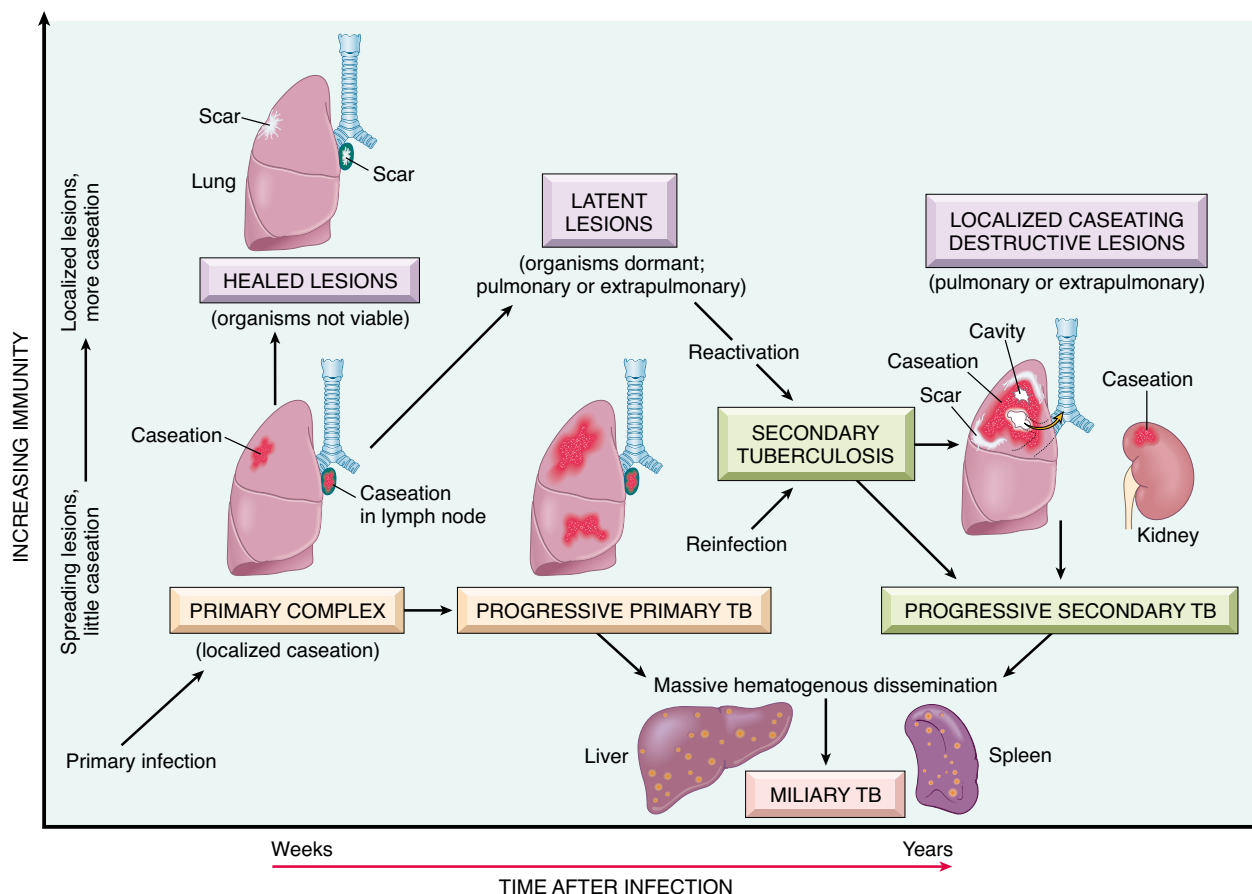


Fig. 8.27 Natural history and spectrum of tuberculosis. TB, Tuberculosis. (Courtesy Professor RK Kumar, School of Pathology, University of New South Wales, Sydney, Australia.)

hypersensitivity, the bacilli elicit a prompt and marked tissue response that tends to wall off the focus of infection. As a result, the regional lymph nodes are less prominently involved early in secondary disease than they are in primary tuberculosis. If untreated, cavitation occurs readily in the secondary form. Erosion of the cavities into an airway is an important source of spread of infection because the infected person coughs sputum that contains bacteria.

Localized secondary tuberculosis may be asymptomatic. When manifestations appear, they are usually insidious in onset. Systemic symptoms, probably related to cytokines released by activated macrophages (e.g., TNF), often appear early in the course and include malaise, anorexia, weight loss, and fever. Commonly, the fever is low grade and remittent (appearing late each afternoon and then subsiding), and night sweats occur. With progressive pulmonary involvement, increasing amounts of sputum, at first mucoid and later purulent, appear. Some degree of hemoptysis is present in about one-half of all cases of pulmonary tuberculosis. Pleuritic pain may result from extension of the infection to the pleural surfaces. Extrapulmonary manifestations of tuberculosis are legion and depend on the organ system involved.

Multidrug resistance is seen more commonly than it was in past years; hence, all newly diagnosed cases in the United States are treated with four drugs, unless the susceptibility of the bacterium from the source case is known. The prognosis is generally good if infections are localized to the lungs, except when they are caused by drug-resistant strains or occur in debilitated individuals.

Tests for tuberculosis detect either the bacteria or the T lymphocyte-mediated host response to the bacteria. There are several tests available for detection of *M. tuberculosis* in patients with active disease. Acid-fast smears and cultures of the sputum of patients suspected of having tuberculosis should be performed. Culture on solid agar media shows growth at 3 to 6 weeks, but culture in liquid media can provide an answer within 2 weeks. Comprehensive antibiotic susceptibility testing can only be performed by culture. Whole-genome sequencing of *M. tuberculosis* can predict susceptibility to first-line drugs with over 90% accuracy more quickly than culture but is not yet widely available. PCR amplification of *M. tuberculosis* DNA allows for even more rapid diagnosis. An FDA-approved PCR test is available that identifies both the presence of *M. tuberculosis* and, if the organism is detected, whether it is resistant to rifampin. An extended PCR assay recommended by the WHO, but not FDA-approved, detects resistance to three first-line drugs and three second-line drugs. PCR assays are as sensitive as culture in acid-fast smear-positive samples, but slightly less sensitive in smear-negative tuberculosis, and substantially less sensitive in children. Although culture remains the gold standard, PCR

should also be performed if active tuberculosis is suspected because results are available quickly.

Testing for latent tuberculosis is done by detecting T cells specific for *M. tuberculosis* antigens. This can be done by either IFN- γ release assays (IGRAs) or the tuberculin (purified protein derivative [PPD], or Mantoux) skin test. IGRAs are in vitro tests in which lymphocytes from the patient are stimulated with protein antigens from *M. tuberculosis*. Production of IFN- γ by the T cells is measured to assess the level of T-cell immunity to the organism. The tuberculin skin test is performed by intracutaneous injection of purified protein derivative of *M. tuberculosis*, which induces a visible and palpable induration that peaks in 48 to 72 hours. A positive IGRA or tuberculin test signifies T cell-mediated immunity to mycobacterial antigens but does not differentiate between infection (current or past) and active disease. False-negative reactions may occur in the setting of certain viral infections, sarcoidosis, malnutrition, Hodgkin lymphoma, immunosuppression, and (notably) active tuberculous disease. A false-positive tuberculin skin test may result from infection by atypical mycobacteria or prior vaccination with BCG, an attenuated strain of *M. bovis* that is used as a vaccine in some countries. False-positive results are uncommon with IGRAs.

Tuberculosis is a leading cause of death in people with AIDS. **All stages of HIV infection are associated with an increased risk of tuberculosis.** The use of antiretroviral therapy (ART) reduces the risk of tuberculosis in people with HIV infection, but even with ART, people infected with HIV are more likely to develop symptomatic tuberculosis. A low CD4+ T-cell count before starting ART is an important risk factor for development of tuberculosis, which underscores the role of the immune response in keeping reactivation of *M. tuberculosis* in check. Pulmonary manifestations of tuberculosis in HIV-infected individuals are variable, ranging from focal lesions to multifocal infiltrates to localized apical disease with cavitation. The extent of immunodeficiency also determines the frequency of extrapulmonary involvement, rising from 10% to 15% in mildly immunosuppressed people to greater than 50% in those with severe immune compromise. Other features of tuberculosis in HIV-positive people include an increased frequency of false-negative sputum smears and tuberculin tests (the latter sometimes referred to as “anergy”), and the absence of characteristic granulomas in tissues, particularly in the late stages of HIV. The increased frequency of sputum smear-negativity is paradoxical because these immunosuppressed patients typically have higher bacterial loads. The likely explanation is that cavitation and bronchial damage are more severe in immunocompetent individuals, resulting in more bacilli in expelled sputum. In contrast, the absence of bronchial wall destruction due to reduced T cell-mediated hypersensitivity results in the excretion of fewer bacilli in sputum.

MORPHOLOGY

Primary Tuberculosis. In countries where consumption of infected milk has been eliminated, primary tuberculosis almost always begins in the lungs. Typically, the inhaled bacilli implant in the distal airspaces of the lower part of the upper lobe or the upper part of the lower lobe, usually close to the pleura. As sensitization develops, a 1- to 1.5-cm area of gray-white inflammation with consolidation emerges, known as the *Ghon focus*. In most cases, the center of this focus undergoes caseous necrosis. Tubercle bacilli, either free or within phagocytes, drain to the regional nodes, which also often caseate. **This combination of parenchymal lung lesion and nodal involvement is referred to as the Ghon complex** (Fig. 8.28). During the first few weeks, there is also lymphatic and hematogenous dissemination to other parts of the body. In approximately 95% of cases, development of cell-mediated immunity controls the infection. Hence, the Ghon complex undergoes progressive fibrosis, often followed by radiologically detectable calcification, and despite seeding of other organs, no lesions develop.

Histologically, sites of active involvement are marked by a characteristic granulomatous inflammatory reaction that forms both caseating and noncaseating tubercles (Fig. 8.29A–C). Individual tubercles are microscopic; it is only when multiple granulomas coalesce that they become macroscopically visible. The granulomas are usually enclosed within a fibroblastic rim punctuated by lymphocytes. Multinucleated giant cells are present in the granulomas. Immunocompromised people do not form the characteristic granulomas, and their macrophages contain many bacilli (Fig. 8.29D).

Secondary (Postprimary) Tuberculosis. The initial lesion is usually a small focus of consolidation, less than 2 cm in diameter, within 1 to 2 cm of the apical pleura. Such foci are sharply circumscribed, firm and gray-white to yellow in color, and have variable degrees of central caseation and peripheral fibrosis (Fig. 8.30). In immunocompetent individuals, the initial parenchymal focus undergoes progressive fibrous encapsulation, leaving only fibrocalcific scars. Histologically, the active lesions show characteristic coalescent tubercles with central

caseation. Tubercle bacilli can often be identified with acid-fast stains in early exudative and caseous phases of granuloma formation but are usually too few to be found in the late, fibrocalcific stages. Localized, apical, secondary pulmonary tuberculosis may heal with fibrosis either spontaneously or after therapy, or the disease may progress and extend along several different pathways.

Progressive pulmonary tuberculosis may ensue in older adults and immunosuppressed people. The apical lesion expands into adjacent lung and eventually erodes into bronchi and vessels. This evacuates the caseous center, creating a ragged, irregular cavity that is poorly walled off by fibrous tissue. Erosion of blood vessels results in hemoptysis. With adequate treatment the process may be arrested, although healing by fibrosis often distorts the pulmonary architecture. The cavities, now free of inflammation, may persist or become fibrotic. If the treatment is inadequate or if host defenses are impaired, the infection may spread via airways, lymphatic channels, or the vascular system. **Miliary pulmonary disease** occurs when organisms draining through lymphatics enter the venous blood and circulate back to the lung. Individual lesions are either microscopic or small, visible (2-mm) foci of yellow-white consolidation scattered through the lung parenchyma (the adjective “miliary” is derived from the resemblance of these foci to millet seeds). Miliary lesions may expand and coalesce, resulting in consolidation of large regions or even whole lobes of the lung. With progressive pulmonary tuberculosis, the pleural cavity is invariably involved, and serous **pleural effusions, tuberculous empyema, or obliterative fibrous pleuritis** may develop. Progressive primary tuberculosis that occurs in immunosuppressed individuals spreads in a similar manner.

Endobronchial, endotracheal, and laryngeal tuberculosis may develop by spread through lymphatic channels or from expectorated infectious material. The mucosal lining may be studded with minute granulomatous lesions that may only be apparent microscopically.

Systemic miliary tuberculosis occurs when bacteria disseminate through the systemic arterial system. Miliary tuberculosis is most prominent in the liver, bone marrow, spleen, adrenals, meninges, kidneys, fallopian tubes, and epididymis, but can involve any organ (Fig. 8.31).

Isolated tuberculosis may appear in any organ or tissue that is seeded hematogenously and may be the presenting manifestation. Organs that are commonly involved include the meninges (tuberculous meningitis), kidneys (renal tuberculosis), adrenals (formerly an important cause of Addison disease), bones (osteomyelitis), and fallopian tubes (salpingitis). When the vertebrae are affected, the disease is referred to as **Pott disease**. Paraspinal “cold” abscesses in these patients may track along tissue planes and present as an abdominal or pelvic mass.

Lymphadenitis is the most frequent presentation of extrapulmonary tuberculosis, usually occurring in the cervical region (“scrofula”). In HIV-negative individuals, lymphadenitis tends to be unifocal and localized. HIV-positive people, on the other hand, almost always have multifocal disease, systemic symptoms, and either pulmonary or other organ involvement by active tuberculosis.

As previously mentioned, **intestinal tuberculosis** contracted by the drinking of contaminated milk is common in countries where bovine tuberculosis is present and milk is not pasteurized. In countries where milk is pasteurized, intestinal tuberculosis is more often caused by the swallowing of coughed-up infective material in patients with advanced pulmonary disease. Typically, the organisms are seeded to mucosal lymphoid aggregates of the small and large bowel, which then undergo granulomatous inflammation that can lead to ulceration of the overlying mucosa (mostly superficial transverse ulcer), particularly in the ileum. Healing creates strictures.

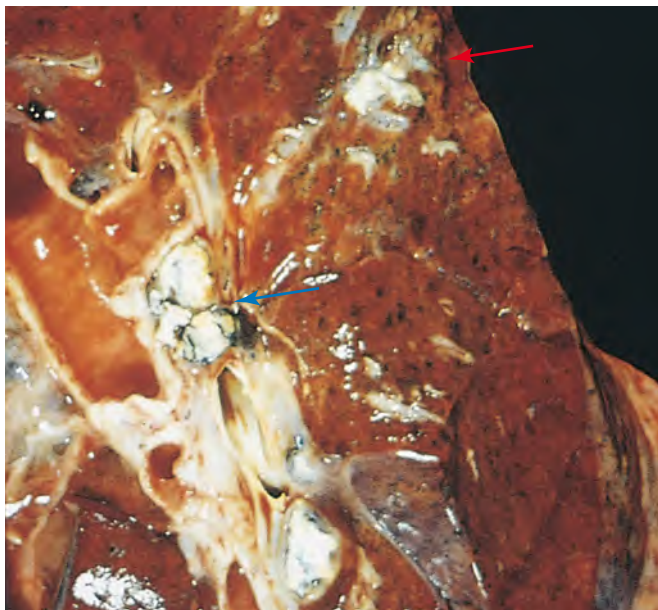


Fig. 8.28 Primary pulmonary tuberculosis, Ghon complex. The gray-white parenchymal focus is under the pleura in the lower part of the upper lobe (red arrow). Hilar lymph nodes with caseation are seen on the left (blue arrow).

Nontuberculous Mycobacterial Infections

Nontuberculous mycobacteria (NTM) refer to mycobacteria other than *M. tuberculosis* and the very fastidious *M. leprae*. The prevalence of NTM disease has increased worldwide. Of the 150 NTM species, the most frequent human pathogens are *Mycobacterium avium* complex

(MAC), *M. abscessus* complex, and *M. kansasii*. Treatment differs for these pathogens, so identifying the specific organism is important. It has been difficult to distinguish *M. avium* and *M. intracellulare* using traditional physical and biochemical tests, and the clinical features of these infections are similar, so they are grouped into a complex. Newer molecular methods are better able to distinguish these two species, as well as *M. chimaera* in the same complex. NTM are ubiquitous in the environment, water, and soil, and there is little human-to-human transmission, except for a small number of cases in patients with cystic fibrosis.

Clinical presentations of NTM infections include chronic pulmonary disease (most common), lymphadenitis, cutaneous disease, and disseminated disease. Normal host defense mechanisms usually prevent infection, so vulnerable individuals include those with structural lung damage, cystic fibrosis, bronchiectasis, primary ciliary dyskinesia, chronic obstructive pulmonary disease, or pneumoconiosis. Additional predisposing factors include immunosuppression from HIV, transplantation, treatment with TNF inhibitors, and rare cases of anti-IFN- γ autoantibodies or inherited defects in the IL-12/IFN- γ pathways. Rapidly growing mycobacterial infections are often associated with postsurgical or posttraumatic infections. Neutrophils play a critical role in host defense against mycobacterial infections. MAC infection induces neutrophil extracellular trap formation (a network of extracellular strings of DNA and proteins that bind pathogens, Chapter 3), IL-8 release, and release of matrix metalloproteinases from neutrophils, leading to worsening lung inflammation. In patients with marked T-cell immunodeficiency, MAC causes widely disseminated infections, and organisms proliferate abundantly in many organs, including the lungs and gastrointestinal system. Patients are feverish, with drenching night sweats and weight loss. In cases of MAC infection in a person without HIV or another severe immunodeficiency, the organisms primarily infect the lung, causing a productive

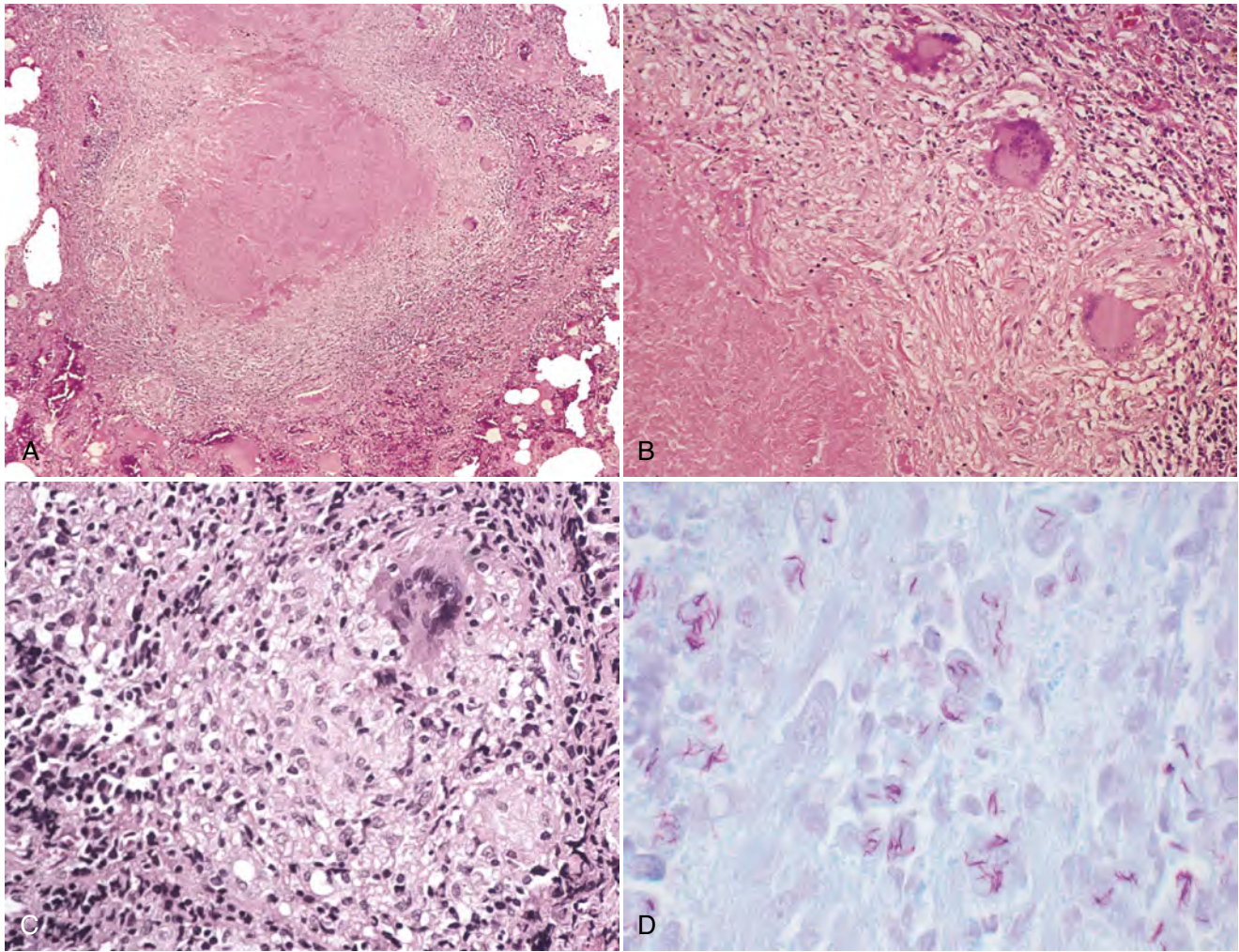


Fig. 8.29 Morphologic spectrum of tuberculosis. Characteristic tubercle at low magnification (A) and high magnification (B) shows central granular caseation surrounded by epithelioid and multinucleate giant cells. This is the usual response in people who have developed cell-mediated immunity to the organism. (C) Occasionally, even in immunocompetent patients, tubercular granulomas may not show central caseation; hence regardless of the presence or absence of caseous necrosis, use of special stain for acid-fast organisms is indicated when granulomas are present. (D) In this specimen from an immunocompromised patient, macrophages with large numbers of mycobacteria are seen (acid-fast stain).

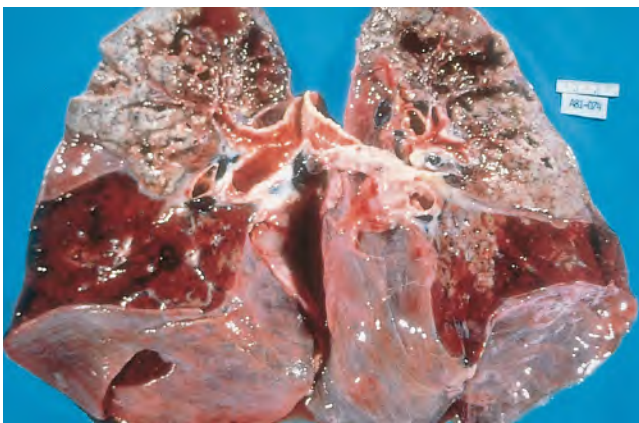


Fig. 8.30 Secondary pulmonary tuberculosis. The upper parts of both lungs are riddled with gray-white areas of caseation and multiple areas of softening and cavitation.

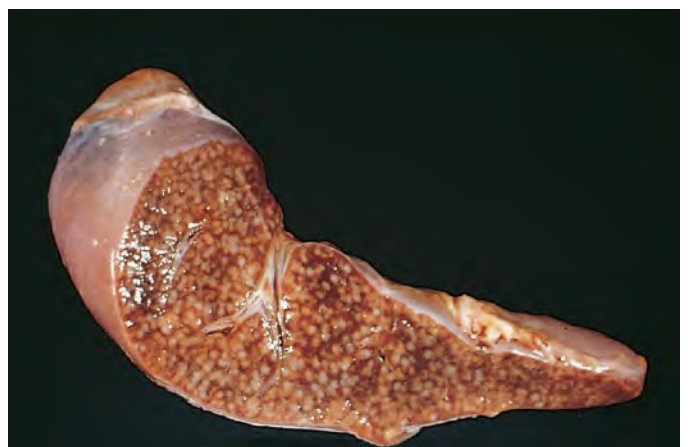


Fig. 8.31 Miliary tuberculosis of the spleen. The cut surface shows numerous gray-white tubercles.

cough and sometimes fever and weight loss. Radiographic characteristics of disease may be fibrocavitary lesions primarily in the upper lobes or nodular bronchiectasis with multifocal clusters of small nodules. NTM biofilm development on medical devices is a well-documented source of infection.

MORPHOLOGY

The hallmark of MAC infections in patients with HIV is abundant acid-fast bacilli within macrophages (Fig. 8.32). Depending on the severity of immunodeficiency, MAC infections can be widely disseminated throughout the mononuclear phagocyte system, causing enlargement of involved lymph nodes, liver, and spleen, or localized to the lungs. There may be a yellowish pigmentation to these organs secondary to the large number of organisms present in swollen macrophages. Granulomas, lymphocytes, and tissue destruction are rare.

Leprosy

Leprosy, or Hansen disease, is a slowly progressive infection caused most often by *M. leprae* and, less frequently, by the more recently discovered *M. lepromatosis* (proposed name since 2008) that mainly affects the skin and peripheral nerves. Despite its low communicability, leprosy remains endemic, with 200,000 new infections per year among people living in multiple low-resource tropical nations, especially in Southeast Asia, East Africa, and Brazil.

Pathogenesis

Transmission is believed to be from human respiratory secretions, with zoonotic transmission from armadillos occurring in rare cases. *M. leprae* is an obligate intracellular pathogen that is taken up by dermal macrophages and disseminates in the blood, but it replicates primarily in relatively cool tissues of the skin and extremities. It cannot be cultured in vitro and proliferates best at 32° to 34°C, the temperature of the human skin. *M. leprae* secretes no toxins, and its virulence is based in part on properties of its cell wall, with the bacterial lipid PGL-1 being critical for invasion of host cells. *M. leprae* is similar enough to *M. tuberculosis* that immunization with BCG confers some protection against *M. leprae* infection. *M. leprae* inhibits mitochondrial energy metabolism to evade the host immune response.

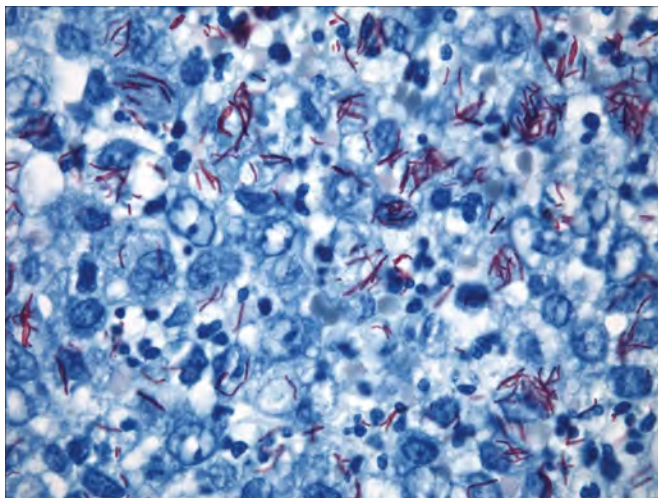


Fig. 8.32 *Mycobacterium avium* infection in a patient with AIDS, showing massive infection with acid-fast organisms. This pattern is more common in patients with acquired immunodeficiencies.

M. leprae causes five different patterns of disease on a spectrum from tuberculoid to lepromatous, determined by the helper T-lymphocyte response to *M. leprae*. People with the less severe tuberculoid leprosy have dry, scaly skin lesions that lack sensation. They often have asymmetric involvement of large peripheral nerves. The more severe form, lepromatous leprosy, includes symmetric skin thickening and nodules. In lepromatous leprosy, widespread invasion of the mycobacteria into Schwann cells and into endoneural and perineural macrophages damages the peripheral nervous system. In advanced cases of lepromatous leprosy, *M. leprae* is present in sputum and blood. People with tuberculoid leprosy have classically activated (M1) macrophages and a Th1 response associated with production of IL-2 and IFN- γ as well as a Th17 response. As with *M. tuberculosis*, IFN- γ functions to mobilize an effective host macrophage response, and hence the microbial burden is low. Also, antibody production is low. In contrast, lepromatous leprosy is associated with a weak Th1 response and perhaps a relative increase in regulatory T cells and the Th2 response. The net result is weak cell-mediated immunity and an inability to control the bacteria, which can be readily visualized in tissue sections. Occasionally, most often in the lepromatous form, antibodies are produced against *M. leprae* antigens. Paradoxically, these antibodies are usually not protective, but they may form immune complexes with free antigens that can lead to erythema nodosum, vasculitis, and glomerulonephritis.

MORPHOLOGY

Tuberculoid leprosy begins with localized flat, red skin lesions that enlarge and develop irregular shapes with indurated, elevated, hyperpigmented margins and depressed pale centers (central healing). Neuronal involvement dominates tuberculoid leprosy. Nerves become enclosed within granulomatous inflammatory reactions and, if small (e.g., the peripheral twigs) are destroyed (Fig. 8.33). Nerve degeneration causes skin anesthetics and skin and muscle atrophy that render the person susceptible to trauma of the affected parts, leading to the development of chronic skin ulcers. Contractures, paralyses, and autoamputation of fingers or toes may ensue. Facial nerve involvement can lead to paralysis of the eyelids, with keratitis and corneal ulcerations. On microscopic examination, all sites of involvement have granulomatous lesions closely resembling those found in tuberculosis. Because of the strong host defense, bacilli are almost never found, hence the name **paucibacillary** leprosy. The presence of granulomas and absence of bacteria reflect strong T-cell immunity. Because leprosy pursues an extremely slow course, spanning decades, most patients die with leprosy rather than of it.

Lepromatous leprosy involves the skin, peripheral nerves, anterior eye chamber, upper airways (down to the larynx), testes, hands, and feet. The vital organs and CNS are rarely affected, presumably because the core temperature is too high for growth of *M. leprae*. Lepromatous lesions contain large aggregates of lipid-laden macrophages (**lepra cells**), often filled with masses ("globi") of acid-fast bacilli (Fig. 8.33C). Because of the abundant bacteria, lepromatous leprosy is referred to as **multibacillary**. Macular, papular, or nodular lesions form on the face, ears, wrists, elbows, and knees. With progression, the nodular lesions coalesce to yield a distinctive leonine facies. Most skin lesions are hypoesthetic or anesthetic. Lesions in the nose may cause persistent inflammation and bacilli-laden discharge. The peripheral nerves, particularly the ulnar and peroneal nerves where they approach the skin surface, are symmetrically invaded with mycobacteria, with minimal inflammation. Loss of sensation and trophic changes in the hands and feet follow the nerve lesions. Lymph nodes contain aggregates of bacteria-filled foamy macrophages in the paracortical (T-cell) areas and reactive germinal centers. In advanced disease, aggregates of macrophages are also present in the splenic red pulp and the liver. The testes are usually extensively involved, leading to destruction of the seminiferous tubules and consequent sterility.

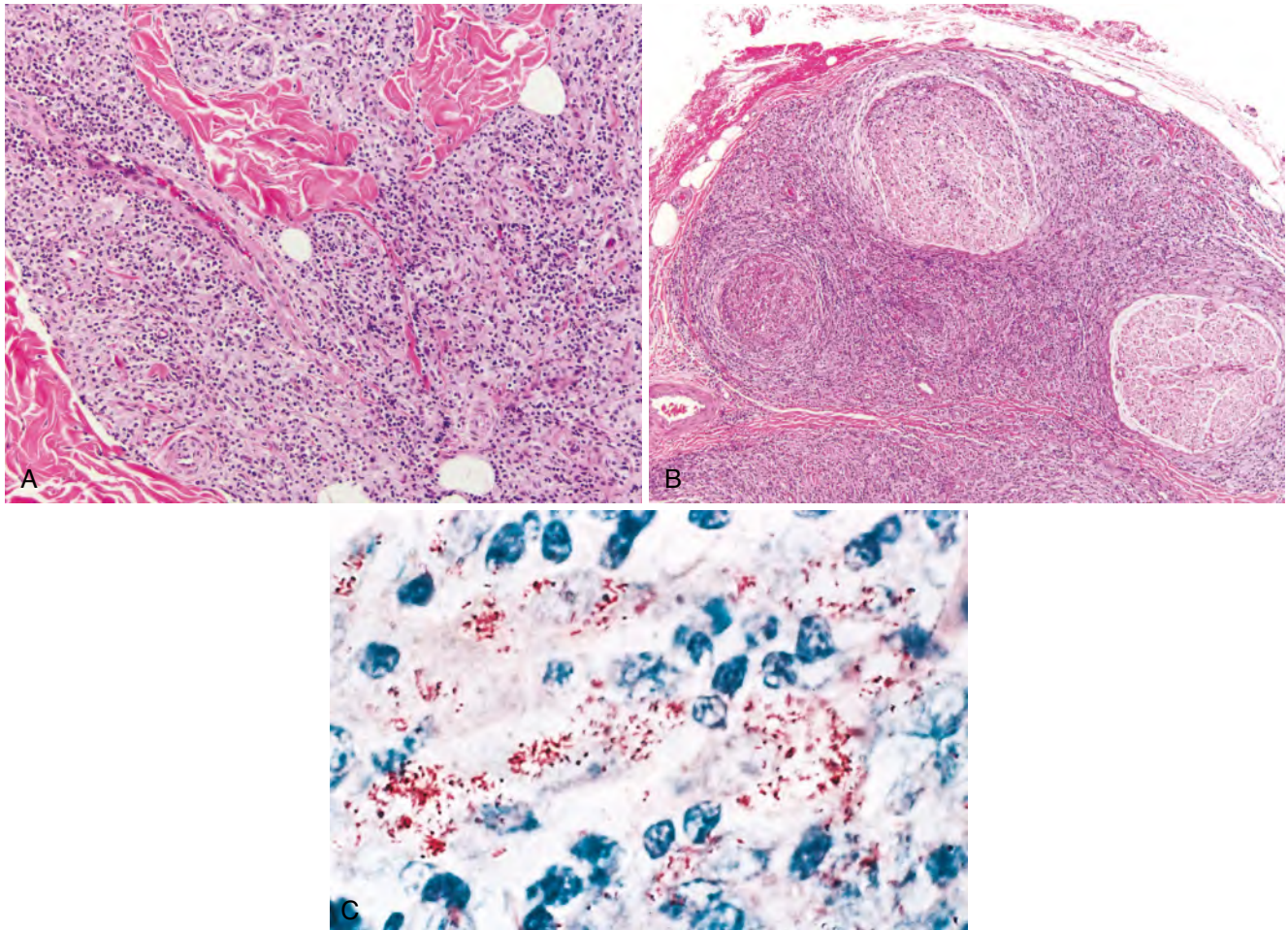


Fig. 8.33 Leprosy. Two types of inflammatory infiltrates commonly seen are (A) dense dermal macrophage infiltration surrounding adnexa, vessels, and nerves (resulting in subcutaneous nodules) in tuberculoid leprosy, and (B) dense chronic lymphocytic and macrophage infiltration into large nerve bundles (resulting in mononeuropathy) in lepromatous leprosy. (C) Acid-fast bacilli within macrophages in the lepromatous form.

Spirochete Infections

Spirochetes are gram-negative, slender, corkscrew-shaped bacteria with axial periplasmic flagella wound around a helical protoplasm. The bacteria are covered in a membrane called an outer sheath, which is thought to mask bacterial antigens from the host immune response. *Treponema pallidum* subsp. *pallidum* is the microaerophilic spirochete that causes syphilis, a chronic STI with multiple clinical presentations. Other closely related treponemes cause yaws (*T. pallidum* subsp. *pertenue*) and pinta (*T. pallidum* subsp. *carateum*).

Syphilis

Syphilis is a chronic STI with varied clinical and pathologic manifestations. The causative spirochete, *T. pallidum* subsp. *pallidum*, hereafter referred to simply as *T. pallidum*, is too slender to be seen in Gram stain, but it can be visualized by silver stains and immunofluorescence techniques (Fig. 8.34). Transplacental transmission of *T. pallidum* occurs readily, and active disease during pregnancy results in congenital syphilis. *T. pallidum* cannot be easily grown in culture. Public health programs and penicillin treatment reduced the number of cases of syphilis in the United States from the late 1940s until the late 1970s. Unfortunately, in North America and western Europe, the incidence of syphilis has increased dramatically more recently, with a

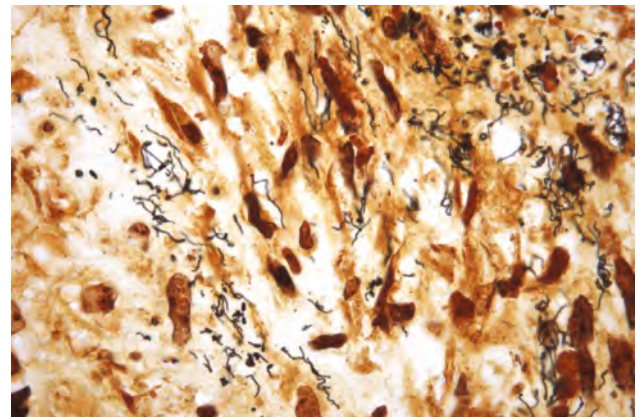


Fig. 8.34 *Treponema pallidum* (Steiner silver stain) showing several spirochetes in histologic sections of placental syphilis.

doubling from 2000 to 2016 in the United States. More than half of the cases are among men who have sex with men, and 42% of those patients have HIV infection. Worldwide, more than 7 million new cases of syphilis are diagnosed every year, with 1.4 million cases in pregnant females, especially in Africa and Eastern Mediterranean regions, and

over 500,000 adverse outcomes in children. Mother-to-child transmission drops significantly with appropriate penicillin treatment, especially in the primary and secondary stages.

Pathogenesis

Proliferative endarteritis affecting small vessels with a surrounding plasma cell-rich infiltrate is characteristic of all stages of syphilis. Much of the pathology of syphilis can be ascribed to the ischemia produced by the vascular lesions. The pathogenesis of endarteritis is unknown.

The immune response to *T. pallidum* reduces the burden of bacteria and can lead to resolution of local lesions but does not reliably eliminate the systemic infection. Superficial sites of infection (chancres and rashes) have an intense inflammatory infiltrate that includes T cells, plasma cells, and macrophages that surround the bacteria. The attachment of *T. pallidum* to activated platelets may play a role in dissemination. The infiltrating CD4⁺ T cells are Th1 cells that appear to provide limited protection. Treponeme-specific antibodies are detectable, and these activate complement in the lesion and opsonize the bacteria for phagocytosis by macrophages. In many patients, the organism persists despite these host responses. A protein in the outer membrane of *T. pallidum*, TprK, can alter its structure during the course of infection through gene conversion (recombination) between silent donor sites and the *tprK* gene, and this may contribute to antigenic diversification that allows the organism to escape from antibodies. The difficulty of developing a continuous in vitro culture system for this organism has limited investigation of pathogenesis.

Clinical Features

Syphilis is divided into three stages, with distinct clinical and pathologic manifestations (Fig. 8.35).

Primary Syphilis. This stage, occurring approximately 2 to 6 weeks after infection, features a single firm, nontender, raised, red lesion (chancre) located at the site of treponemal invasion on the penis, cervix, vaginal wall, or anus. The chancre heals with or without therapy. Spirochetes are plentiful within the chancre and spread from there throughout the body by hematologic and lymphatic dissemination.

Secondary Syphilis. This stage is marked by painless, superficial lesions of the skin and mucosal surfaces. It occurs 2 to 10 weeks after the primary chancre in approximately 75% of untreated people. Skin lesions frequently occur on the palms or soles of the feet and may be maculopapular, scaly, or pustular. Moist areas of the skin, such as the anogenital region, inner thighs, and axillae, may have broad-based elevated plaques called *condylomata lata*. Silvery gray superficial erosions may form on the oral, pharyngeal, and genital mucous membranes. Lymphadenopathy, mild fever, malaise, and weight loss are also common in secondary syphilis. Invasion of the CNS occurs in 40% of patients, more often as asymptomatic neurosyphilis, while symptomatic neurosyphilis with meningitis, visual changes, or hearing changes occurs in 1% to 2%. Secondary syphilis lasts several weeks, and then the person enters the latent stage of the disease.

Tertiary Syphilis. Tertiary syphilis has three main manifestations: cardiovascular syphilis, neurosyphilis, and so-called benign tertiary syphilis. These may occur alone or in combination. Tertiary syphilis occurs in one-third of untreated patients, after a latent period of 2 to 50 years.

- **Cardiovascular syphilis**, in the form of syphilitic aortitis, accounts for more than 80% of cases of tertiary disease. The pathogenesis of this vascular lesion is not known, but the scarcity of treponemes and the intense inflammatory infiltrate suggest that the immune response plays a role. The aortitis leads to slowly progressive

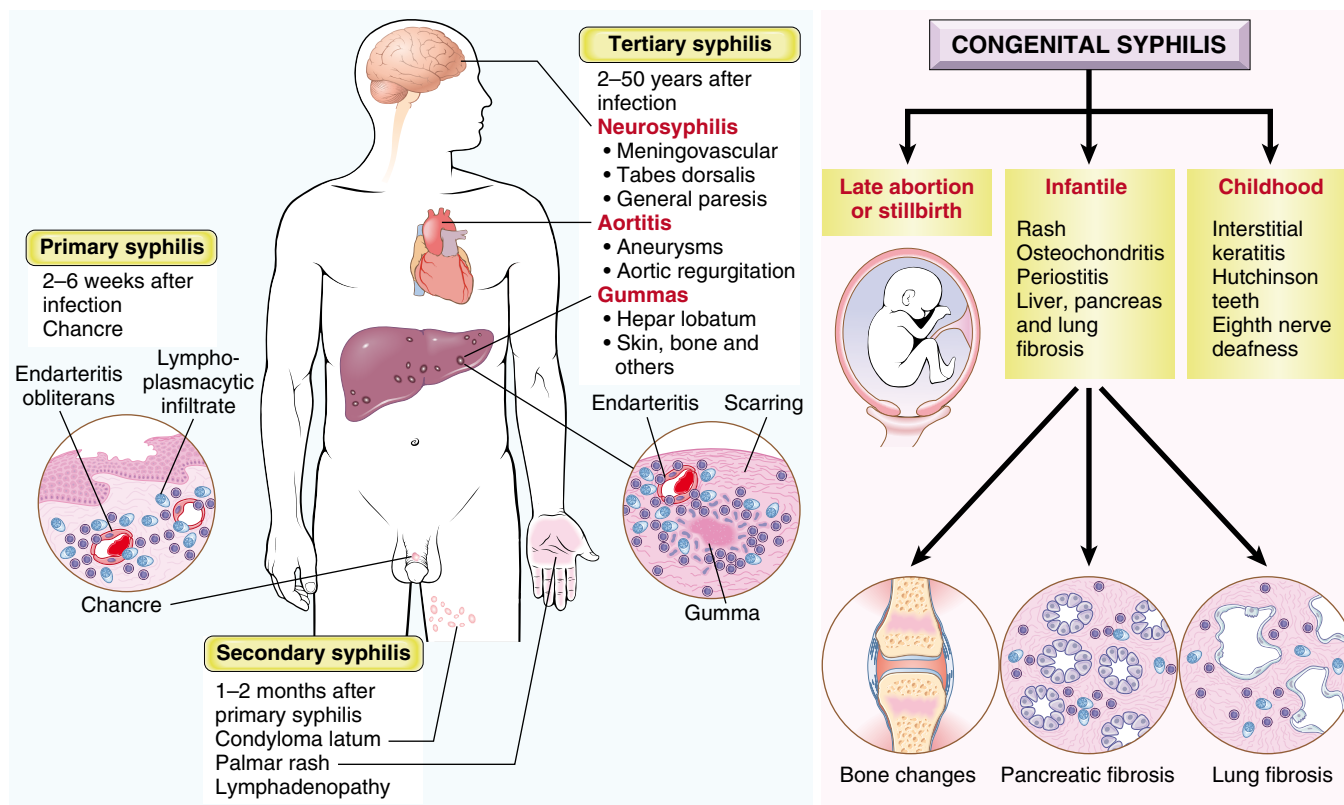


Fig. 8.35 Protean manifestations of syphilis.

dilation of the aortic root and arch, which causes aortic valve insufficiency and aneurysms of the proximal aorta (Chapter 11).

- *Neurosyphilis* may be symptomatic or asymptomatic. Symptomatic neurosyphilis is discussed in Chapter 28. Asymptomatic neurosyphilis, which accounts for about one-third of neurosyphilis cases, is initially suspected based on detection of CSF abnormalities such as pleocytosis (increased numbers of inflammatory cells), elevated protein levels, or decreased glucose, and is confirmed by detection of antibodies against cardiolipin-derived antigens stimulated by the spirochetes (discussed later) in the CSF. Antibiotics are given for a longer time if the spirochetes have spread to the CNS, so patients with tertiary syphilis should be tested for neurosyphilis even if they do not have neurologic symptoms.
- *Benign tertiary syphilis* is characterized by the formation of *gummas* in bone, skin, and the mucous membranes of the upper airway and mouth. Gummas are nodular lesions probably related to the development of delayed hypersensitivity to the bacteria. Skeletal involvement characteristically causes pain, tenderness, swelling, and pathologic fractures. Gummas in the skin and mucous membranes may produce nodular lesions or, rarely, destructive, ulcerative lesions. Gummas are rare because of the use of effective antibiotics.

Congenital Syphilis. Congenital syphilis occurs most frequently during maternal primary or secondary syphilis, when the spirochetes are most numerous. Intrauterine death and perinatal death each occurs in approximately 25% of cases of untreated congenital syphilis.

Manifestations of congenital syphilis are divided into those that occur in the first 2 years of life (infantile syphilis) and those that occur later (tardive syphilis). Infantile syphilis is often manifested by nasal discharge and congestion (snuffles) in the first few months of life. A desquamating or bullous rash can lead to sloughing of the skin, particularly of the hands and feet and around the mouth and anus. Hepatomegaly and skeletal abnormalities are common. Late manifestations develop in nearly one-half of untreated children with neonatal syphilis.

Serologic Tests for Syphilis. Serology remains the mainstay of diagnosis. There are two accepted testing algorithms for diagnosis of syphilis. The traditional approach uses a nontreponemal test to measure antibody to a cardiolipin-cholesterol-lecithin antigen that is present in both host tissues and *T. pallidum*. These antibodies are detected in the rapid plasma reagin (RPR) and Venereal Disease Research Laboratory (VDRL) tests, both of which are nonspecific. Positive results must be confirmed by a treponemal antibody test that measures antibodies specific to *T. pallidum*. These include the fluorescent treponemal antibody absorption test and the *T. pallidum* enzyme immunoassay test. Conversely, the reverse algorithm uses the treponemal test for screening, followed by a nontreponemal test.

The interpretation of these tests is complex because of differences in the antibodies they measure and imperfections in the tests.

- Both treponemal and nontreponemal antibody tests are only moderately sensitive (~70% to 85%) for primary syphilis.
- Both types of test are very sensitive (>95%) for secondary syphilis.
- Treponemal tests are very sensitive for tertiary and latent syphilis. Nontreponemal antibody levels fall with successful treatment of syphilis, and so changes in the quantitative titers should be used to monitor therapy.
- Treponemal tests, which are nonquantitative, remain positive even after successful therapy.
- Confirmatory testing is needed because false-positive results can occur in both nontreponemal and treponemal tests due to conditions including pregnancy, autoimmune diseases, and infections other than syphilis.

MORPHOLOGY

In primary syphilis, a chancre occurs on the penis or scrotum of 70% of men and on the vulva or cervix of 50% of females. The chancre is a slightly elevated, firm, reddened papule, up to several centimeters in diameter, that erodes to create a clean-based shallow ulcer. The contiguous induration creates a buttonlike mass directly adjacent to the eroded skin, providing the basis for the designation hard chancre (Fig. 8.36). On histologic examination, the chancre contains an intense infiltrate of plasma cells, with scattered macrophages and lymphocytes and a proliferative endarteritis. The endarteritis starts with endothelial cell activation and proliferation and progresses to intimal fibrosis (see Fig. 8.5). The regional nodes are usually enlarged due to nonspecific acute or chronic lymphadenitis, plasma cell–rich infiltrates, or granulomas.

In secondary syphilis, widespread mucocutaneous lesions involve the oral cavity, palms of the hands, and soles of the feet. The rash frequently consists of discrete red-brown macules less than 5 mm in diameter, but it may be follicular, pustular, annular, or scaling. Red lesions in the mouth or vagina contain the most organisms and are the most infectious. Histologically, the mucocutaneous lesions of secondary syphilis show the same plasma cell infiltrate and obliterative endarteritis as the primary chancre, although the inflammation is often less intense.

Tertiary syphilis may involve the aorta, CNS, liver, bones, or testes. The aortitis is caused by endarteritis of the vasa vasorum of the proximal aorta. Occlusion of the vasa vasorum results in scarring of the media of the proximal aortic wall, causing a loss of elasticity. There may be narrowing of the coronary artery ostia caused by subintimal scarring with resulting myocardial ischemia. The morphologic and clinical features of syphilitic aortitis are discussed in greater detail with diseases of the blood vessels (Chapter 11).

Neurosyphilis takes one of several forms, designated meningovascular syphilis, tabes dorsalis, and general paresis (Chapter 28).

Syphilitic gummas are white-gray and rubbery, occur singly or multiply, and vary in size from microscopic lesions resembling tubercles to large tumorlike masses. They occur in most organs but particularly in skin, subcutaneous tissue, bone, and joints. In the liver, scarring because of gummas may cause a distinctive hepatic lesion known as *hepar lobatum* (Fig. 8.37). On histologic examination, the gummas have centers of coagulated, necrotic material and margins composed of plump, palisading macrophages and fibroblasts surrounded by large numbers of mononuclear leukocytes, chiefly plasma cells. Treponemes are scant in gummas and are difficult to demonstrate.

The rash of **congenital syphilis** is more severe than that of adult secondary syphilis. It is a bullous eruption of the palms and soles of the feet associated with epidermal sloughing. **Syphilitic osteochondritis and periostitis** affect all bones, but lesions of the nose and lower legs are most distinctive. Destruction of the vomer causes collapse of the bridge of the nose and, later on, the characteristic saddle nose deformity. Periostitis of the tibia leads to excessive new bone growth on the anterior surfaces and anterior bowing, or **saber shin**. There is also widespread disturbance in endochondral bone formation. The epiphyses become widened as the cartilage overgrows, and cartilage is found in displaced islands within the metaphysis.

The **liver** is often severely affected in cases of congenital syphilis. Diffuse fibrosis permeates lobules to isolate hepatic cells into small nests, accompanied by the characteristic lymphoplasmacytic infiltrate and vascular changes. Gummas are occasionally found in the liver, even in early cases. The **lungs** may be affected by a diffuse interstitial fibrosis. In the syphilitic stillborn, the lungs appear pale and airless (pneumonia alba). The generalized spirochetemia may lead to diffuse interstitial inflammatory reactions in virtually any other organ (e.g., the pancreas, kidneys, heart, spleen, thymus, endocrine organs, and CNS).

The late manifestations of congenital syphilis include a distinctive **triad of interstitial keratitis, Hutchinson teeth, and eighth-nerve deafness**. In addition to interstitial keratitis, the ocular changes include choroiditis and abnormal retinal pigmentation. Hutchinson teeth are small incisors shaped like a screwdriver or a peg, often with notches in the enamel. Eighth-nerve deafness and optic nerve atrophy develop secondary to meningovascular syphilis.



Fig. 8.36 Syphilitic chancre in the scrotum. (Courtesy Dr. Richard Johnson, Beth Israel-Deaconess Hospital, Boston.)

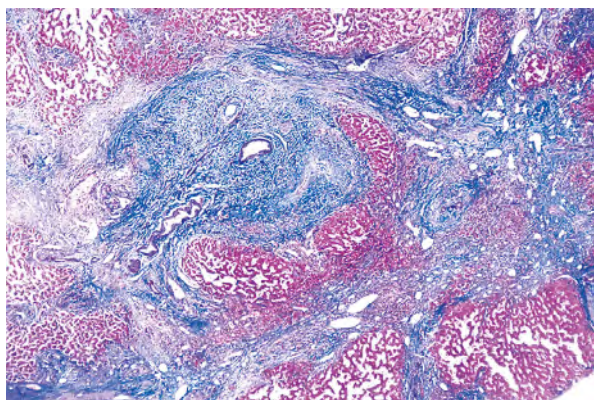


Fig. 8.37 Syphilis in liver. Trichrome stain of liver shows a gumma (scar), stained blue, caused by tertiary syphilis (the hepatic lesion is also known as hepar lobatum).

Lyme Disease

Lyme disease is a common arthropod-borne illness caused by spirochetes in the genus *Borrelia* that tend to cause persistent chronic arthritis. It is named for the Connecticut town of Lyme, where there was an epidemic of arthritis associated with skin erythema in the mid-1970s. The common species of *Borrelia* that cause Lyme diseases are *Borrelia burgdorferi* in the United States, and additionally *B. afzelii* and *B. garinii* in Europe and Asia. These are each transmitted from rodents or, for *B. garinii*, from birds, to people by *Ixodes* deer ticks. Lyme disease is endemic in the United States, Europe, and Asia. In the United States, there were between 30,000 and 40,000 cases reported each year for several years; however, this may represent only 10% of people treated for Lyme disease. Most cases occur in the northeastern states and the upper Midwest. In endemic areas, *B. burgdorferi* infects up to 50% of ticks, which may also be infected with *Ehrlichia* spp. and *Babesia* spp. Serology is the main method of diagnosis, but PCR can be done on infected tissue.

Lyme disease involves multiple organ systems and is divided into three stages (Fig. 8.38).

- In *early localized disease*, spirochetes multiply and spread in the dermis at the site of a tick bite, causing an expanding area of redness, often with a pale center. This lesion, called *erythema migrans*, is present in approximately 80% of patients and may be accompanied by fever and lymphadenopathy. The rash spontaneously disappears in 4 to 12 weeks.
- In *early disseminated disease*, spirochetes spread hematogenously throughout the body and cause secondary skin lesions, lymphadenopathy, migratory joint and muscle pain, cardiac arrhythmias, and meningitis often associated with cranial nerve involvement. In Europe, *B. afzelii* is associated with a rare manifestation, borreliar lymphocytoma, in which there is blue to red swelling of the earlobe, in children, or nipple, in adults, with lymphocytic infiltration.
- The third stage, *late disseminated disease*, may appear many months after the tick bite. It manifests as chronic arthritis, sometimes with severe damage to large joints. Less often, patients have polyneuropathy and encephalitis that vary from mild to debilitating. Lyme encephalopathy is less common in Europe than in the United States.

Pathogenesis

B. burgdorferi does not produce endotoxin or exotoxins that damage the host. Much of the pathology associated with the infection is thought to be secondary to the immune response against the bacteria and the inflammation that accompanies it. The initial immune response is stimulated by binding of bacterial lipoproteins to TLR1 and TLR2 on macrophages. In response, these cells release pro-inflammatory cytokines (IL-6 and TNF) and generate bactericidal reactive nitrogen intermediates, reducing but usually not eliminating the infection.

The inflammatory lesions are likely triggered by T cells and cytokines. *Borrelia*-specific antibodies, made 2 to 4 weeks after infection, trigger complement-mediated phagocytosis and killing of the bacteria; however, *B. burgdorferi* can escape the antibody response through antigenic variation and downregulation of outer surface protein C, which is needed during establishment, but not persistence, of infection. *B. burgdorferi* has a plasmid with a single promoter sequence and approximately 15 coding sequences for an antigenic surface protein, VlsE (variable major protein-like sequence, expressed), each of which can shuttle into position next to the promoter and be expressed. Thus, as the host mounts an antibody response to one VlsE, bacteria expressing an alternate VlsE protein emerge and can escape immune recognition. Chronic manifestations of Lyme disease, such as late arthritis, may be caused by the immune response against some, as yet unknown, bacterial antigen that cross-reacts with a self antigen.

MORPHOLOGY

Skin lesions caused by *B. burgdorferi* are characterized by edema and a lymphocytic-plasma cell infiltrate. In early Lyme arthritis, the synovium resembles early rheumatoid arthritis, with villous hypertrophy, lining-cell hyperplasia, and abundant lymphocytes and plasma cells in the subsynovium. A distinctive feature of Lyme arthritis is an arteritis, which produces onion skin-like lesions resembling those seen in lupus (Chapter 6). In late Lyme disease, there may be extensive erosion of the cartilage in large joints. In Lyme meningitis, the CSF is hypercellular due to a marked lymphoplasmacytic infiltrate, and it contains antispirochete IgGs.

Anaerobic Bacterial Infections

Many anaerobic bacteria are normal microbiota in sites of the body that have low oxygen levels. The anaerobic microbiota cause disease (such as abscesses or peritonitis) when they are introduced into normally sterile sites or when the balance of organisms is disturbed and pathogenic anaerobes overgrow (e.g., *C. difficile* colitis with antibiotic treatment). Environmental anaerobes also cause disease (tetanus, botulism, and gas gangrene).

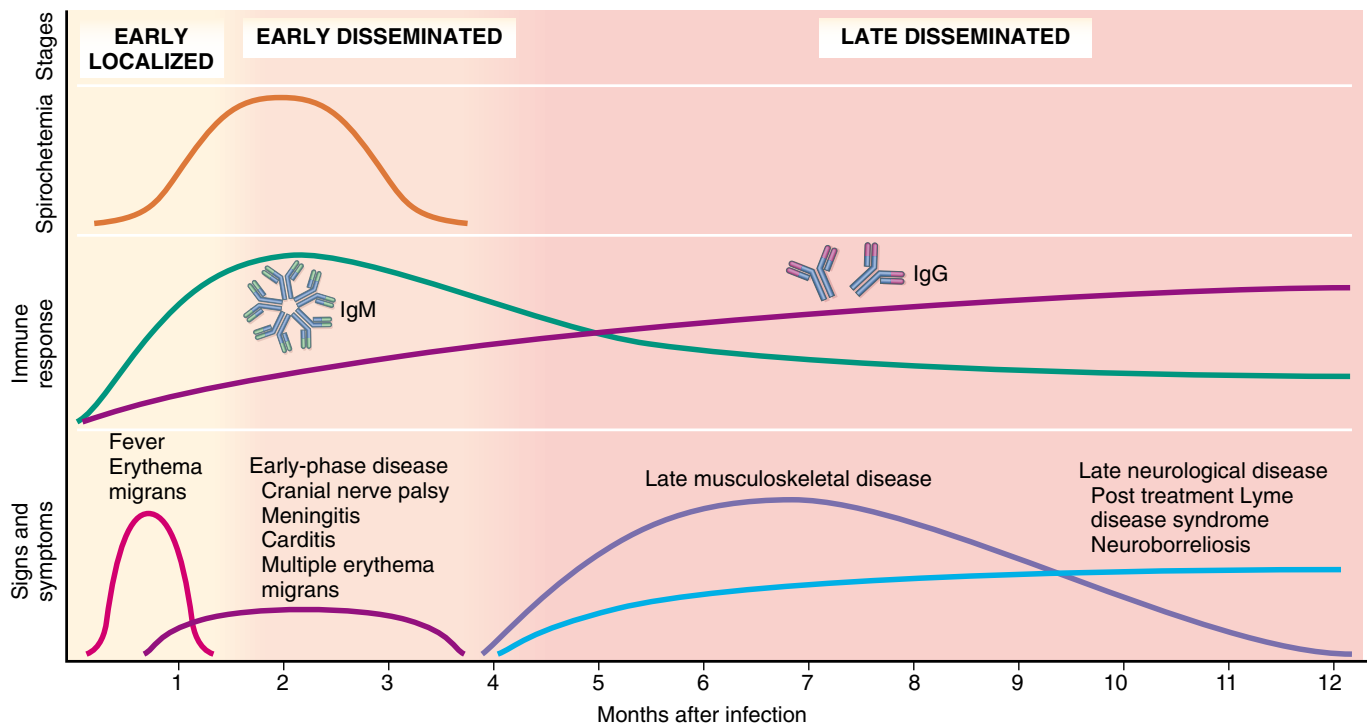


Fig. 8.38 Lyme disease progresses through three clinically recognizable phases: early localized, early disseminated, and late disseminated. Although initial manifestations result directly from spirochete infection, later signs and symptoms are likely immune-mediated. (Courtesy Dr. Charles Chiu, University of California, San Francisco.)

Abscesses

Commensal bacteria from adjacent sites (oropharynx, intestine, and female genital tract) are the usual cause of abscesses, so the species found in an abscess often reflect the normal microbiota in that site. Abscesses are usually caused by mixed anaerobic and facultative aerobic bacterial infections. Because most anaerobes that cause abscesses are part of the normal microbiota, it is not surprising that these organisms do not usually produce toxins.

The bacteria found in head and neck abscesses reflect oral and pharyngeal microbiota. Common anaerobes at this site include the gram-negative bacilli *Prevotella* spp. and *Porphyromonas* spp., often mixed with the facultative *S. aureus* and *S. pyogenes*. *Fusobacterium necrophorum*, an oral commensal, causes *Lemierre syndrome*, characterized by infection of the lateral pharyngeal space and septic jugular vein thrombosis. Abdominal abscesses are caused by the anaerobes of the gastrointestinal tract, including gram-positive *Peptostreptococcus* spp. and *Clostridium* spp., as well as gram-negative *Bacteroides fragilis* and *E. coli*. Genital tract infections in females (e.g., Bartholin cyst abscesses and tubo-ovarian abscesses) are caused by anaerobic gram-negative bacilli such as *Prevotella* spp., often mixed with *E. coli* or *S. agalactiae*.

MORPHOLOGY

Abscesses caused by anaerobes contain discolored, foul-smelling pus that is often poorly walled off. Otherwise, these lesions pathologically resemble those of the common pyogenic infections. Gram stain reveals mixed infection with gram-positive and gram-negative rods and gram-positive cocci mixed with neutrophils.

Clostridial Infections

Clostridium spp. are gram-positive bacilli that grow under anaerobic conditions and produce spores that are present in the soil. Several types of disease can occur:

- *C. perfringens*, *C. septicum*, and other species may variously cause cellulitis and myonecrosis of traumatic and surgical wounds (*gas gangrene*); uterine myonecrosis often associated with abortions; mild food poisoning; and infection of the small bowel associated with ischemia or neutropenia that often leads to severe sepsis. Spontaneous gas gangrene due to *C. septicum* is highly associated with an underlying malignancy.
- *C. tetani*, the cause of *tetanus*, proliferates in puncture wounds and in the umbilical stump of newborn infants and releases a potent neurotoxin that causes increased muscle tone and generalized spasms of skeletal muscles (lockjaw). Tetanus toxoid (formalin-fixed tetanus toxin) is part of the DPT (diphtheria, pertussis, and tetanus) immunization, and this has greatly decreased the incidence of tetanus worldwide.
- *C. botulinum*, the cause of *botulism*, grows in inadequately cooked foods and releases a potent neurotoxin that blocks synaptic release of acetylcholine and causes flaccid paralysis of respiratory and skeletal muscles.
- *Clostridioides difficile* (previously *Clostridium difficile*) proliferates in the intestines when dysbiosis occurs, most often due to antibiotic treatment. The bacteria release toxin and cause *pseudomembranous colitis* (Chapter 17). There are 460,000 cases leading to 20,000 deaths per year in the United States. Differentiating asymptomatic carriage from infection is challenging and there are various testing algorithms using a combination of PCR screening for the *C. difficile* toxin gene, immunoassays for *C. difficile* antigens, and clinical presentation to determine if treatment is needed.

In general, other clostridial infections can be diagnosed by culture (cellulitis, myonecrosis), PCR assays for toxin, or both (botulism).

Pathogenesis

C. perfringens does not grow in the presence of oxygen, so tissue death is a prerequisite for growth of the bacteria in the host. These bacteria release collagenase and hyaluronidase that degrade extracellular matrix proteins and contribute to bacterial invasiveness, but their most powerful virulence factors are the many toxins they produce. *C. perfringens* secretes 14 toxins, the most important of which is α -toxin. This toxin has multiple actions. It is a phospholipase that degrades lecithin, a major component of cell membranes, and so destroys red cells, platelets, and muscle cells, causing myonecrosis. It also has a sphingomyelinase activity that contributes to nerve sheath damage.

Ingestion of food contaminated with *C. perfringens* causes a brief episode of diarrhea. Spores, usually in contaminated meat, survive cooking, and the organism proliferates in cooling food. *C. perfringens* enterotoxin forms pores in the epithelial cell membranes, lysing the cells and disrupting tight junctions between epithelial cells.

The neurotoxins produced by *C. botulinum* and *C. tetani* both inhibit release of neurotransmitters, resulting in paralysis. Botulism toxin, eaten in contaminated foods or absorbed from wounds infected with *C. botulinum*, binds gangliosides on motor neurons and is transported into the cell. In the cytoplasm, the A fragment of botulism toxin cleaves a protein called synaptobrevin that mediates fusion of neurotransmitter-containing vesicles with the neuron membrane. By blocking vesicle fusion, botulism toxin prevents the release of acetylcholine at the neuromuscular junction, resulting in flaccid paralysis. If the respiratory muscles are affected, botulism can lead to death. The widespread use of botulism toxin (Botox) in cosmetic surgery is based on its ability to cause paralysis of strategically chosen muscles on the face. Tetanus toxin causes a violent spastic paralysis by blocking release of γ -aminobutyric acid, a neurotransmitter that inhibits motor neurons.

C. difficile produces toxin A, an enterotoxin that stimulates chemokine production and thus attracts leukocytes, and toxin B, a cytotoxin that causes distinctive cytopathic effects in cultured cells. Both toxins are glucosyl transferases and are part of a pathogenicity island that is absent from the chromosomes of nonpathogenic strains of *C. difficile*. There has been significant success in treatment of recurrent *C. difficile* infection with antibiotics along with fecal transplantation, emphasizing the role of normal commensal microbiota in defense against pathogens.

MORPHOLOGY

The most significant lesions are caused by *C. perfringens*; these are described next. **Clostridial cellulitis**, which originates in wounds, can be differentiated from infection caused by pyogenic cocci by its foul odor, its thin, discolored exudate, and the rapid onset of widespread tissue destruction. On microscopic examination, the amount of tissue necrosis is disproportionate to the number of neutrophils and gram-positive bacteria present (Fig. 8.39). Clostridial cellulitis, which often has granulation tissue at its borders, is treatable by debridement and antibiotics.

In contrast, **clostridial gas gangrene** is life-threatening and is characterized by marked edema and enzymatic necrosis of involved muscle cells 1 to 3 days after injury. An extensive fluid exudate, which lacks inflammatory cells, causes swelling of the affected region and the overlying skin, producing large bullous vesicles that rupture. Gas bubbles caused by bacterial fermentation appear within the gangrenous tissues. As the infection progresses, the inflamed muscles become soft, blue-black, friable, and semifluid because of the

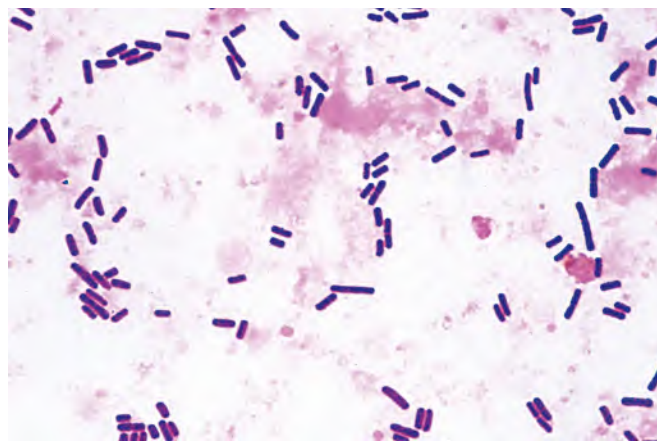


Fig. 8.39 Boxcar-shaped gram-positive *Clostridium perfringens* intermingled with necrotic debris in gangrenous tissue.

proteolytic action of the released bacterial enzymes. On microscopic examination, there is severe **myonecrosis**, extensive hemolysis, and marked vascular injury with thrombosis. *C. perfringens* is also associated with dusky, wedge-shaped infarcts in the small bowel, particularly in neutropenic people. Regardless of the site of entry, when *C. perfringens* disseminates hematogenously, there is widespread formation of gas bubbles.

Despite the severe neurologic damage caused by botulinum and tetanus toxins, the neuropathologic changes are subtle and nonspecific.

Obligate Intracellular Bacterial Infections

Obligate intracellular bacteria proliferate only within host cells, although some may survive outside cells. These organisms are well adapted to the intracellular environment, with membrane pumps to capture amino acids and glucose-6-phosphate from the host cell for synthesis of proteins and ATP.

Chlamydial Infections

C. trachomatis is a small gram-negative bacterium that is an obligate intracellular pathogen. *C. trachomatis* exists in two forms during its unique life cycle. The infectious form, called the elementary body, has reduced metabolic activity. Elementary bodies bind host cells and then use a type III secretion system to inject several effectors into host cells, leading to actin remodeling at the site of bacterial entry into the host cell by phagocytosis. In the presence of intracellular metabolites, such as glucose-6-phosphate and glutamine, the elementary body differentiates into a metabolically active form called the reticulate body within a membrane-bound inclusion derived from the endosome. The bacteria use the type III secretion system to modify the inclusion by adding an array of inclusion membrane proteins (Incs). Incs and other bacterial molecules prevent the inclusion from maturing into an inhospitable phagolysosome and protect the bacteria from autophagy, as well as facilitating the entry of nutrients into the inclusion. The reticulate body uses nutrients from the host cell to replicate and forms new infectious elementary bodies that lyse or are extruded from the infected cell.

The nature of diseases caused by *C. trachomatis* infection varies with different serotypes of the bacteria: urogenital infections and inclusion conjunctivitis are caused by serotypes D to K, lymphogranuloma venereum by serotypes L1, L2, and L3, and an ocular infection of children, trachoma, by serotypes A, B, and C. The venereal infections caused by *C. trachomatis* are discussed here.

Genital infection by *C. trachomatis* is the most common bacterial STI in the world. In 2021, over 1.6 million cases of chlamydia were reported to the CDC. After rising steadily for several years, the rate of genital chlamydia in the United States declined in 2019 and 2020, but subsequently rose in 2021. Genital *C. trachomatis* infections (other than lymphogranuloma venereum, discussed later) are associated with clinical features that are similar to those caused by *N. gonorrhoeae*. Patients may develop epididymitis, prostatitis, pelvic inflammatory disease, pharyngitis, conjunctivitis, perihepatic inflammation, and proctitis. Unlike *N. gonorrhoeae* urethritis, *C. trachomatis* urethritis in men is commonly asymptomatic and so may go untreated. Both *N. gonorrhoeae* and *C. trachomatis* frequently cause asymptomatic infections in females. *C. trachomatis* urethritis can be diagnosed using amplified nucleic-acid tests performed on genital swabs or urine specimens.

Genital infection with the L serotypes of *C. trachomatis* causes *lymphogranuloma venereum*, a chronic, ulcerative disease. Lymphogranuloma venereum is a sporadic cause of proctitis in men who have sex with men in the United States and Western Europe, but it is endemic in parts of Asia, Africa, the Caribbean region, and South America. The infection initially manifests as a small, often unnoticed, papule on the genital mucosa or nearby skin. Between 2 and 6 weeks later, growth of the organism and the host response in draining lymph nodes produce swollen, tender lymph nodes, which may coalesce and rupture. If not treated, the infection can subsequently cause fibrosis and strictures in the anogenital tract. Rectal strictures are particularly common in females.

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The features of *C. trachomatis* **urethritis** are virtually identical to those of gonorrhea. The primary infection is characterized by a mucopurulent discharge containing a predominance of neutrophils. Organisms are not visible in Gram-stained smears or sections.

The lesions of **lymphogranuloma venereum** contain a mixed granulomatous and neutrophilic inflammatory response. Variable numbers of chlamydial inclusions are seen in the cytoplasm of epithelial cells or inflammatory cells. Regional lymphadenopathy is common, usually occurring within 30 days of infection. Lymph node involvement is characterized by a granulomatous inflammatory reaction associated with irregularly shaped foci of necrosis containing neutrophils (stellate abscesses). With time, the inflammatory reaction is dominated by nonspecific chronic inflammatory infiltrates and extensive fibrosis. The latter, in turn, may cause local lymphatic obstruction, lymphedema, and strictures. In active lesions, the diagnosis of lymphogranuloma venereum may be made by demonstration of the organism in biopsy sections or smears of exudate. In more chronic cases, the diagnosis rests with the demonstration of antibodies to the appropriate chlamydial serotypes in the patient's serum.

Infections Caused by Other Intracellular Bacteria

Rickettsia spp., *Orientia* spp., *Ehrlichia* spp., and *Anaplasma* spp. are vector-borne obligate intracellular bacteria that cause epidemic and scrub typhus, spotted fevers (*Rickettsia rickettsii* and others), ehrlichiosis, and anaplasmosis. These organisms have the structure of gram-negative, rod-shaped bacteria, although they stain poorly with Gram stain.

- **Epidemic typhus** (caused by *Rickettsia prowazekii*) is transmitted from person-to-person by body and head lice. It occurs in wars and states of poverty, when individuals live in close contact with poor hygiene. Manifestations include a rash that is initially macular, progressing to a petechial, maculopapular rash on the entire body except the face, palms, and soles.

- **Scrub typhus** (caused by *Orientia tsutsugamushi*) is transmitted by chiggers (mites). It is endemic in areas of rural Asia and Australia, and it has recently emerged in South Korea, Chile, and areas of China. Fever, headache, myalgia, and cough are usual symptoms, sometimes accompanied by a characteristic eschar and associated lymphadenopathy from the chigger bite.
- **Rocky Mountain spotted fever** (caused by *Rickettsia rickettsii*) is transmitted to humans by dog ticks. It is most common in the southeastern and south-central United States but occurs in other regions of the United States as well as Canada, Mexico, Central America, and South America. It begins as a nonspecific severe illness with fever, myalgias, and gastrointestinal distress, and then progresses to a widespread macular then petechial rash that can involve the palms and soles.
- **Ehrlichiosis** (caused mainly by *Ehrlichia chaffeensis*) and **anaplasmosis** (*Anaplasma phagocytophilum*) are transmitted by the lone star tick and deer tick, respectively. These bacteria predominantly infect monocytes (*E. chaffeensis*) or neutrophils (*A. phagocytophilum*). Ehrlichiosis and anaplasmosis are characterized by abrupt onset of fever, headache, and malaise, and may progress to respiratory insufficiency, renal failure, and shock. Rash occurs in approximately 40% of people with *E. chaffeensis* infections.

Rickettsial diseases are usually diagnosed clinically and confirmed by serology or immunostaining of the organisms.

Pathogenesis

The severe manifestations of rickettsial infections are primarily due to infection of endothelial cells and the consequent endothelial dysfunction and injury. The rickettsiae that cause typhus and spotted fevers preferentially infect endothelial cells in the lungs and brain. The bacteria enter cells by endocytosis but escape from the endosome into the cytoplasm using hemolysins to disrupt phagosomal membranes. The organisms proliferate in the cytoplasm and then either lyse the cell (*typhus group*) or spread from cell to cell using actin-based motility (*spotted fever group*). The widespread endothelial dysfunction can cause shock, peripheral and pulmonary edema, and DIC, as well as renal failure and a variety of CNS manifestations that can include coma.

The innate immune response to rickettsial infection is complex, but includes NK cells, which produce IFN- γ that stimulates production of bactericidal nitric oxide derivatives in macrophages. Later in the infection, CTLs lyse infected cells, reducing bacterial proliferation.

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Typhus fever. In mild cases, the gross changes are limited to a rash and small hemorrhages due to the vascular lesions. In more severe cases, there may be areas of necrosis of the skin and gangrene of the tips of the fingers, nose, earlobes, scrotum, penis, and vulva. In such cases, irregular ecchymotic hemorrhages may be found internally, principally in the brain, heart muscle, testes, serosal membrane, lungs, and kidneys.

The most prominent microscopic changes are small-vessel lesions and focal areas of hemorrhage and inflammation in various organs and tissues. Endothelial swelling in capillaries, arterioles, and venules may narrow the lumens of these vessels. A cuff of mononuclear inflammatory cells usually surrounds the affected vessel. The vascular lumens are sometimes thrombosed. Necrosis of the vessel wall is unusual in typhus (compared with Rocky Mountain spotted fever). Vascular thromboses lead to ischemic necrosis of the skin and other structures in a minority of cases. In the brain, characteristic typhus nodules are composed of focal microglial proliferations with an infiltrate of mixed T lymphocytes and macrophages (Fig. 8.40).

Scrub typhus, or mite-borne infection, is usually a milder version of typhus fever. The rash is usually transitory or might not appear. Vascular necrosis or thrombosis is rare, but there may be a prominent inflammatory lymphadenopathy.

Rocky Mountain Spotted Fever. A hemorrhagic rash that extends over the entire body, including the palms of the hands and soles of the feet, is the hallmark of Rocky Mountain spotted fever. An eschar at the site of the tick bite is uncommon with Rocky Mountain spotted fever. The vascular lesions that underlie the rash often lead to acute necrosis, fibrin extravasation, and occasionally thrombosis of the small blood vessels, including arterioles (Fig. 8.41). In severe Rocky Mountain spotted fever, foci of necrotic skin appear, particularly on the fingers, toes, elbows, ears, and scrotum. The perivascular inflammatory response, similar to that of typhus, is seen in the brain, skeletal muscle, lungs, kidneys, testes, and heart muscle. The vascular lesions in the brain may involve larger vessels and produce microinfarcts. Noncardiogenic pulmonary edema causing adult respiratory distress syndrome is the major cause of death in patients with Rocky Mountain spotted fever.

Ehrlichiosis and **anaplasmosis** have similar presentations. The rash is nonspecific and can be macular, maculopapular, or petechial. Characteristic cytoplasmic inclusions (morulae), composed of masses of bacteria that occasionally take the shape of a mulberry, are present in leukocytes (Fig. 8.42).

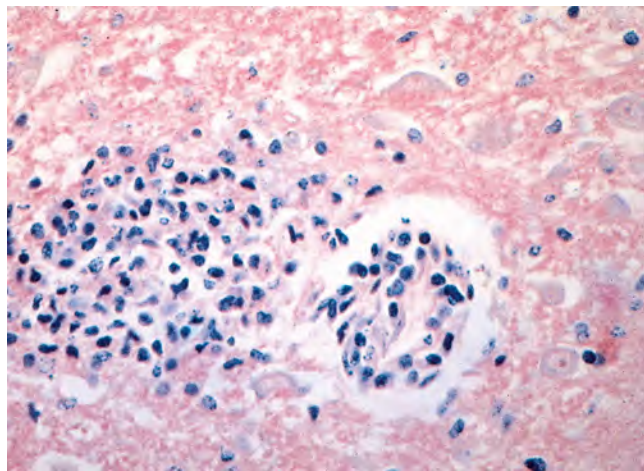


Fig. 8.40 Typhus nodule in the brain.

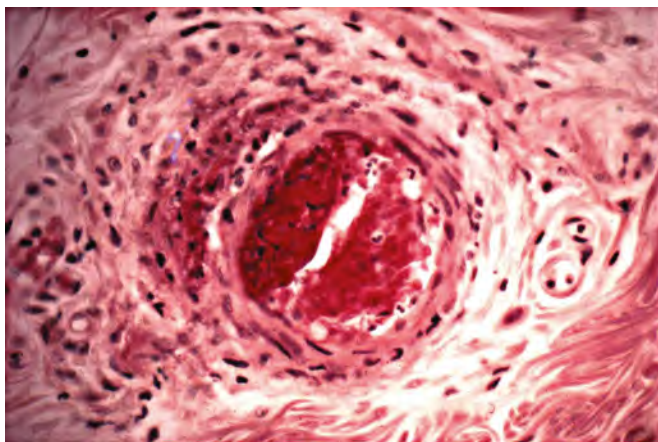


Fig. 8.41 Rocky Mountain spotted fever with a thrombosed vessel and vasculitis.

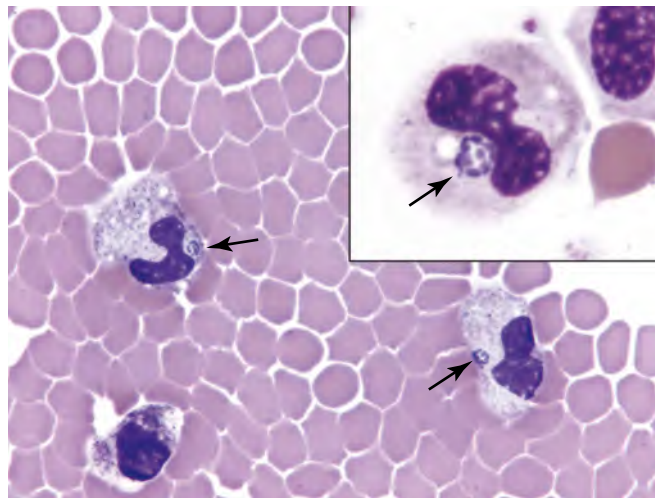


Fig. 8.42 Anaplasmosis. Peripheral blood granulocytes (band neutrophils) containing *Anaplasma* inclusions (arrows). Inset shows high-power detail of an inclusion. (Courtesy Dr. Tad Weiczorek, Faulkner Hospital, Boston.)

FUNGAL INFECTIONS

Fungi are eukaryotes that grow as individual cells alone or in chains (yeast) or multicellular filaments (mold). Cell walls give fungi their shape. Yeasts are round to oval and mainly reproduce by budding. Some yeasts, such as *C. albicans*, can produce buds that fail to detach and become elongated, producing a chain of elongated yeast cells called *pseudohyphae*. Molds consist of threadlike filaments (hyphae) that grow and divide at their tips. They can produce round cells called *conidia* that easily become airborne, disseminating the fungus. Many medically important fungi are *dimorphic*, existing as yeast or molds, depending on environmental conditions (yeast forms at human body temperature and mold forms at room temperature). Fungal infections can be diagnosed by histologic examination, although definitive identification of some species requires culture.

Fungal infections, also called *mycoses*, are of four major types:

- *Superficial and cutaneous mycoses* are common and limited to the superficial or keratinized layers of skin, hair, and nails.
- *Subcutaneous mycoses* involve the skin, subcutaneous tissues, and lymphatics and rarely disseminate systemically.
- *Endemic mycoses* are caused by dimorphic fungi that can produce serious systemic illness in healthy individuals.
- *Opportunistic mycoses* can cause life-threatening systemic diseases in individuals who are immunosuppressed or who carry implanted prosthetic devices or vascular catheters. Some of the fungi that cause opportunistic mycoses are discussed in the following sections; those involving specific organs are discussed in other chapters.

Yeast Infections

Candidiasis

C. albicans is the most prevalent fungal pathogen of humans. Although there are more than 200 species of *Candida*, there are about 15 to 20 species that are frequently seen in human infections, with *C. albicans*, *C. auris*, *C. glabrata*, *C. tropicalis*, *C. parapsilosis*, and *C. krusei* being the most frequent. This section will focus on *C. albicans* and *C. auris*.

Most *C. albicans* infections originate when these members of the commensal microbiota breach the skin or mucosal barriers. Residing normally in the skin, mouth, gastrointestinal tract, and vagina, *Candida* spp. usually live as benign commensals and seldom produce disease in

healthy people. These infections may be confined to the skin or mucous membranes or may disseminate widely. In otherwise healthy people, *C. albicans* causes vaginitis and diaper rash. Individuals with diabetes and burn patients are particularly susceptible to superficial candidiasis. In individuals with indwelling intravenous lines or catheters or undergoing peritoneal dialysis, *C. albicans* can spread to the bloodstream. Severe disseminated candidiasis most commonly occurs in patients who are neutropenic due to leukemia, chemotherapy, or hematopoietic stem cell transplantation, and may cause shock and DIC.

Pathogenesis

There is no known environmental reservoir for *C. albicans*, unlike other *Candida* spp. *C. albicans* has developed multiple adaptive mechanisms to complete its life cycle fully within the human host. *C. albicans* can shift between nine distinct cell shapes. Phenotypic switching involves coordinated transcriptional regulation of phase-specific genes and provides a way for *C. albicans* to adapt to changes in the environment such as temperature, nutrient availability, antibiotic therapy, or the immune response. These variants can exhibit altered colony morphology, cell shape, antigenicity, and virulence.

C. albicans produces a large number of functionally distinct adhesion molecules involved in binding to fibrinogen, fibronectin, laminin, epithelial cells, and endothelial cells. *C. albicans* also produces enzymes that contribute to invasiveness, including at least nine secreted aspartyl proteinases, which may promote tissue invasion by degrading extracellular matrix proteins, and catalases, which may enable the organism to resist oxidative killing by phagocytic cells.

The ability of *C. albicans* to grow as biofilms also contributes to its capacity to cause disease, with at least 30 factors identified to play a role in adhesion, maturation, and dispersion that affect biofilm formation. The biofilms are microbial communities consisting of mixtures of yeast, filamentous forms, and fungal-derived extracellular matrix. *C. albicans* can form biofilms on implanted medical devices that reduce the organism's susceptibility to immune responses and antifungal drug therapy.

Neutrophils, macrophages, and Th17 cells are important for protection against *Candida* infection.

- Neutrophils and macrophages phagocytose *C. albicans*, and oxidative killing by these phagocytes is a first line of host defense. The important role of neutrophils and macrophages is illustrated by the increased risk of *C. albicans* infection in individuals with neutropenia or defects in phagocyte oxidase or myeloperoxidase. Filamentous forms, but not yeast, can escape from phagosomes, enter the cytoplasm, and proliferate.
- *C. albicans* yeast forms activate dendritic cells through multiple pathways, more so than do the filamentous forms of the fungi. For example, β -1,3-glucan expressed by the yeast engages the lectin called dectin on dendritic cells and induces IL-6 and IL-23 production, which promotes Th17 responses. The Th17 responses elicited by *C. albicans* promote the recruitment of neutrophils and monocytes (Chapter 6). These responses are critical for protection against *C. albicans* infection, as shown by recurrent mucocutaneous candidiasis in individuals with either low T-cell counts due to HIV infection or inherited or acquired defects in Th17 cell development and function.

may be present together in the same tissue. The organisms may be seen in routine hematoxylin and eosin stains, but a variety of special fungal stains (Gömöri methenamine silver, periodic acid-Schiff) are commonly used to better visualize them.

Most commonly, candidiasis takes the form of a superficial infection on mucosal surfaces of the oral cavity (**thrush**). Florid proliferation of the fungi creates gray-white, dirty-looking pseudomembranes composed of matted organisms and inflammatory debris. Deeper parts of the lesion show mucosal congestion and inflammation. This form of candidiasis is seen in newborns, debilitated people, children receiving oral steroids for asthma, and following a course of broad-spectrum antibiotics that destroys competing normal bacterial microbiota. The other major risk group includes HIV-positive patients; people with oral thrush for no obvious reason should be evaluated for HIV infection.

C. albicans **esophagitis** is commonly seen in AIDS patients and in those with hematologic malignancies. These patients present with dysphagia (painful swallowing) and retrosternal pain; endoscopy demonstrates white plaques and pseudomembranes resembling oral thrush on the esophageal mucosa (see Fig. 8.43).

C. albicans **vaginitis** is common, especially in females who are diabetic, pregnant, or on oral contraceptive pills. It is usually associated with intense itching and a thick, curd-like discharge.

Cutaneous candidiasis can present in many different forms, including infection of the nail proper (onychomycosis); nail folds (paronychia); hair follicles (folliculitis); moist, intertriginous skin, such as armpits or webs of the fingers and toes (intertrigo); and penile skin (balanitis). Diaper rash is a cutaneous candidal infection seen in the perineum of infants, the region in contact with wet diapers.

Invasive candidiasis is caused by blood-borne dissemination of organisms to various tissues or organs. Common patterns include (1) renal abscesses, (2) myocardial abscesses and endocarditis, (3) brain microabscesses and meningitis, (4) endophthalmitis (virtually any eye structure can be involved), and (5) hepatic abscesses. In any of these locations, depending on the immune status of the infected person, the fungus may evoke little inflammation, cause the usual suppurative response, or occasionally produce granulomas. People with acute leukemias who are profoundly neutropenic after chemotherapy are particularly prone to developing systemic disease. *Candida* spp. endocarditis is the most common fungal endocarditis, usually occurring in the setting of prosthetic heart valves or in intravenous drug abusers. In the latter group, the tricuspid valve is involved.

***Candida auris* Infections.** *C. auris* is an emerging pathogen affecting patients who are critically ill or immunocompromised, with a mortality rate for invasive infection of 30% to 60%. *C. auris* was identified as a cause of ear infection in Japan in 2009. Multiple clades of *C. auris* subsequently emerged as causes of infections independently in different parts of the world, leading the Centers for Disease Control and Prevention to issue a clinical alert in 2016. *C. auris* has now been reported in 55 countries on six continents. In 2022, there were 2377 cases associated with illness reported in the United States, with serious infections affecting the bloodstream, gut, wounds, and other body sites. Resistance to fluconazole, amphotericin B and echinocandins is 90%, 30%, and 5%, respectively, and there are reports of pan-resistant strains. Colonization with *C. auris* has been reported in the nostrils, groin, axilla, and rectum and most colonized patients do not become sick. Increased colonization and infection with non-*Candida albicans* spp. are attributed in part to increased use of prophylactic antifungal agents. There have been multiple outbreaks of *C. auris* infection in intensive care units in patients with central venous or urinary catheters, so biofilm formation is a suggested virulence mechanism of this organism. This fungus has been demonstrated to survive on dry and moist

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In tissue sections, *C. albicans* can appear as yeast, pseudohyphae, and, less commonly, true hyphae, defined by the presence of septae, such as under reduced oxygen tension (Fig. 8.43). Pseudohyphae, an important diagnostic clue, are a chain of budding yeast cells joined end to end at constrictions. All forms

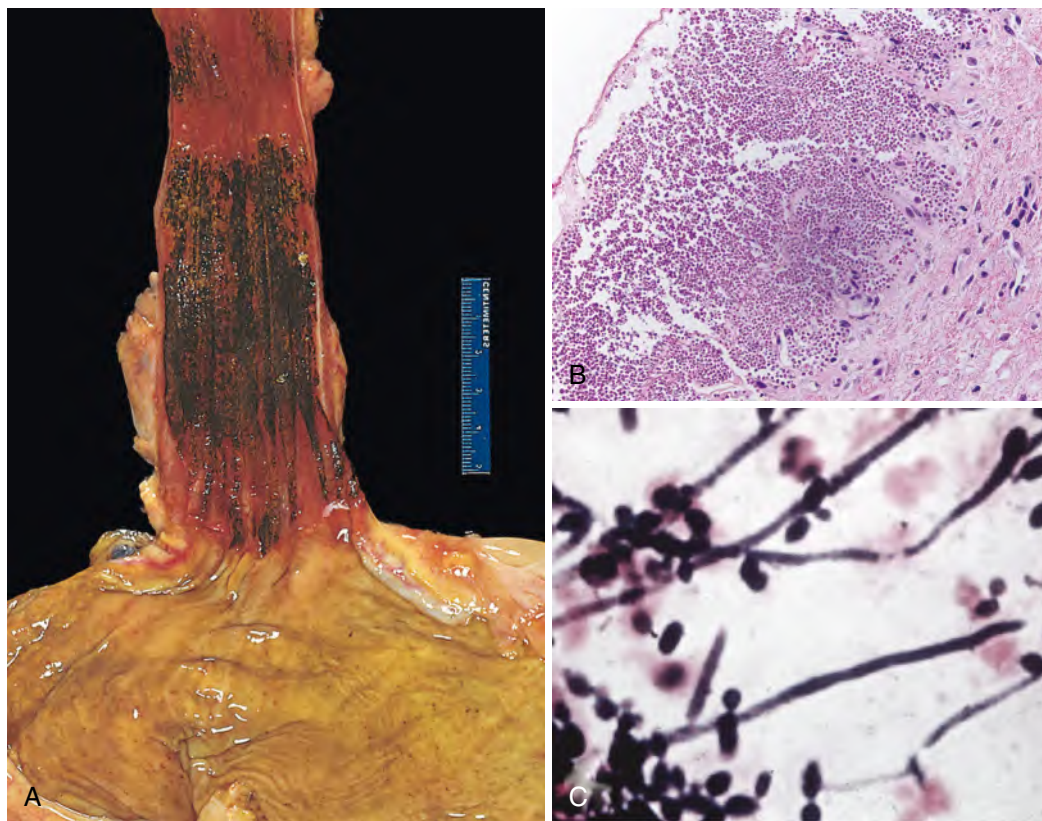


Fig. 8.43 The morphology of *Candida* infections. (A) Severe candidiasis of the distal esophagus. (B) Hematoxylin and eosin stain of esophageal candidiasis reveals the dense mat of *Candida* spp. (C) Characteristic pseudohyphae and budding yeast of *Candida* spp. (C, Courtesy Dr. Dominick Cuvuoti, Department of Pathology, University of Texas Southwestern Medical School.)

surfaces for up to 14 days, so environmental cleaning to eliminate a source of nosocomial infections is a challenge.

Cryptococcosis

Two species of cryptococcus are known to cause disease in humans, *C. neoformans* and *C. gattii*, both of which grow as encapsulated yeasts. It has long been recognized that although *C. neoformans* may cause meningoencephalitis in otherwise healthy individuals, it more frequently presents as an opportunistic infection in people with AIDS, leukemia, lymphoma, systemic lupus erythematosus, or sarcoidosis, as well as in immunosuppressed transplant recipients. Many of these patients receive high-dose corticosteroids, a major risk factor for *C. neoformans* infection. In 2020, it was estimated that in HIV-infected individuals, there were 152,000 cases of cryptococcal meningitis worldwide, with 112,000 associated deaths, accounting for 19% of AIDS-related mortality. *C. neoformans* is present in the soil, especially soil containing pigeon guano, and infects people when it is inhaled.

C. gattii was an obscure infectious agent that was viewed as a tropical or subtropical fungus until 1999, and it is now primarily reported in Australia, British Columbia, other regions in Canada, the Pacific Northwest of the United States, South America, and Africa. It is responsible for 11% to 33% of the global cryptococcal infections. Although *C. gattii* infection can occur in immunocompetent hosts, evidence suggests possible immune and genetic predisposing factors, so immune function should be evaluated in those with infection. *C. gattii* may present with large lesions that produce mass effects or that mimic the radiologic appearance of a neoplasm.

Arboreal species from oak to boabobs, and especially eucalypts, have been identified as an environmental reservoir. It is also found in soil and is acquired by inhalation.

Pathogenesis

Cryptococcus spp. have several virulence factors that enable it to evade host defenses:

- **Polysaccharide capsule.** A component of the capsule, glucuronoxylomannan, inhibits phagocytosis by alveolar macrophages, leukocyte migration, and recruitment of inflammatory cells. *Cryptococcus* spp. can block dendritic cell functions by reducing MHC class II-dependent antigen presentation and inhibiting the production of IL-12 and IL-23. *Cryptococcus* spp. can make large cells, called *Titan cells* (10 to 100 μm), and have a thick cell wall. *Cryptococcus* spp. also produce small (micro) cells of 2 to 4 μm that may be adapted for growth in macrophages.
- **Melanin production.** Laccase in the yeast catalyzes the formation of melanin which (1) has antioxidant properties, (2) decreases antibody-mediated phagocytosis, (3) counteracts the effects of antifungal agents, (4) binds iron, (5) and provides cell wall integrity.
- **Enzymes.** Phospholipases degrade cell wall components and may aid tissue invasion. Urease helps neutralize the reactive oxygen species and pH of the phagocytic cell.
- **Chitin** is essential for stabilizing the cell wall, and chitin mutants are less virulent pathogens.
- **CPL1** is a secreted cysteine-rich protein that drives macrophage activation.

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Cryptococcus spp. have yeast forms, but not pseudohyphal or hyphal forms, in human hosts. **The typically 5- to 10- μ m cryptococcal yeast form has a highly characteristic thick gelatinous capsule containing a polysaccharide that stains intense red with periodic acid-Schiff and mucicarmine in tissues (Fig. 8.44) and can be detected in blood or CSF with various immunoassays.** Although the lung is the primary site of infection, pulmonary involvement is usually mild and asymptomatic, even while the fungus is spreading to the CNS. *C. gattii* appears to be particularly likely to form a solitary pulmonary granuloma similar to the circumscribed (coin) lesions caused by *Histoplasma* spp.

The major lesions caused by *Cryptococcus* spp. are in the CNS, involving the meninges, cortical gray matter, and basal nuclei. The host response to cryptococci is extremely variable. In immunosuppressed people, organisms may evoke virtually no inflammatory reaction, so gelatinous masses of fungi grow in the meninges or expand the perivascular Virchow-Robin spaces within the gray matter, producing so-called soap-bubble lesions (see Fig. 8.44). In severely immunosuppressed persons, *C. neoformans* may disseminate widely to the skin, liver, spleen, adrenals, and bones. In immunocompetent people or in those with prolonged disease, the fungi induce a chronic granulomatous reaction composed of macrophages, lymphocytes, and foreign body-type giant cells. Suppuration may also occur, as well as granulomatous arteritis of the circle of Willis.

Pneumocystis Infections

P. jirovecii is a yeast-like fungus that primarily causes lung infections and is a significant opportunistic infection in immunocompromised patients. The organism can cause a rapidly progressive, bilateral pneumonia, and pneumocystis pneumonia was prominent in patients with AIDS in the 1980s. The incidence is significantly reduced due to effective antiretroviral therapy, but there were still >10,000 US hospitalizations due to *Pneumocystis* infections in 2017. The organism was originally classified as a protozoal parasite until 1988. Descriptions of developmental forms reflect this historical classification. Three forms of the organism include trophozoites of 1 to 4 μ m, sporocytes of 5 to 6 μ m, and cysts of 5 to 8 μ m. The cysts have a characteristic cup-shaped appearance, or they are oval with a central dot. Clusters of organisms in bronchoalveolar lavage fluid may be 200 μ m in diameter. The sporocyst

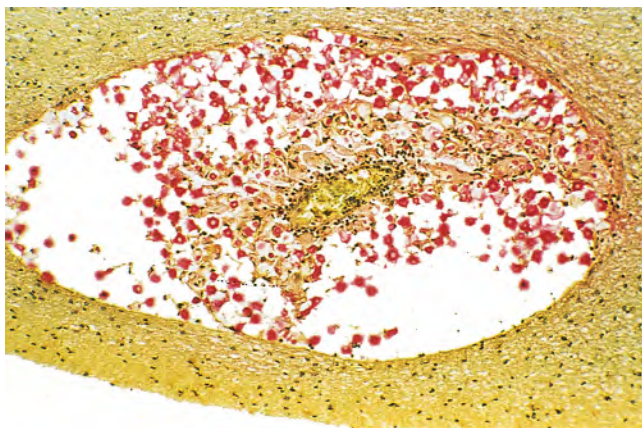


Fig. 8.44 Mucicarmine stain of cryptococci (staining red) in a Virchow-Robin perivascular space of the brain (soap-bubble lesion).

and ascus forms are visible with a cell wall stain such as methenamine silver, whereas the trophic forms are not (Fig. 8.45). Histopathologic findings include alveolar interstitial thickening and eosinophilic honey-combed exudate in the lumen of the lung. Fluorescein-conjugated antibody stains are commonly used to diagnose these infections. Beta-D-glucan is elevated with infection, although it is not specific for *Pneumocystis* spp., and sensitive and specific PCR tests are available for diagnosis.

No environmental source or external reservoir outside of humans has been identified for *P. jirovecii*, and the lack of a continuous in vitro culture method has hindered study of its pathogenesis. Transmission is believed to be via an airborne route, and there is evidence that healthy individuals might contribute to the reservoir. Most people are infected transiently early in life, with subsequent effective clearance. The average mortality rate of symptomatic infection is 10% to 14%, but it is higher in low-resource countries and urban communities. Extrapulmonary findings with *P. jirovecii* infection are not common, but can include involvement of lymph nodes, spleen, bone marrow, liver, and other sites. Infection with *P. jirovecii* induces both a humoral and cellular immune response.

Mold Infections

Thousands of types of molds are found indoors and outdoors. The most important hyaline (transparent) molds are discussed below.

Aspergillus

***Aspergillus* is a ubiquitous mold that causes allergic bronchopulmonary aspergillosis in otherwise healthy people and serious sinusitis, pneumonia, and invasive disease in immunocompromised individuals.** The major conditions that predispose to *Aspergillus* infection are neutropenia and use of corticosteroids. *Aspergillus fumigatus* is the most common pathogenic species of the fungus. *A. fumigatus* strains resistant to azole antifungal drugs are increasing in prevalence.

Pathogenesis

Aspergillus spp. are transmitted as airborne conidia, and the lung is the major portal of entry. The small size of *A. fumigatus* spores, approximately 2 to 3 μ m, enables them to reach alveoli. Conidia are coated in hydrophobic proteins that mask the microbial molecules from innate immune recognition. As conidia grow and form hyphae, these molecules are exposed. Alveolar macrophages recognize *Aspergillus* through TLR2 and the lectin dectin-1, which senses β -1,3-glucan in the fungal cell wall. Both receptors activate phagocytes to ingest and kill the conidia. In immunosuppressed states, conidia can germinate into hyphae, which then invade tissues. TLRs can recognize products of the fungal hyphae and trigger the release of pro-inflammatory mediators, including TNF, IL-1, and chemokines. Neutrophils produce reactive oxygen intermediates that kill hyphae. Invasive aspergillosis is highly associated with neutropenia and impaired neutrophil defenses.

***Aspergillus* produces several virulence factors, including adhesion molecules, antioxidants, enzymes, and toxins.** The antioxidant defenses include melanin pigment, mannitol, catalases, and superoxide dismutases. This fungus also produces phospholipases, proteases, and toxins, but their roles in pathogenicity are not clear. Aflatoxin is made by *Aspergillus* spp. that grow on the surface of some crops, including corn and peanuts, particularly in warm regions if the crops are not stored or inspected appropriately. Aflatoxin causes acute and chronic hepatotoxicity and is associated with increased risk of liver cancer. Sensitization to *Aspergillus* spores produces an allergic alveolitis (Chapter 15). Allergic bronchopulmonary aspergillosis, associated with hypersensitivity arising from superficial colonization of the bronchial mucosa, often occurs in asthmatic people.

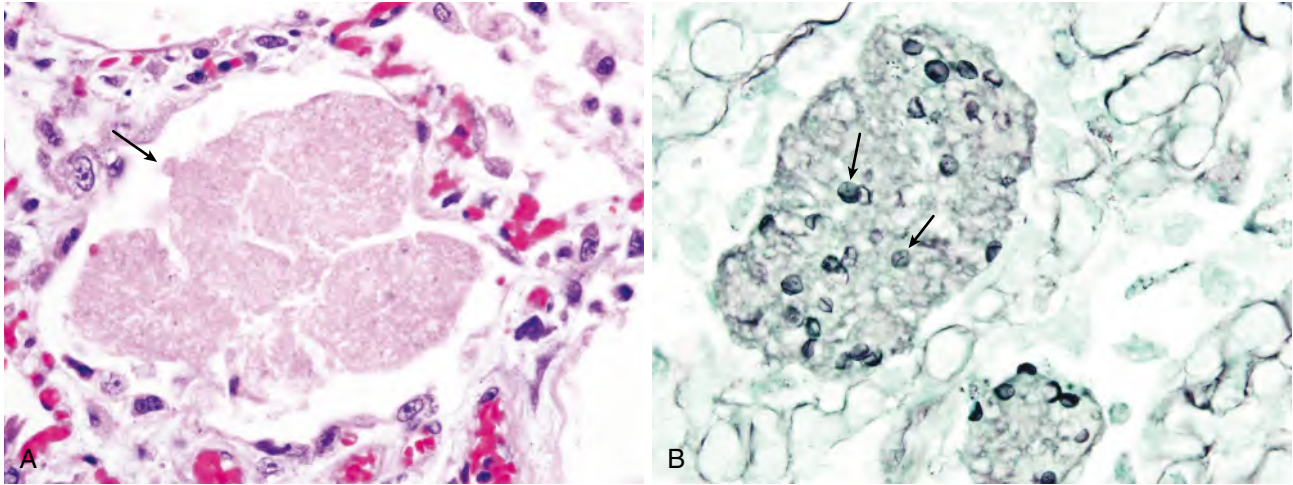


Fig. 8.45 Pneumonia caused by *Pneumocystis jirovecii*. (A) Alveoli of lung filled with foamy exudate (arrow) and interstitial inflammation. (B) Silver stain of alveoli, showing cysts of *Pneumocystis jirovecii* (crushed ping-pong ball appearance) stained black (arrows). (From Procop GW, Pritt BS. *Pathology of Infectious Diseases*. Elsevier; 2015; Courtesy Dr. Ann M. Nelson, Joint Pathology Center, Silver Spring, MD.)

MORPHOLOGY

Colonizing aspergillosis (aspergilloma) refers to growth of the fungus in the respiratory tract with minimal or no invasion of the tissues. Colonized lung cavities are usually the result of prior tuberculosis, bronchiectasis, old infarcts, or abscesses. Proliferating masses of hyphae within proteinaceous debris form brownish “fungus balls” within the cavities. The surrounding inflammatory reaction may be sparse, or there may be chronic inflammation and fibrosis. People with aspergillomas usually have recurrent hemoptysis.

Invasive aspergillosis is an opportunistic infection that is seen only in immunosuppressed hosts. *Aspergillus* has a proclivity for **vascular invasion**. The primary lesions are usually in the lung, but widespread hematogenous dissemination with involvement of the skin, brain, eyes, liver, and kidneys is common. The pulmonary lesions take the form of necrotizing pneumonia with sharply delineated, rounded, gray foci and hemorrhagic borders; they are often referred to as target lesions (Fig. 8.46A). *Aspergillus* forms **fruiting bodies (usually in lung cavities) and septate filaments, 5 to 10 μ m thick, branching at acute angles (40 degrees)** (see Fig. 8.46B). *Aspergillus*

hyphae without the distinct fruiting body cannot be distinguished from *Pseudallescheria boydii* and *Fusarium* spp. by morphology alone. Because of vascular invasion, areas of hemorrhage and infarction are usually superimposed on the necrotizing, inflammatory tissue reactions. Rhinocerebral *Aspergillus* infection in immunosuppressed individuals resembles that caused by Mucormycoses (e.g., *Mucor* spp., *Rhizopus* spp.).

Mucormycosis (Zygomycosis)

Mucormycotina are widely distributed in nature and cause no harm to immunocompetent individuals, but they infect immunosuppressed people, causing mucormycosis. Mucormycosis (formerly zygomycosis) is an opportunistic infection caused by environmental fungi, including *Mucor* spp., *Rhizopus* spp., *Rhizomucor* spp. and *Cunninghamella* spp. Major predisposing factors are neutropenia, corticosteroid use, diabetes mellitus, iron overload, and breakdown of the cutaneous barrier (e.g., because of burns, surgical wounds, or trauma).

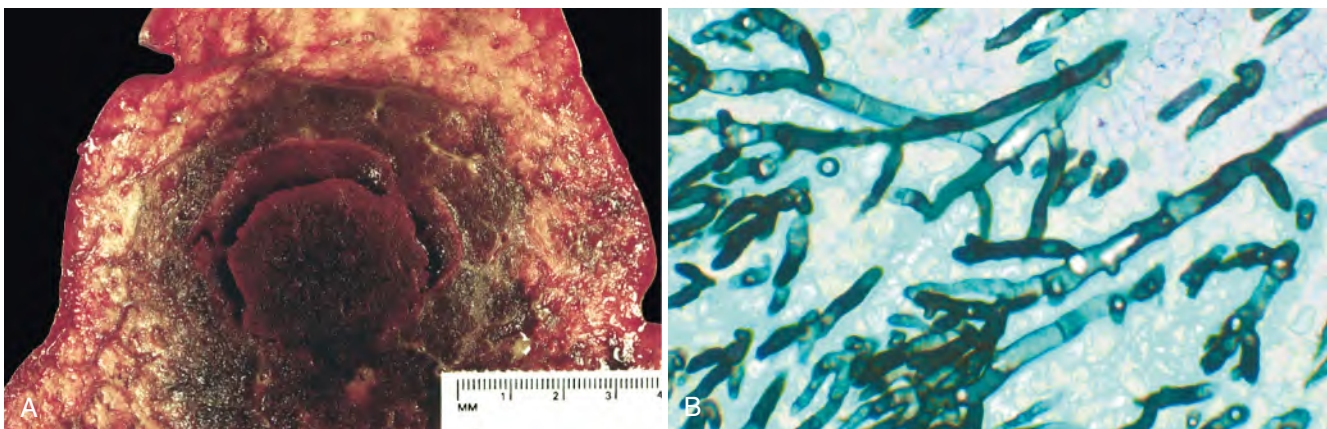


Fig. 8.46 *Aspergillus* infection. (A) Invasive aspergillosis of the lung in a hematopoietic stem cell transplant patient. (B) Gömöri methenamine silver stain shows septate hyphae with acute-angle branching, consistent with *Aspergillus*.

Pathogenesis

Mucormycotina are transmitted by airborne asexual spores. Inhalation of spores is the most common route of entry into the body, but percutaneous exposure of wounds or surgical sites or ingestion can also lead to infection. Macrophages provide the initial defense by phagocytosis and nonoxidative killing of germinating sporangiospores. *Mucormycotina* hyphal components are recognized by TLR2, which results in a pro-inflammatory cascade of cytokines including IL-6 and TNF. Neutrophils have a key role in killing hyphae after germination by directly damaging the walls of hyphae. If the macrophages or neutrophils are compromised in number or function, the probability of an established and then invasive infection is greatly increased. For unknown reasons, different *Mucor* spp. vary in resistance to phagocytosis of spores and in neutrophil damage to hyphae; thus, some infections can appear more aggressive than others despite a similar host milieu. The availability of free iron (a promoter of *Mucormycotina* growth) increases the probability of infection, as seen in people with diabetes (increased free iron due to ketoacidosis and/or glycosylation-induced poor iron affinity) and patients on chronic iron chelation treatment (where deferoxamine acts as a siderophore for the fungi).

MORPHOLOGY

Mucormycetes form nonseptate hyphae of variable width (6 to 50 μm) with frequent right-angle branching, distinct from *Aspergillus* hyphae, which are readily demonstrated by hematoxylin and eosin or special fungal stains (Fig. 8.47). The three primary sites of invasion are the nasal sinuses, lungs, and gastrointestinal tract, depending on whether the spores (which are widespread in dust and air) are inhaled or ingested. Most commonly in individuals with ketoacidosis due to diabetes, the fungus may spread from nasal sinuses to the orbit and brain, giving rise to **rhinocerebral mucormycosis**. The *Mucormycotina* cause local tissue necrosis, invade arterial walls, and penetrate the periorbital tissues and cranial vault. Meningoencephalitis follows, sometimes complicated by cerebral infarctions when fungi invade arteries and induce thrombosis.

Lung involvement with *Mucormycotina* may be secondary to rhinocerebral disease, or it may be primary in people with severe immunodeficiency. The lung lesions combine areas of hemorrhagic pneumonia with vascular thrombi and distal infarctions.

Dimorphic Fungal Infections

These fungi are endemic to specific geographic regions and they are dimorphic, meaning that they grow as molds with hyphae in the

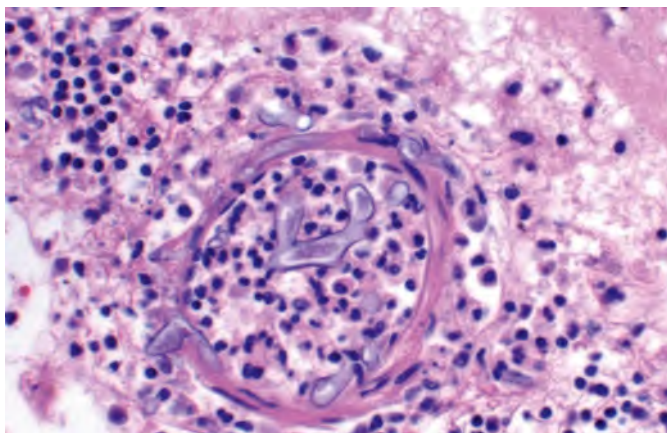


Fig. 8.47 Meningeal blood vessels with angioinvasive *Mucor* spp. Note the irregular width and near right-angle branching of the hyphae. Compare with *Aspergillus*, Fig. 8.46.

environment and as round yeast in the body. The medically important dimorphic fungi are discussed here.

Histoplasmosis

H. capsulatum infection is acquired by inhalation of dust particles from soil contaminated with bird or bat droppings containing small spores (microconidia), the infectious form of the fungus. It is endemic in the Ohio and Mississippi River Valleys and in the Caribbean. It is also found in Mexico, Central and South America, parts of eastern and southern Europe, Africa, eastern Asia, and Australia. Like *M. tuberculosis*, *H. capsulatum* is an intracellular pathogen that is found mainly in phagocytes. The clinical presentations and morphologic lesions of histoplasmosis resemble those of tuberculosis, including (1) a self-limited and often latent primary pulmonary involvement, which may result in coin lesions on chest radiography; (2) chronic, progressive, secondary lung disease, which is localized to the lung apices and causes cough, fever, and night sweats; (3) spread to extrapulmonary sites, including mediastinum, adrenal glands, liver, or meninges; and (4) widely disseminated disease in immunocompromised patients. Histoplasmosis can occur in immunocompetent individuals but is more severe in those with depressed cell-mediated immunity.

The pathogenesis of histoplasmosis is incompletely understood. The portal of entry is virtually always the lung. Macrophages ingest but cannot kill the organism without T-cell help, and this allows the organism to multiply within phagolysosomes and disseminate prior to the development of T-cell immunity, which takes 10 to 14 days. In individuals with adequate cell-mediated immunity, the infection is controlled by Th1 helper T cells that recognize fungal antigens and subsequently secrete IFN- γ , which activates macrophages and enables them to kill intracellular yeasts. In addition, *H. capsulatum* induces macrophages to secrete TNF, which recruits and stimulates other macrophages to kill the fungus.

The diagnosis of histoplasmosis may be established by serologic tests for antibodies or fungal antigens in serum or urine, culture, or identification of the fungus in tissue biopsies. Most cases resolve spontaneously. Progressive disease or disease in immunocompromised patients is treated with antifungal agents.

MORPHOLOGY

In the lungs of otherwise healthy adults, *Histoplasma* infections produce **granulomas**, which usually become necrotic and may coalesce to produce areas of consolidation. With spontaneous resolution or effective treatment, these lesions undergo fibrosis and concentric calcification (tree-bark appearance) (Fig. 8.48A). Histologic differentiation from tuberculosis, sarcoidosis, and coccidioidomycosis requires identification of the 3- to 5- μm thin-walled yeast forms, which may persist in tissues for years. In **fulminant disseminated histoplasmosis**, which occurs in immunosuppressed individuals, granulomas do not form; instead, there are focal accumulations of mononuclear phagocytes filled with fungal yeasts throughout the body (Fig. 8.48B). With disseminated disease, the yeast may be found in monocytes in a peripheral blood film.

Blastomycosis

Blastomyces dermatitidis is a soil-inhabiting dimorphic fungus. It causes disease in the central and southeastern United States; infection also occurs in Canada, Mexico, the Middle East, Africa, and India. There are three clinical forms: *pulmonary blastomycosis*, *disseminated blastomycosis*, and a rare *primary cutaneous form* that results from direct inoculation of organisms into the skin. The pneumonia most often resolves spontaneously, but it may persist or progress to a chronic lesion.

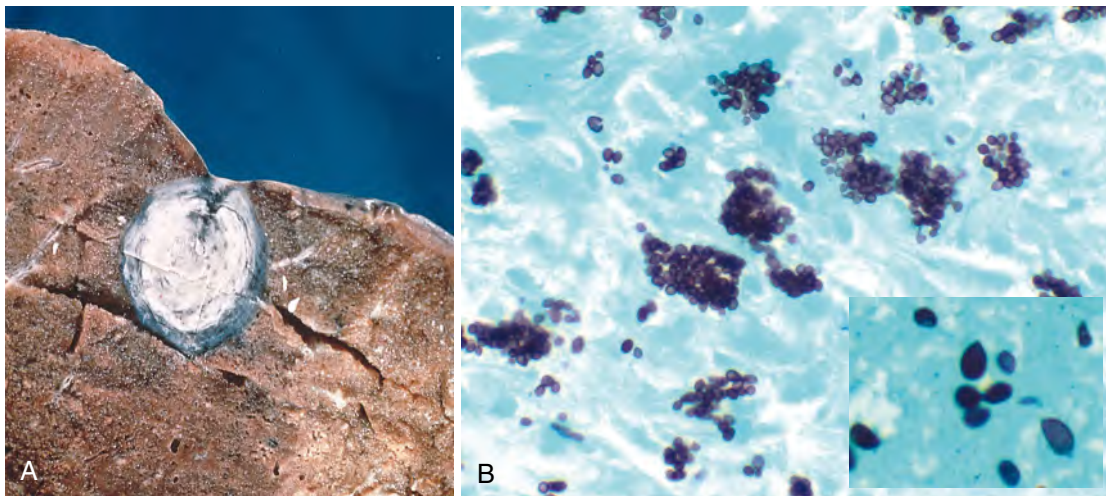


Fig. 8.48 Histoplasmosis. (A) Laminated *Histoplasma* granuloma of the lung. (B) *Histoplasma capsulatum* yeast forms fill phagocytes in the lung of a patient with disseminated histoplasmosis; inset shows high-power detail of pear-shaped, thin-based budding yeasts (silver stain).

MORPHOLOGY

In the normal host, the lung lesions of blastomycosis are suppurative granulomas. Macrophages have a limited ability to ingest and kill *B. dermatitidis*, and the persistence of the yeast cells leads to the recruitment of neutrophils. In tissue, *B. dermatitidis* is a round, 5- to 15- μm yeast cell that divides by broad-based budding. It has a thick, double-contoured cell wall and visible nuclei (Fig. 8.49). Involvement of the skin and larynx is associated with marked epithelial hyperplasia, which may be mistaken for squamous cell carcinoma.

Coccidioidomycosis

Many people who inhale the spores of *Coccidioides immitis* or *Coccidioides posadasii* become infected and develop a delayed-type hypersensitivity reaction to the fungus, but most remain asymptomatic. Indeed, more than 80% of people in endemic areas of the southwestern and western United States and in Mexico have a positive skin test reaction. One reason for the infectivity of *Coccidioides* spp. is that infective arthroconidia, when ingested by alveolar macrophages, block fusion of the phagosome and lysosome and so resist intracellular killing. Approximately 10% of infected people develop lung lesions, and less than

1% develop disseminated *Coccidioides* spp. infection, which frequently involves the skin and meninges. Risk factors for disseminated disease include initial infection during the second or third trimester of pregnancy, and suppressed T-lymphocyte immunity due to HIV infection, treatment for organ transplantation, or other immunosuppressive medications.

MORPHOLOGY

Coccidioides spp. are present as thick-walled, nonbudding spherules 20 to 60 μm in diameter, often filled with many small endospores. A pyogenic reaction is superimposed when the spherules rupture to release the endospores (Fig. 8.50). Rare progressive *Coccidioides* spp. disease involves the lungs, meninges, skin, bones, adrenals, lymph nodes, spleen, or liver. At all these sites, the inflammatory response may be purely granulomatous, pyogenic, or mixed.

PARASITIC INFECTIONS

Pathogenic parasites include protozoa (single-celled organisms) and metazoa (multicellular organisms).

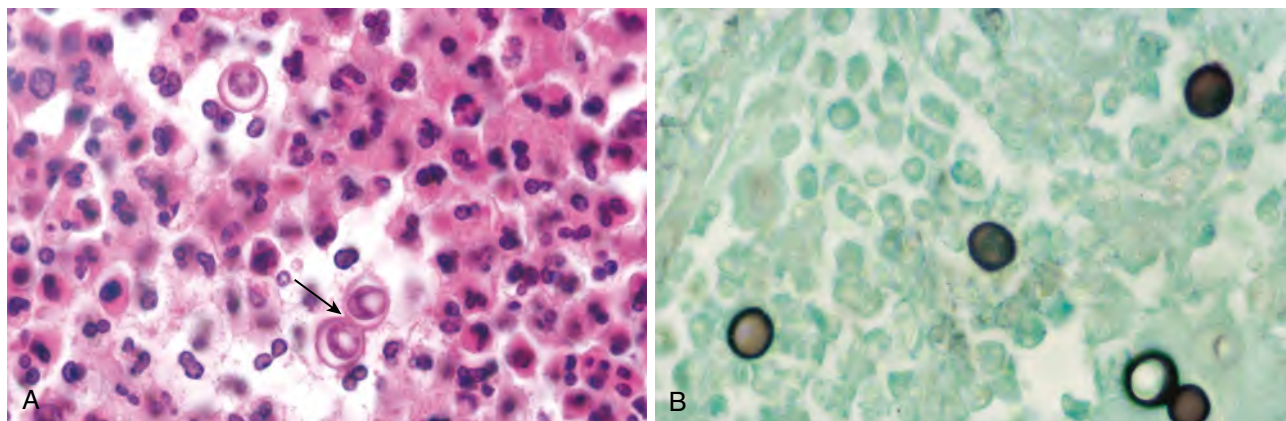


Fig. 8.49 Blastomycosis. (A) Rounded budding yeasts, larger than neutrophils, are present (arrow). Note the characteristic thick wall and nuclei (not seen in other fungi). (B) Silver stain: note the broad-based budding in the right lower corner.

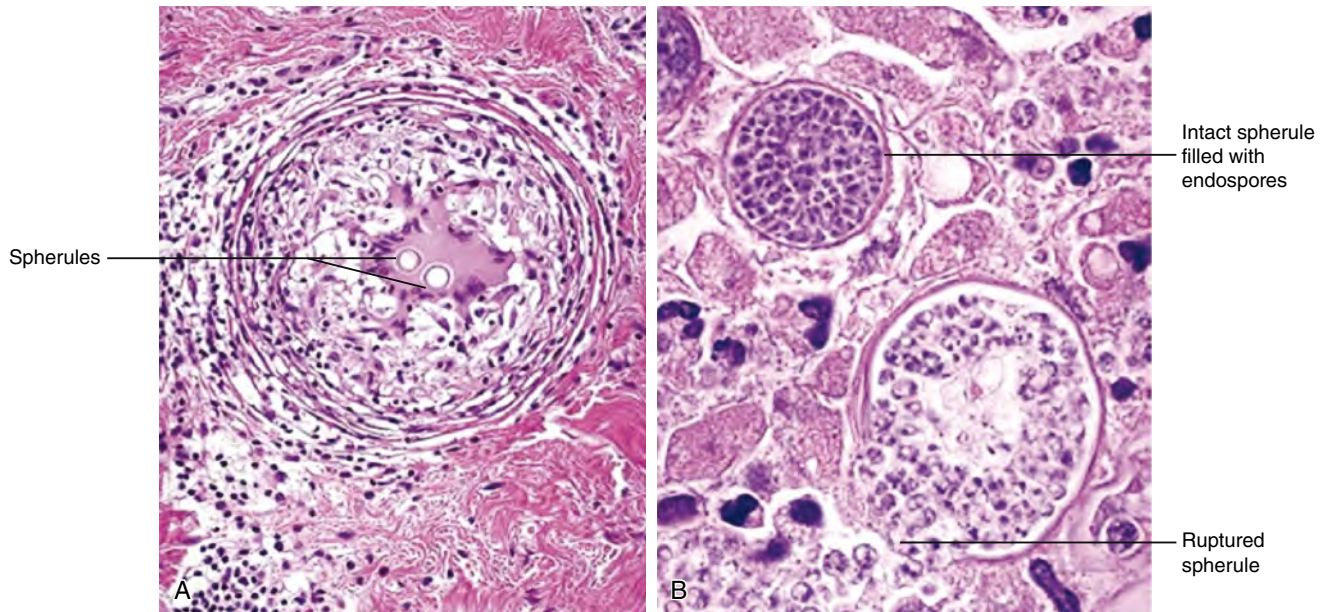


Fig. 8.50 Coccidioidomycosis. (A) Well-defined granuloma with a large giant cell containing two small spherules of *Coccidioides immitis* (arrows). (B) Higher magnification showing both intact and ruptured spherules with endospores. (From Klatt E. *Robbins and Cotran Atlas of Pathology*. 3rd ed. Elsevier; 2015.)

Protozoal Infections

Protozoa are unicellular eukaryotic organisms. The parasitic protozoa are transmitted by insects or by the fecal-oral route and, in humans, mainly reside in the blood or intestine (Table 8.7). Most protozoal infections are diagnosed by microscopic examination of blood smears or lesions, although molecular methods are now an important element of diagnostic testing.

Malaria

Malaria, caused by the intracellular parasite *Plasmodium*, causes significant global morbidity with an estimated 247 million cases in 2021 and 619,000 deaths. According to the WHO, 95% of cases and 96% of deaths occur in the African region, where malaria is a leading cause of death in children younger than 5 years. *Plasmodium falciparum* (the cause of severe cerebral malaria) and the four other malarial parasites that infect humans (*P. vivax*, *P. ovale*, *P. knowlesi*, and *P. malariae*) are all transmitted by female *Anopheles* mosquitoes, which are widely distributed throughout Africa, Asia, and Latin America. Transmission can also occur via blood transfusion or contaminated needles. Nearly all the approximately 2000 new cases of malaria each year in the United States occur in travelers or immigrants, with only a few infections acquired locally. Symptoms of infection include fever, chills, headache, muscle aches, fatigue, and possibly nausea, vomiting, and diarrhea. Severe malaria manifests as kidney failure, seizures, mental confusion, and coma. In regions with moderate to high *P. falciparum* transmission, the RTS,S/AS01 vaccine is recommended in children, as it reduces the incidence and severity of disease. The vaccine targets the circumsporozoite surface protein, expressed on the surface of *Plasmodium* before the infection of liver cells. Worldwide public health efforts to control malaria today face the challenges of insecticide-resistant mosquitoes and artemisinin-resistant *Plasmodium* spp.

Pathogenesis

The life cycles of the *Plasmodium* spp. are similar, although *P. falciparum* differs in ways that contribute to its greater virulence. *P. vivax*, *P. ovale*, *P. knowlesi*, and *P. malariae* cause low levels of parasitemia,

TABLE 8.7 Selected Human Protozoal Diseases

Location	Species	Disease
Luminal or epithelial	<i>Entamoeba histolytica</i>	Amoebic dysentery; liver abscess
	<i>Balantidium coli</i>	Colitis
	<i>Giardia duodenalis</i>	Diarrheal disease, malabsorption
	<i>Cystoisospora belli</i> <i>Cryptosporidium</i> spp.	Chronic enterocolitis or malabsorption or both
	<i>Trichomonas vaginalis</i>	Urethritis, vaginitis
Central nervous system	<i>Naegleria fowleri</i>	Meningoencephalitis
	<i>Acanthamoeba</i> spp.	Meningoencephalitis or ophthalmitis
Bloodstream	<i>Plasmodium</i> spp.	Malaria
	<i>Babesia</i> spp.	Babesiosis
	<i>Trypanosoma</i> spp.	African sleeping sickness
Intracellular	<i>Trypanosoma cruzi</i>	Chagas disease
	<i>Leishmania donovani</i>	Kala-azar
	<i>Leishmania</i> spp.	Cutaneous and mucocutaneous leishmaniasis
	<i>Toxoplasma gondii</i>	Toxoplasmosis

mild anemia, and, in very rare instances, splenic rupture and nephrotic syndrome. *P. falciparum* infection is associated with high levels of parasitemia that may lead to severe anemia, cerebral symptoms, renal failure, pulmonary edema, and death, depending on the susceptibility of the host.

The life cycle of *Plasmodium* spp. involves only humans and mosquitoes. The infectious stage of *Plasmodium*, the *sporozoite*, is found in

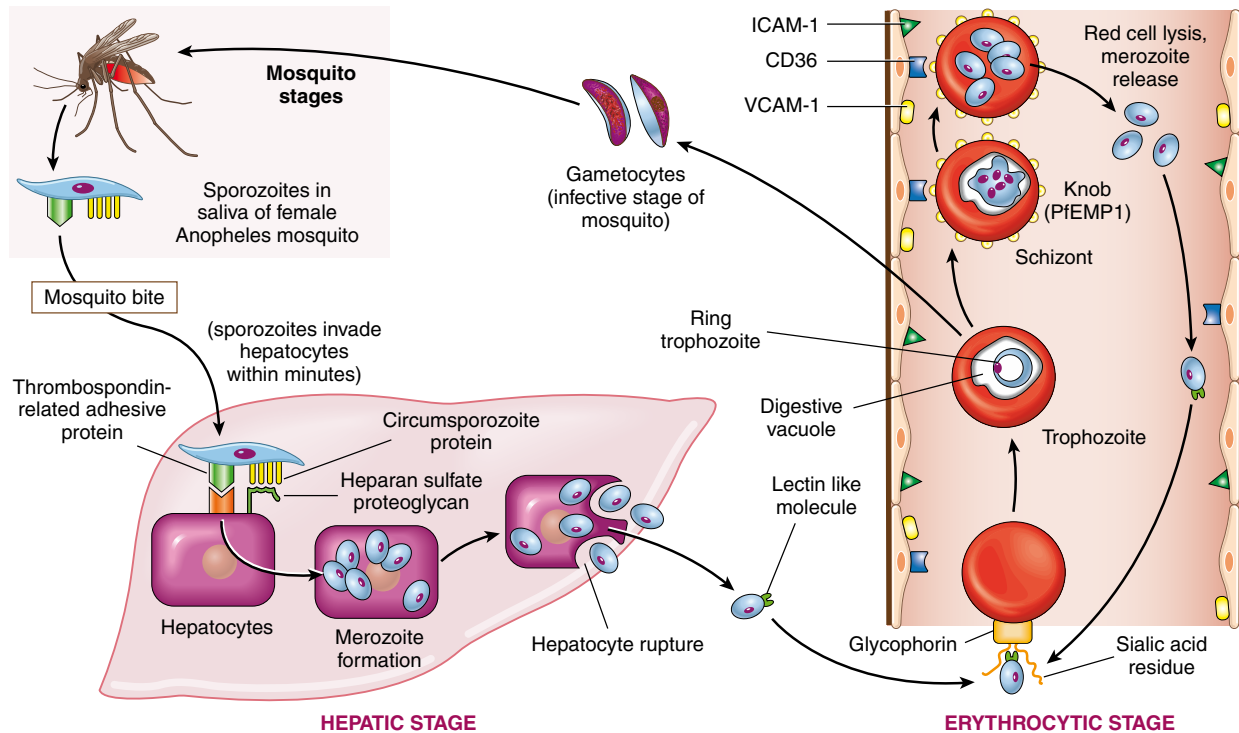


Fig. 8.51 Life cycle of *Plasmodium falciparum*. ICAM-1, Intercellular adhesion molecule-1; PfEMP1, *Plasmodium falciparum* erythrocyte membrane protein 1; VCAM-1, vascular cell adhesion molecule-1.

the salivary glands of female mosquitoes. When the mosquito takes a blood meal, sporozoites are released into the human's blood and, within minutes, attach to and invade liver cells by binding to the hepatocyte receptor for the serum proteins thrombospondin and properdin (Fig. 8.51). Within liver cells, malarial parasites multiply, releasing as many as 30,000 merozoites (asexual, haploid forms) when each infected hepatocyte ruptures. During *P. falciparum* infection, rupture usually occurs within 8 to 12 weeks. In contrast, *P. vivax* and *P. ovale* form latent hypnozoites in hepatocytes, which cause relapses of malaria weeks to months after initial infection. The infection of the liver and development of merozoites is called the *exoerythrocytic stage*. This stage is asymptomatic. Once released from the liver, *Plasmodium* merozoites use a lectin-like molecule to bind to sialic acid residues on glycophorin molecules on the surface of red cells and invade by active membrane penetration. Within the red cells (*erythrocytic stage*), the parasites grow in a membrane-bound digestive vacuole, hydrolyzing hemoglobin through secreted enzymes. The *trophozoite* is the first stage of the parasite in the red cell and is defined by the presence of a single chromatin mass. The next stage, the *schizont*, has multiple chromatin masses, each of which develops into a merozoite. Upon lysis of the red cell, the new merozoites infect additional red cells. Paroxysmal fever, chills, and rigors characteristic of malaria occur when the merozoites are released into the blood. As discussed later, release of merozoites induces the host cells to produce cytokines such as TNF that cause fever. The periodicity of such paroxysms (every 24 to 72 hours) varies with the species of the malaria parasite. Although most malaria parasites within the red cells develop into merozoites, some parasites develop, under specific conditions, into sexual forms called *gametocytes* that infect the mosquito when it takes its blood meal.

Several features of *P. falciparum* account for its greater pathogenicity:

- *P. falciparum* can infect red cells of any age, whereas other species infect only young or old red cells, which are a smaller fraction of the red cell pool.
- *P. falciparum* causes infected red cells to clump together (rosette) and to stick to endothelial cells lining small blood vessels (sequestration), which blocks blood flow. Adhesive polypeptides, including *P. falciparum* erythrocyte membrane protein 1 (PfEMP1), associate and form knobs on the surface of red cells (see Fig. 8.51). PfEMP1 binds to ligands on endothelial cells, including CD36, thrombospondin, VCAM-1, ICAM-1, and E-selectin. Red cell sequestration decreases tissue perfusion and leads to ischemia, which is responsible for the manifestations of cerebral malaria, the major cause of death in children with malaria.
- In *P. falciparum* infection, GPI-linked proteins, including merozoite surface antigens, are released from infected red cells and induce cytokine production by host cells. These cytokines increase fever, stimulate the production of reactive nitrogen species (leading to tissue damage), and induce expression of endothelial receptors for PfEMP1 (increasing sequestration).

Host resistance to *Plasmodium* can be intrinsic or acquired. Intrinsic resistance stems from inherited alterations that reduce the susceptibility of red cells to productive *Plasmodium* infections. Resistance may also be acquired following repeated or prolonged exposure to *Plasmodium* spp., which stimulates a partially protective immune response.

Several types of mutations affecting red cells in the heterozygous state may inhibit *Plasmodium* infection and are highly prevalent in parts of the world where malaria is endemic and are absent in other parts of the world. Most of these mutations are pathogenic in homozygous form, suggesting that they are maintained in populations due to a selective advantage for heterozygous carriers against malaria. The mutations fall into four broad classes.

- Point mutations in globin genes—sickle cell disease (HbS), HbC disease (hemoglobinopathies)
- Mutations leading to globin deficiencies— α - and β -thalassemia
- Mutations affecting red cell enzymes—glucose-6-phosphate dehydrogenase (G6PD) deficiency

- Mutations causing red cell membrane defects—absence of DARC (Duffy surface blood group), band 3, spectrin

P. vivax enters red cells by binding to the Duffy blood group antigen, and most of the population of West Africa is not susceptible to infection by *P. vivax* because they do not have the Duffy antigen. The mechanisms of the protective effects of the other three types of mutations are less well understood.

Individuals living where *Plasmodium* is endemic often gain partial immune-mediated resistance to malaria, evidenced by reduced illness despite infection. Antibodies and T lymphocytes specific for *Plasmodium* reduce disease manifestations, although the parasite has developed strategies to evade the host immune response. *P. falciparum* uses antigenic variation to escape from antibody responses to PfEMP1 and other surface proteins. Each haploid *P. falciparum* genome has multiple genes encoding variants of these parasite proteins. At least a percentage of the parasites switch genes each generation, producing antigenically new surface proteins. CTLs may also be important in resistance to *P. falciparum*.

MORPHOLOGY

The diagnostic test for malaria infection is examination of a Giemsa-stained peripheral blood smear, which permits the asexual stages of the parasite to be identified within infected red cells. The immunochromatographic detection of antigen is a useful rapid diagnostic test that must be confirmed by smear but allows for a quick response. PCR and other molecular assays are more sensitive than the smear, but the detection of low levels of parasitemia does not improve identification of those with significant symptoms. Molecular assays are useful for confirmation of species, which is important for appropriate therapy. Insertion of parasite proteins into the red cell membrane leads to recognition by macrophages, particularly in the spleen. *Plasmodium falciparum* infection leads to splenomegaly, due to both congestion and hyperplasia of the red pulp, and the spleen may eventually exceed 1000 g in weight. In chronic infections, the spleen becomes increasingly fibrotic and brittle, with a thick capsule and fibrous trabeculae. The parenchyma is gray or black because the phagocytes contain granular, brown-black, faintly birefringent hemozoin pigment. Macrophages with engulfed parasitized red cells are also numerous.

With progression of malaria, the liver becomes enlarged and pigmented. Kupffer cells are heavily laden with malarial pigment, parasites, and cellular debris, and some pigment is also present in the parenchymal cells. Pigmented phagocytic cells may be found dispersed throughout the bone marrow, lymph nodes, subcutaneous tissues, and lungs. The kidneys are often enlarged and congested with a dusting of pigment in the glomeruli and hemoglobin casts in the tubules.

In **cerebral malaria** caused by *P. falciparum*, brain vessels are plugged with parasitized red cells (Fig. 8.52). Around the vessels, there are ring hemorrhages that are probably related to local hypoxia incident to the vascular stasis and small focal inflammatory reactions (called *malarial* or *Dürk* *granuloma*). With more severe hypoxia, there is degeneration of neurons, focal ischemic softening, and occasionally scant inflammatory infiltrates in the meninges.

Nonspecific focal hypoxic lesions in the heart may be induced by the progressive anemia and circulatory stasis in chronically infected people. In some, the myocardium shows focal interstitial infiltrates. Finally, in the nonimmune patient, pulmonary edema or shock with DIC may cause death, sometimes in the absence of other characteristic lesions.

Babesiosis

Babesia microti and *Babesia divergens*, the primary causes of babesiosis, are malaria-like protozoans transmitted in a manner similar to Lyme disease and granulocytic ehrlichiosis, by ticks, including *Ixodes*

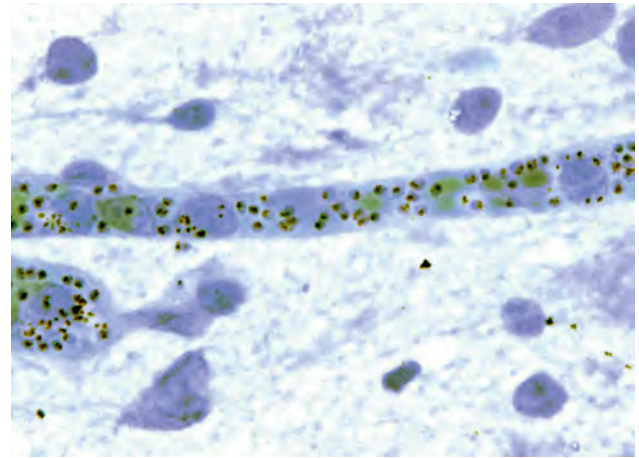


Fig. 8.52 Field stain of *Plasmodium falciparum*-infected red cells marginating within a capillary in cerebral malaria.

scapularis (deer tick), *I. ricinus* (sheep tick), *I. pacificus*, and *I. persulcatus*, with additional cases due to *B. duncani* and *B. venatorum*. Babesiosis is found in Europe, Northwestern China, Northeastern United States, and northern Midwestern United States. The incidence in the United States has significantly increased in the past 2 decades, along with expansion of the tick population and geographic range. The white-footed mouse is the reservoir for *B. microti*, and in some areas, nearly all mice have a persistent low-level parasitemia. *B. microti* survives well in refrigerated blood, and a number of transfusion-acquired babesiosis cases have been reported, leading to the addition of specific tests for screening of donated blood in endemic areas. *Babesia* spp. parasitize red cells and cause fever, jaundice, hemoglobinuria, and hemolytic anemia. Although most infections cause no symptoms or nonspecific flulike symptoms, infection in debilitated or splenectomized individuals can cause severe, potentially fatal parasitemias.

MORPHOLOGY

In blood smears, *Babesia* spp. superficially resemble *P. falciparum* ring stages, but lack hemozoin pigment, exhibit greater pleomorphism, and some species form characteristic tetrads of merozoites (Maltese cross), which are diagnostic, if found (Fig. 8.53). The level of *B. microti* parasitemia is a good indication of the severity of infection (about 1% in mild cases and up to 30% in splenectomized persons). In fatal cases, the anatomic findings are related to shock and hypoxia, and include jaundice, hepatic necrosis, acute renal tubular necrosis, adult respiratory distress syndrome, erythrophagocytosis, and visceral hemorrhage.

Leishmaniasis

Leishmaniasis is a chronic inflammatory disease of the skin, mucous membranes, or viscera caused by obligate intracellular, kinetoplast-containing (kinetoplastid) protozoan parasites transmitted through the bite of infected sandflies. Leishmaniasis is endemic throughout the Middle East, South Asia, Africa, and Latin America. Leishmania is associated with conditions of poverty and epidemics occur with forced migration of people with low immunity into areas of high transmission. There are an estimated 0.6 million to 1 million cases per year of cutaneous leishmaniasis, and 50,000 to 90,000 cases of visceral leishmaniasis with more than 20,000 deaths. Leishmanial

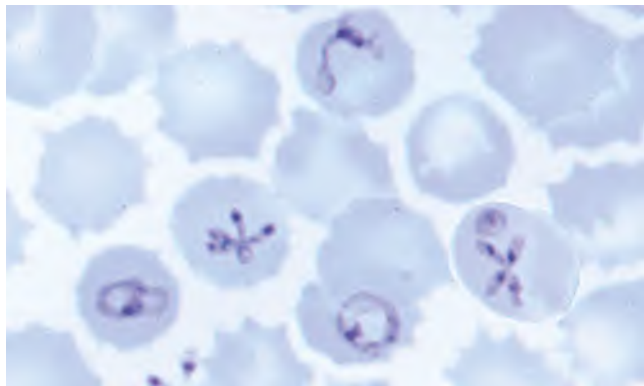


Fig. 8.53 Erythrocytes with *Babesia*, including the distinctive Maltese cross form. (Courtesy Lynne Garcia, LSG and Associates, Santa Monica, CA.)

infection, like infections by other intracellular organisms (mycobacteria, *Histoplasma* spp., *Toxoplasma* spp., and trypanosomes), is exacerbated by conditions that interfere with T-cell function. There are effective treatments for leishmaniasis for those with access to the drugs, but a competent immune system is essential to clear the infection, and many individuals are coinfecting with HIV. Culture, PCR, and histologic examination are used to diagnose the infection.

Pathogenesis

The life cycle of *Leishmania* spp. involves two forms: the promastigote, which develops and lives extracellularly in the sandfly vector, and the amastigote, which multiplies in host macrophages. There are about 21 species of *Leishmania* that infect humans and more than 90 species of sandflies that serve as vectors. Mammals, including rodents, dogs, and foxes, are reservoirs of *Leishmania* spp. When sandflies bite infected humans or animals, macrophages harboring amastigotes are ingested. The amastigotes differentiate into promastigotes, multiply within the digestive tract of the sandfly, and migrate to the salivary gland, where they are poised for transmission. When the infected fly bites a person, the slender, flagellated infectious promastigotes are released into the host dermis along with sandfly saliva, which potentiates parasite infectivity. The promastigotes are phagocytosed by macrophages, and the acidity within the phagolysosome induces them to transform into round amastigotes that lack flagella but contain a single mitochondrion with its DNA massed into a unique suborganelle, the kinetoplast. Amastigotes proliferate within macrophages, and dying macrophages release progeny amastigotes that can infect additional macrophages.

How far the amastigotes spread throughout the body depends on the specific *Leishmania* spp. and host and determines the extent of disease. There are several forms of disease that are caused by different *Leishmania* spp. Old World (*L. major*, *L. tropica*, *L. donovani*, and *L. infantum*) refers to the Eastern Hemisphere (parts of Asia, the Middle East, Africa [particularly the tropical region] and North Africa), and southern Europe. New World (*L. mexicana*, *L. braziliensis*, and *L. chagasi*) refers to the Western Hemisphere (parts of Mexico, Central America, and South America). A few cases have occurred in Texas and Oklahoma.

***Leishmania* spp. manipulate innate host defenses to facilitate their entry and survival in phagocytes.**

- Promastigotes produce two abundant surface glycoconjugates that contribute to their virulence: lipophosphoglycan interferes with the complement pathway and the zinc-dependent proteinase Gp63 cleaves critical host enzymes.

- To escape killing by neutrophils, *Leishmania* spp.: (1) interfere with the formation of phagolysosomes and fusion with granules, (2) localize to cellular compartments where they are insusceptible to lytic mechanisms, (3) resist toxicity by reactive oxygen species, and (4) resist neutrophil extracellular trap (NET) formation by producing endonucleases that digest the NETs and expressing protease-resistant molecules.
- *Leishmania* spp. amastigotes protect themselves from the hostile environment of the macrophage by maintaining the phagolysosome pH at 6.5, instead of the normal pH 4.5.
- *Leishmania* amastigotes use the iron transporter from the ZIP (ZRT, IRT-like protein) family of membrane proteins to stimulate iron entry into phagolysosomes. The host counters by using pro-inflammatory cytokines to repress iron absorption, by downregulation of the transferrin receptor, and by removal of iron from the phagosome.
- Another virulence mechanism of *Leishmania* parasites utilizes microRNAs that regulate gene expression posttranscriptionally. The miRNAs are expressed in the parasite and the host cells, and the parasites affect cellular signaling, leading to disruption of immunity.

The primary mechanisms of resistance and susceptibility to *Leishmania* spp. are determined by Th1 and Th2 responses. Parasite-specific CD4⁺ Th1 cells are needed to control *Leishmania* spp. in mice and humans. *Leishmania* spp. evade host immunity by impairing the development of the Th1 response. In animal models, mice that are resistant to *Leishmania* infection produce high levels of Th1-derived IFN- γ , which activates macrophages to kill the parasites. By contrast, mouse strains that are susceptible to leishmaniasis mount a dominant Th2 response. Th2 cytokines prevent effective killing of *Leishmania* spp. by inhibiting the microbicidal activity of macrophages.

MORPHOLOGY

Leishmania spp. produce four different types of lesions in humans: visceral, cutaneous, mucocutaneous, and diffuse cutaneous. **Visceral leishmaniasis** is the most serious and is almost always fatal without treatment. The parasites invade and activate macrophages throughout the mononuclear phagocyte system (Fig. 8.54) and cause a systemic inflammatory disease marked by hepatosplenomegaly, lymphadenopathy, pancytopenia, fever, and weight loss. The spleen may weigh as much as 3 kg. Phagocytic cells are enlarged and filled with *Leishmania* spp., many plasma cells are present, and the normal architecture of the spleen is obscured. In the late stages, the liver becomes increasingly fibrotic. Phagocytic cells crowd the bone marrow and may also be found in the lungs, gastrointestinal tract, kidneys, pancreas, and testes. Often there is hyperpigmentation of the skin in individuals of South Asian ancestry, which is why the disease is called *kala-azar* (black fever in Hindi). In the kidneys there may be an immune complex-mediated mesangioproliferative glomerulonephritis, and in advanced cases there may be amyloid deposition. People with advanced leishmaniasis can develop life-threatening secondary bacterial infections, such as pneumonia, sepsis, or tuberculosis. Hemorrhages related to thrombocytopenia may also be fatal.

Cutaneous leishmaniasis, the most common form, is a relatively mild, localized disease consisting of ulcers on exposed skin. The lesion begins as a papule surrounded by induration, changes into a shallow and slowly expanding ulcer, often with heaped-up borders, and usually heals by involution within 6 to 18 months without treatment. On microscopic examination, the lesion shows granulomatous inflammation, usually with many giant cells and few parasites.

Mucocutaneous leishmaniasis is most common in Brazil, Ethiopia, and Peru. Moist, ulcerating, or nonulcerating lesions develop in the nasopharyngeal

areas and, with progression, may be highly destructive and disfiguring. Microscopic examination reveals a mixed inflammatory infiltrate composed of parasite-containing macrophages with lymphocytes and plasma cells. Later, the tissue inflammatory response becomes granulomatous, and the number of parasites declines. Eventually, the lesions remit and scar, although reactivation may occur after long intervals by mechanisms that are not currently understood.

Diffuse cutaneous leishmaniasis is a rare form of dermal infection found in Ethiopia, adjacent areas of East Africa, and Central and South America. Diffuse cutaneous leishmaniasis begins as a single skin nodule, which continues spreading until the entire body is covered by nodular lesions. Microscopically, they contain aggregates of foamy macrophages stuffed with *Leishmania* organisms.

African Trypanosomiasis

African trypanosomes are kinetoplastid parasites (containing a large mass of DNA called a *kinetoplast*) that proliferate as extracellular forms in the blood and cause sustained or intermittent fevers, lymphadenopathy, splenomegaly, progressive brain dysfunction (sleeping sickness), cachexia, and death. *Trypanosoma brucei rhodesiense* infection, which occurs in eastern and southern Africa and is often acute and virulent, is a zoonotic infection that is best combated by reducing infected fly (vector) populations. Fewer than 50 cases were reported in 2022. *Trypanosoma brucei gambiense* infection, occurring in western and central Africa, spreads from human-to-human via fly bites and requires active case detection and treatment. There has been a dramatic decrease in incidence over the past 20 years with about 800 cases reported in 2022, many in the Democratic Republic of Congo. Tsetse flies (genus *Glossina*) transmit African *Trypanosoma* to humans either from wild and domestic animals (*T. brucei rhodesiense*) or from other humans (*T. brucei gambiense*). Within the fly, the parasites multiply in the stomach and then in the salivary glands before developing into nondividing trypomastigotes, which are transmitted to humans and animals with the next blood meal. The parasites live entirely extracellularly in the blood and interstitial spaces in tissues. Diagnosis is made by microscopic examination of blood smears, lymph node, or chancre. Cerebrospinal fluid must be examined to determine whether there is infection of the CNS, as this determines the course of treatment.

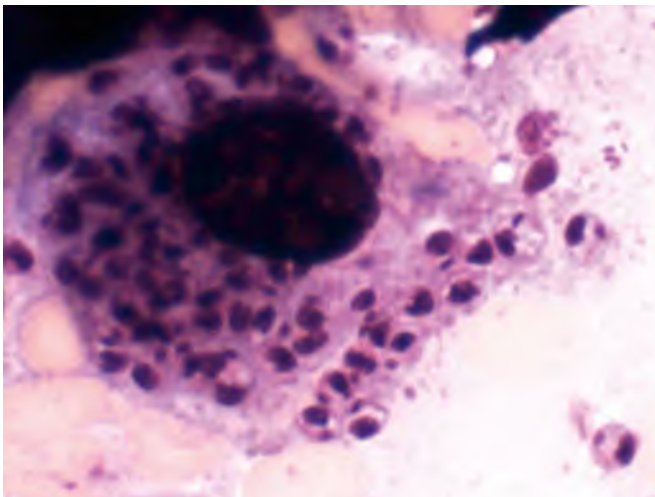


Fig. 8.54 Giemsa stain of a tissue macrophage with *Leishmania donovani* parasites.

Pathogenesis

African trypanosomes are covered by a single, abundant, glycolipid-anchored protein called the *variant surface glycoprotein* (VSG). As parasites proliferate in the bloodstream, the host produces antibodies to the VSG, which, in association with phagocytes, kill most of the organisms, causing a spike of fever. A small number of parasites, however, undergo a genetic rearrangement and produce a different VSG on their surface and so escape the host immune response. These successor trypanosomes multiply until the host mounts a new anti-VSG response and kills most of them, but then another clone with a distinct VSG takes over. In this way, African trypanosomes cause waves of fever before they finally invade the CNS. Trypanosomes have many VSG genes, only one of which is expressed at a time. The parasite uses an elegant mechanism to turn VSG genes on and off. Although VSG genes are scattered throughout the trypanosome genome, only VSG genes found near the ends of chromosomes are transcribed by parasites in the blood. New VSG genes are periodically moved into these sites, mainly by homologous recombination, generating a new VSG. A specialized RNA polymerase that transcribes VSG genes associates with only a single bloodstream expression site, limiting expression to one VSG at a time.

MORPHOLOGY

A large, red, rubbery chancre forms at the site of the insect bite and contains numerous parasites surrounded by a dense, predominantly mononuclear, inflammatory infiltrate. With chronicity, the lymph nodes and spleen enlarge due to infiltration by lymphocytes, plasma cells, and macrophages, which are filled with dead parasites. Trypanosomes (Fig. 8.55) concentrate in capillary loops, such as the choroid plexus and glomeruli. When parasites breach the blood-brain barrier and invade the CNS, leptomeningitis develops that extends into the perivascular Virchow-Robin spaces, and eventually a demyelinating panencephalitis occurs. Plasma cells containing cytoplasmic globules filled with immunoglobulins are frequent and are referred to as **Mott cells**. Chronic disease leads to progressive cachexia; patients, devoid of energy and normal mentation, literally waste away.

Chagas Disease

Trypanosoma cruzi is a kinetoplastid, intracellular protozoan parasite that causes American trypanosomiasis (Chagas disease). Chagas disease occurs in Mexico, and Central and South America,

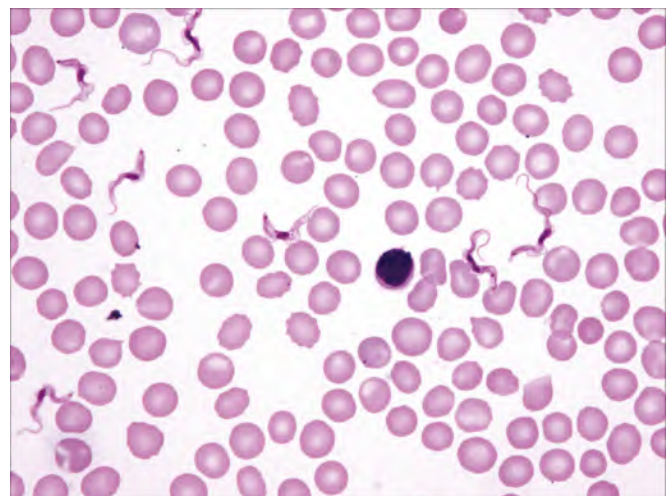


Fig. 8.55 Slender bloodstream parasites of African trypanosomiasis.

with an estimated 6 to 7 million people infected globally. Disease from contact with the insect vector is rare in the United States. There are an estimated 12,000 deaths per year and 28,000 new cases per year, a significant decrease over the past 30 years. *T. cruzi* parasites infect many animals, including cats, dogs, and rodents. The parasites are transmitted between animals and to humans by triatomine bugs (also known as kissing bugs or reduviids), which hide in the cracks of loosely constructed houses, feed on the sleeping inhabitants, and pass the parasites in feces; the infectious parasites enter the host through damaged skin or through mucous membranes. At the site of skin entry, there may be a transient, erythematous nodule. Congenital transmission is still significant with a small proportion of transmission due to consumption of contaminated foods or receipt of infected blood products or organ transplantation. Diagnosis can be made by a blood smear in the acute case but is made more commonly by serology. Antiparasitic drugs are effective if given soon after infection.

Pathogenesis

T. cruzi has a superfamily of glycosylphosphatidylinositol-anchored surface glycoproteins that interact with multiple ligands, such as laminin, fibronectin, collagen, cytokeratin, and other extracellular proteins, and are involved in cell adhesion and invasion. Although most intracellular pathogens avoid the toxic contents of lysosomes, after ingestion into macrophages, *T. cruzi* requires brief exposure to the acidic phagolysosome for development of amastigotes, the intracellular stage of the parasite. To ensure exposure to lysosomes, *T. cruzi* trypomastigotes elevate the concentration of cytoplasmic calcium in host cells, which promotes fusion of the phagosome and lysosome. In addition to enhancing amastigote development, the low pH of the lysosome activates pore-forming proteins that disrupt the lysosomal membrane, releasing the parasite into the cell cytoplasm. Parasites reproduce as rounded amastigotes in the cytoplasm of host cells and then develop flagella, lyse host cells, enter the bloodstream, and penetrate smooth, skeletal, and heart muscles. *T. cruzi* evades complement system activation by expressing complement regulatory proteins.

Chagas disease primarily affects the heart, and in endemic areas it is a major cause of sudden death due to cardiac arrhythmia. In acute Chagas disease, which is mild in most individuals, cardiac damage results from direct invasion of myocardial cells by the organisms and subsequent inflammation. Rarely, acute Chagas disease presents with high parasitemia, fever, or progressive cardiac dilation and failure, often with generalized lymphadenopathy or splenomegaly. In chronic Chagas disease, which occurs in 20% to 30% of people 5 to 15 years after initial infection, the mechanism of cardiac damage has two components.

- The presence of persistent *T. cruzi* parasites leads to a continued immune response with inflammatory infiltration of the myocardium, even though only scant organisms may be present.
- The parasite also may induce autoimmune responses, such that antibodies and T cells that recognize parasite proteins cross-react with host myocardial cells, nerve cells, and extracellular proteins such as laminin. For example, cross-reactive antibodies may induce electrophysiologic dysfunction of the heart.

Damage to myocardial cells and to conductance pathways causes dilated cardiomyopathy and cardiac arrhythmias. In addition, damage to the myenteric plexus causes dilation of the colon (megacolon) and esophagus. This is particularly common in endemic areas in Brazil, where as many as 50% of the patients with lethal carditis also have colonic and esophageal disease.

MORPHOLOGY

In lethal **acute myocarditis**, the changes are diffusely distributed throughout the heart. Clusters of amastigotes cause swelling of individual myocardial fibers and create intracellular pseudocysts. There is focal myocardial cell necrosis accompanied by extensive, dense, acute interstitial inflammatory infiltration throughout the myocardium, often associated with four-chamber cardiac dilation (Chapter 12).

In **chronic Chagas disease**, the heart is typically dilated and increased in size and weight. Often, there are mural thrombi that, in about one-half of autopsy cases, have given rise to pulmonary or systemic emboli or infarctions. On histologic examination, there are interstitial and perivascular inflammatory infiltrates composed of lymphocytes, plasma cells, and monocytes. There are scattered foci of myocardial cell necrosis and interstitial fibrosis, especially toward the apex of the left ventricle, which may undergo aneurysmal dilation and thinning. Even with dilation of the esophagus and colon, parasites cannot be found within ganglia of the myenteric plexus. Chronic Chagas cardiomyopathy often must be treated by cardiac transplantation.

Toxoplasmosis

Although millions of people carry the ubiquitous parasitic protozoa *T. gondii* without symptoms due to control by their immune systems, *T. gondii* causes significant disease in the immunocompromised and in pregnant females and their offspring. It is estimated that 11% of the US population 6 years of age and older has been infected and one-third of the global population is infected. In some regions of the world, 95% of the population has been infected. *T. gondii* infection occurs by eating undercooked meat (pork, lamb, venison) or shellfish contaminated with the tissue cysts, or by acquiring oocysts through drinking contaminated water, or by accidental ingestion following cleaning of a cat's litter box or handling contaminated soil. Another route of infection is vertical transmission due to active infection during pregnancy. Rarely, infection occurs via blood transfusion or organ transplantation. Some infected individuals (<10%) may experience ocular disease, or mononucleosis-like symptoms including lymphadenopathy, with a benign, self-limited course of weeks to months, followed by a latent infection. The organism can be reactivated following immunosuppression, such as after organ transplantation or in HIV infection. The symptoms of congenital toxoplasmosis may not become evident for months or many years after birth, and prompt treatment at birth may reduce the ultimate sequelae. Screening during gestation, followed by treatment of the acute maternal infection, can prevent vertical transmission and reduce the severity of disease in the offspring. Although congenital chorioretinitis is bilateral, ocular toxoplasmosis may be acquired and then is often unilateral. *Toxoplasma* encephalitis, due to reactivation, is a common opportunistic pathogen of the CNS in patients with AIDS. Myocarditis and pneumonitis are two other common manifestations in patients with AIDS. Investigators have hypothesized a possible connection between toxoplasmosis and schizophrenia and other neurologic conditions, but studies are inconclusive.

Pathogenesis

The cat is the only definitive host, and humans are a dead-end host. Unsporulated oocysts are shed in cat feces, which sporulate in the environment. Intermediate hosts (birds, rodents) become infected from ingestion of contaminated soil, water, or plants. Ingested oocysts release sporozoites that invade the human intestinal epithelium, disseminating throughout the body. The sporozoites transform into actively replicating tachyzoites that localize to tissues and reversibly differentiate into bradyzoites. Cats become infected after ingestion of

intermediate hosts. *T. gondii* is an intracellular pathogen that can invade any nucleated cell through a unique mechanism involving an actin-myosin moving junction and forms a parasitophorous vacuole (PV) protected from lysosomal fusion. To avoid clearance of the tachyzoites, *T. gondii* also inhibits autophagy of the parasite-containing vacuole. The parasite uses host factors to reprogram the host gene expression to counteract the immune response. The parasite enters immune-privileged sites, such as the brain, via infected immune cells.

MORPHOLOGY

In humans, *T. gondii* form cysts in skeletal muscle, myocardium, visceral organs, brain, and eyes. Although diagnosis is usually through serologic testing, cysts may be observed in biopsies (Fig. 8.56), or **tachyzoites** may be observed in stained fluid specimens, such as a bronchoalveolar lavage fluid from an immunocompromised host. Tachyzoites may be observed with periodic acid-Schiff, Giemsa, or hematoxylin and eosin stains. Tissue cysts can vary in size from 5 to 100 μm . Intact cysts can remain for the life of the host without causing inflammation. The cyst wall is thin and contains crescent-shaped bradyzoites of approximately 1.5 by 7 μm . PCR of blood or cerebrospinal fluid is a sensitive assay and permits less invasive testing in many cases. In those with lymphadenopathy, there is a hyperplasia of B cell follicles associated with clusters of epithelioid macrophages that impinge on the reactive follicles, an appearance that is so characteristic as to strongly suggest the diagnosis. **Congenital toxoplasmosis** can result in fetal death and abortion, or hydrocephalus, microcephaly, cerebral calcifications, neurocognitive deficits, and chorioretinitis.

Metazoal Infections

Metazoa are multicellular, eukaryotic organisms with organ systems. Parasitic metazoal infections are contracted by consuming the parasite, often in undercooked meat, or by direct invasion of the host through the skin or via insect bites. Metazoa dwell in many sites of the body, including the intestine, skin, lung, liver, muscle, blood vessels, and lymphatics. The infections are diagnosed by microscopic identification of larvae or ova in excretions or tissues, and by serology.

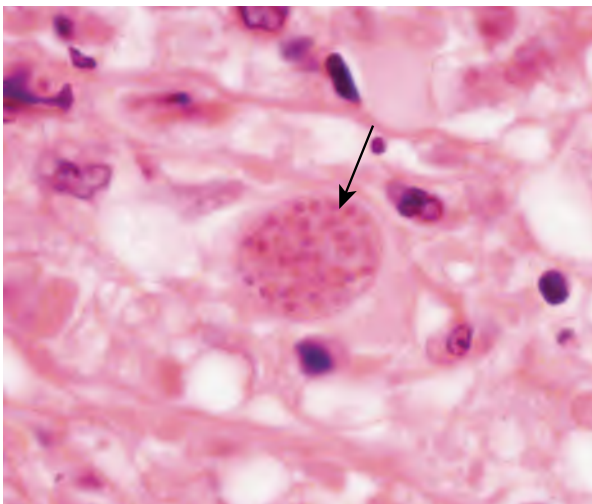


Fig. 8.56 Brain biopsy showing a cyst of *Toxoplasma gondii* (arrow). (From Procop GW, Pritt BS. *Pathology of Infectious Diseases*. Elsevier; 2015; Courtesy Dr. Bobbi S. Pritt, Department of Laboratory Medicine and Pathology, Mayo Clinic, Rochester, MN.)

Strongyloidiasis

Strongyloides stercoralis infects 30 to 100 million people worldwide and is endemic in tropical and subtropical regions of South America, sub-Saharan Africa, and Southeast Asia. It occurs sporadically in the United States, including in Appalachia. **The worms live in the soil and infect humans when larvae penetrate the skin, travel in the circulation to the lungs, and then travel up the trachea to be swallowed.** Female worms reside in the mucosa of the small intestine, where they produce eggs by asexual reproduction (parthenogenesis). Most of the larvae are passed in the stool and then may contaminate soil to continue the cycle of infection; however, some develop and become infectious within the host, leading to autoinfection.

In immunocompetent hosts, *S. stercoralis* is often asymptomatic, but may cause diarrhea, bloating, and occasionally malabsorption. Immunocompromised hosts, particularly people receiving prolonged corticosteroid therapy or who are infected with human T-lymphotropic virus type I, can have very high worm burdens (hyperinfection) throughout the body due to uncontrolled autoinfection, leading to fatal disease. Corticosteroids inhibit the functions of eosinophils and other host immune cells that accumulate in tissues in response to infection and stimulate female *Strongyloides* spp. to increase infective larvae production. In addition, other disease states that perturb immune control mechanisms (e.g., organ transplantation, lymphoma, HTLV-1 infection) are associated with increased risk. Hyperinfection can be complicated by sepsis caused by intestinal bacteria, which enter the blood following damage to the intestinal wall by the invading larvae.

MORPHOLOGY

In mild strongyloidiasis, worms, mainly larvae, are present in the duodenal crypts but are not seen in the underlying tissue. There is an eosinophil-rich infiltrate in the lamina propria with mucosal edema. Hyperinfection with *S. stercoralis* results in invasion of larvae into the colonic submucosa, lymphatics, and blood vessels, with an associated mononuclear infiltrate. There are many adult worms, larvae, and eggs in the crypts of the duodenum and ileum (Fig. 8.57). Worms of all stages may be found in other organs, including skin and lungs, and may even be found in large numbers in sputum.

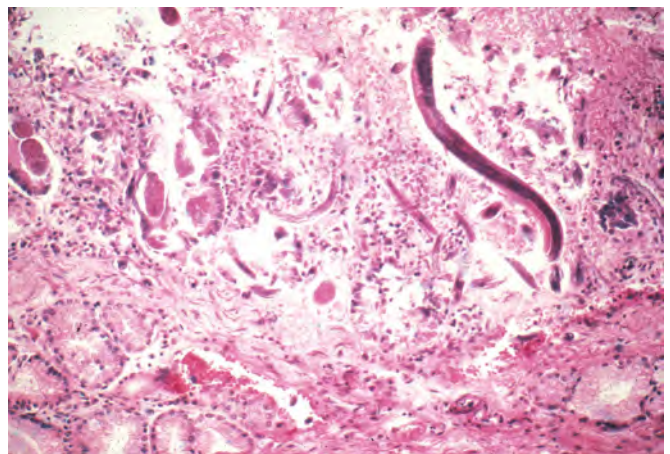


Fig. 8.57 *Strongyloides* hyperinfection in a patient treated with high-dose cortisone. A female, her eggs, and rhabditoid larvae are in the duodenal crypts; filariform larvae are entering the blood vessels and muscularis mucosa. (Courtesy Dr. Franz C. Von Lichtenberg, Brigham and Women's Hospital, Boston.)

Ascariasis

Most cases of ascariasis are caused by *Ascaris lumbricoides* and a small number of cases are caused by *Ascaris suum*. A billion people are estimated to be infected, with most cases occurring in Asia, Africa, or South America. Both species of *Ascaris* are acquired by ingestion of eggs in soil that were passed in stool, either from humans (*A. lumbricoides*) or from pigs (*A. suum*). Once ingested, the eggs release larvae in the small intestine and the larvae then penetrate the mucosa of the proximal colon and cecum and travel to the lungs in the blood or lymph. In the lungs, the larvae mature for 10 to 14 days, after which they are coughed up, swallowed, and return to the intestine where they mature into adult worms 15 to 35 cm in length. The adults live throughout the intestine, mainly in the jejunum, where each female worm releases approximately 200,000 eggs per day into the stool.

Symptoms of ascariasis are variable. Most patients are asymptomatic. The initial pulmonary infection can cause *Loeffler syndrome*, a self-limited eosinophilic pneumonitis, with dry cough, dyspnea, and fever. Gastrointestinal manifestations vary with the burden of worms. Light infections are often asymptomatic, but heavy worm burdens are associated with nonspecific abdominal symptoms including pain, loss of appetite, nausea, vomiting, and diarrhea. Complications include malnutrition and abdominal obstruction, which may require surgery. The worms can also cause obstructions of the ileocecal valve and the appendix. If worms migrate into the biliary tree, they can cause a variety of complications, including acute pancreatitis and cholangitis.

Pulmonary infections may be diagnosed clinically, but a stool examination may not reveal the parasites if they have not yet matured in the intestine. Intestinal infections can be diagnosed by the detection of eggs in the stool or by identification of the adult worms, which may be passed from the rectum or, less often, coughed up.

Enterobiasis (Pinworm)

Enterobiasis or pinworm is caused by the worm *Enterobius vermicularis*, which is thin and only 6 to 13 mm long. It occurs worldwide and is estimated to infect 40 million people in the United States alone, where it is the most common helminthic infection. The infection is acquired by ingestion of eggs, which are found on the perianal skin or passed in stool, and then spread from the hands or environment (e.g., bed linens). The eggs hatch in the small intestine and the adults then mature and live in the cecum and appendix, from which the females migrate out of the anus to deposit eggs on the skin, usually at night. The eggs mature rapidly and are infectious in approximately 6 hours.

Pinworm infections are most common in children between 5 and 10 years old. Most cases are asymptomatic. The most common symptom is perianal itching (**pruritis ani**), due to local inflammation caused by the worms and eggs. Very high worm burdens can cause abdominal pain, nausea, and vomiting. Migration of the worms into the vagina can cause vulvovaginitis. Rarely, high burdens of *E. vermicularis* are associated with rectal prolapse.

The diagnosis can be made readily by microscopic examination of clear tape or sticky plastic pinworm paddles used to collect the worms and eggs from the perianal skin. The eggs have a characteristic elongated oval shape with one of the long sides of the oval flattened.

Cysticercosis and Hydatid Disease (Tapeworm Infections)

Taenia solium and *Echinococcus granulosus* are cestode parasites (tapeworms) that cause cysticercosis and hydatid infections, respectively. Both diseases are caused by larvae that develop after ingestion of tapeworm eggs. These tapeworms have a complex life cycle requiring two mammalian hosts: a definitive host, in which the worm reaches sexual maturity, and an intermediate host, in which the worm does not reach sexual maturity.

***T. solium* causes disease because of deposition of cysts (cysticerci) in organs and the resulting inflammatory response.** The parasite is endemic in South America, much of Africa, and Asia. These tapeworms consist of a head (scolex) that has suckers and hooklets that attach to the intestinal wall, a neck, and many flat segments called proglottids that contain both male and female reproductive organs. New proglottids develop behind the scolex. The most distal proglottids are mature and contain many eggs, and they can detach and be shed in feces. *T. solium* can be transmitted to humans in two ways, ingestion of larval cysts or eggs, with distinct outcomes.

- Larval cysts, called *cysticerci*, are ingested in undercooked pork and attach to the intestinal wall, where they develop into mature adult tapeworms. These can grow to several meters in length and produce mild abdominal symptoms. The parasite life cycle is completed with this mode of infection, and disseminated cysticercosis does not develop.
- When intermediate hosts (pigs or humans) ingest eggs in food or water contaminated with human feces, the egg becomes an oncosphere that hatches and penetrates the gut wall, disseminating hematogenously, followed by transition to a cysticercus that can encyst in many organs, giving rise to clinical symptoms of cysticercosis. The most serious manifestations result from cysts in the brain (*neurocysticercosis*). Convulsions, increased intracranial pressure, and other neurologic disturbances may occur. Adult tapeworms are not produced by this mode of infection because larval cysts lodged in various tissues other than the intestine cannot develop into mature worms. Viable *T. solium* cysts often do not produce symptoms and can evade host immune defenses by producing factors called taeniaestatin and paramyosin, which seem to inhibit complement activation. When the cysticerci die and degenerate, an inflammatory response develops.

Taenia saginata, the beef tapeworm, and *Diphyllobothrium latum*, the fish tapeworm, are acquired by eating undercooked meat or fish. In humans, these parasites live only in the gut and do not form cysticerci.

Hydatid disease is caused by ingestion of eggs of *Echinococcus* spp and formation of cysts in organs where the parasite larvae are deposited. For *E. granulosus* the definitive host is the dog, and the usual intermediate hosts are sheep. For *E. multilocularis* the fox is the most important definitive host, and rodents are intermediate hosts. Humans are accidental intermediate hosts, infected by ingestion of food contaminated with eggs shed by dogs or foxes. Eggs hatch in the duodenum and larvae invade the liver, lungs, or bones. Infection in humans is usually asymptomatic, but large cysts in the liver can cause abdominal pain or obstruction, and pulmonary cysts can cause pain, cough, and hemoptysis. Caution is warranted if surgical removal of the cyst is performed, as anaphylaxis and/or dissemination of organisms can result from spillage of the cyst contents.

MORPHOLOGY

Cysticerci may be found in any organ, but the more common locations include the brain, muscles, skin, and heart. Cerebral symptoms depend on the precise location of the cysts, which may be intraparenchymal, attached to the arachnoid, or freely floating in the ventricular system. The cysts are ovoid and white to opalescent, often grape-sized, and contain an invaginated scolex with hooklets that are bathed in clear cyst fluid (Fig. 8.58). The cyst wall is more than 100 µm thick, rich in glycoproteins, and evokes little host inflammatory response when it is intact. When cysts degenerate, however, there is inflammation, followed by focal scarring, and calcifications, which may be visible by radiography.

***E. granulosus* cysts** are found in the liver in about two-thirds of cases, with 5% to 15% in the lung, and the rest in bones and brain, or other organs.

In the various organs, the larvae lodge within the capillaries and first incite an inflammatory reaction composed principally of mononuclear leukocytes and eosinophils. Many such larvae are destroyed, but others encyst. The cysts begin at microscopic levels and progressively increase in size, so that in 5 years or more they may have achieved dimensions of more than 10 cm in diameter. Enclosing an opalescent fluid is an inner, nucleated, germinative layer and an outer, opaque, nonnucleated layer. The outer nonnucleated layer is distinctive and has innumerable delicate laminations. Outside this opaque layer, there is a host inflammatory reaction that produces a zone of fibroblasts, giant cells, and mononuclear and eosinophilic cells. In time, a dense fibrous capsule forms. Daughter cysts often develop within the large mother cyst. These appear first as minute projections of the germinative layer that develop central vesicles and thus form tiny brood capsules. Degenerating scolices of the worm produce a fine, sand-like sediment within the hydatid fluid (**hydatid sand**).

Trichinellosis

Trichinella spp. are nematode parasites that are acquired by ingestion of larvae in undercooked meat from infected animals (usually pigs, boars, or horses) that have themselves been infected by eating rats or meat products containing *T. spiralis*, *T. nativa*, or *T. britovi*. There are an estimated 10,000 cases in the world each year. In the United States the number of *T. spiralis*-infected pigs has been greatly reduced by laws requiring proper cooking of food fed to hogs; the number of reported human infections in the United States is usually about 20 cases each year. Still, trichinosis remains widespread in other parts of the world, where undercooked meat, including noncommercial livestock and game (e.g., bear), is commonly eaten.

Pathogenesis

The life cycle of *T. spiralis* begins in the human intestine but ends within muscle as humans are dead-end hosts. In the human gut, *T. spiralis* larvae develop into adults that mate and release new larvae, which penetrate the tissues. Larvae disseminate hematogenously and penetrate muscle cells, causing fever, myalgias, marked eosinophilia, and periorbital edema. Less commonly, the larvae lodge in the heart, lungs, and brain, potentially leading to cardiac failure, dyspnea, and encephalitis. In striated skeletal muscle, *T. spiralis* larvae become intracellular parasites, increase dramatically in size, and modify the host muscle cell so that it loses its striations, gains a collagenous capsule, and develops a plexus of new blood vessels around itself.

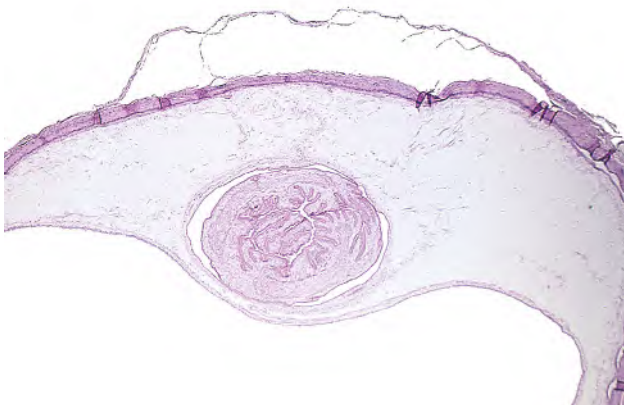


Fig. 8.58 Portion of a cysticercus cyst in the skin.

The cell-parasite complex is largely asymptomatic, and the worm may persist for years before it dies and calcifies. Antibodies to larval antigens, which include an immunodominant carbohydrate epitope called **tyvelose**, may reduce reinfection and are useful for serodiagnosis of the disease.

T. spiralis and other invasive nematodes stimulate a Th2 response, with production of IL-4, IL-5, IL-10, and IL-13. The cytokines produced by Th2 cells activate eosinophils and mast cells, both of which are associated with the inflammatory response to these parasites. In animal models of *T. spiralis* infection, the Th2 response is associated with increased contractility of the intestine, which expels adult worms from the gut and subsequently reduces the number of larvae in the muscles. Although the Th2 response indirectly reduces the number of larvae in muscle by eliminating adults from the intestine, it is not clear whether the intramuscular inflammatory response, which is composed of mononuclear cells and eosinophils, is effective against the larvae.

MORPHOLOGY

During the invasive phase of trichinosis, cell destruction can be widespread with heavy infections and may be lethal. In the heart, there is a patchy interstitial myocarditis characterized by many eosinophils and scattered giant cells. Myocarditis can lead to scarring. Larvae in the heart do not encyst and are difficult to identify, because they die and disappear. In the lungs, trapped larvae cause focal edema and hemorrhages, sometimes with an allergic eosinophilic infiltrate. In the CNS, larvae cause a diffuse lymphocytic and eosinophilic infiltrate, with focal gliosis in and about small capillaries of the brain.

T. spiralis preferentially encysts in striated skeletal muscles with the richest blood supply, including the diaphragm and the extraocular, laryngeal, deltoid, gastrocnemius, and intercostal muscles (Fig. 8.59). Coiled larvae are approximately 1 mm long and are surrounded by intracellular membrane-bound vacuoles, which in turn are surrounded by new blood vessels and an eosinophil-rich mononuclear cell infiltrate. This infiltrate is greatest around dying parasites, which eventually calcify and leave behind scars that are sufficiently characteristic to suggest the diagnosis.

Schistosomiasis

Schistosomiasis is estimated to infect more than 200 million persons and causes significant morbidity and mortality. The affected organs

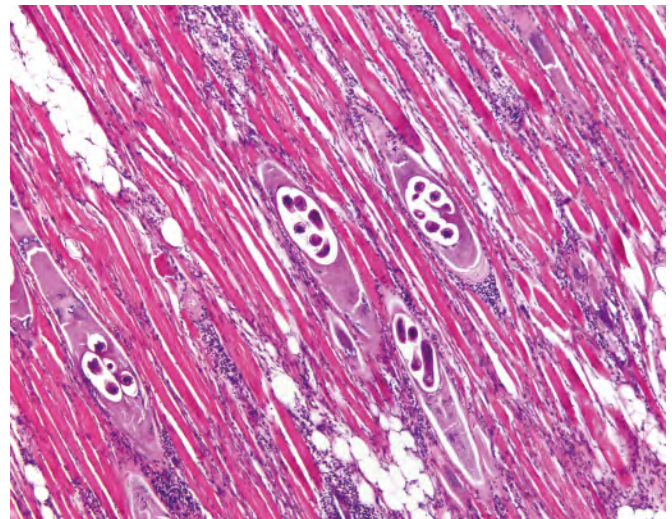


Fig. 8.59 Multiple coiled *Trichinella spiralis* larvae within skeletal muscle cells.

and hence the site of major disease vary with the species. *Schistosoma mansoni* and *S. japonicum* affect the liver and the gut predominantly. Most deaths are due to hepatic cirrhosis, which is caused by *S. mansoni* in Latin America, Africa, and the Middle East and by *S. japonicum* and *S. mekongi* in East Asia. By contrast, *S. haematobium*, found in Africa, causes chronic granulomatous bladder inflammation that may lead to hematuria, obstructive uropathy, and carcinoma. *Schistosoma* flukes require passage through freshwater snails that live in the slow-moving water of tropical rivers, lakes, and irrigation ditches, ironically linking agricultural development with spread of the disease. Acute schistosomiasis in humans can be a severe febrile illness that peaks about 2 months after infection. Severe hepatic fibrosis is a serious manifestation of chronic schistosomiasis.

Pathogenesis

Much of the pathology of schistosomiasis is caused by host inflammatory reactions to different stages of the parasite. The life cycle of *Schistosoma* spp. involves stepwise infection of several human tissues, each associated with host inflammatory responses. After release from snails, ciliated miracidium larvae mature into infectious larvae (cercariae) that swim through fresh water and penetrate human skin with the aid of powerful proteolytic enzymes that degrade the keratinized layer. There is minimal skin reaction. Schistosomes migrate through the skin into the peripheral vasculature and lymphatics, travel to the lungs and heart, from where they are disseminated widely, including the mesenteric, splanchnic, and portal circulation, ultimately reaching the hepatic vessels, where they mature (*S. mansoni* and *S. japonicum*). Mature male-female worm pairs then migrate once again and settle in the venous system (commonly the portal or pelvic veins). Females produce hundreds to thousands of eggs per day that secrete proteases and elicit localized inflammatory reactions. This inflammatory response to egg migration is necessary for passive transfer across the intestine and, in the case of *S. haematobium*, bladder walls, allowing the eggs to be shed in stool or urine, respectively. Infection of freshwater snails completes the life cycle.

Eggs that are carried by the portal circulation into the hepatic parenchyma can cause severe chronic inflammation in the liver. This immune response to *S. mansoni* and *S. japonicum* eggs is responsible for the most serious complication of schistosomiasis, liver fibrosis. The helper T-cell response in the early stage is dominated by Th1 cells that produce IFN- γ , which stimulates macrophages to secrete high levels of the cytokines TNF, IL-1, and IL-6 that cause fever. Chronic schistosomiasis is associated with a dominant Th2 response, associated with the presence of activated profibrotic macrophages. Both types of helper T cells contribute to the formation of granulomas surrounding eggs in the liver. Hepatic fibrosis is a serious manifestation of chronic schistosomiasis in which Th2 cells and activated macrophages may play the major role.

MORPHOLOGY

In early *S. mansoni* or *S. japonicum* infections, white, pinhead-sized granulomas are scattered throughout the gut and liver. At the center of the granuloma is the schistosome egg, which contains a miracidium; this degenerates over time and calcifies. The granulomas are composed of macrophages, lymphocytes, neutrophils, and eosinophils, which are distinctive for helminth infections (Fig. 8.60). The liver is darkened by regurgitated heme-derived pigments from the schistosome gut, which, like malaria pigments, are iron-free and accumulate in Kupffer cells and splenic macrophages.

In late *S. mansoni* or *S. japonicum* infections, inflammatory patches or pseudopolyps may form in the colon. The surface of the liver is bumpy, and cut surfaces reveal granulomas and widespread fibrosis and portal enlargement without intervening regenerative nodules. Because these fibrous tracts resem-

ble the stem of a clay pipe, the lesion is named **pipestem fibrosis** (Fig. 8.61). The fibrosis often obliterates the portal veins, leading to portal hypertension, severe congestive splenomegaly, esophageal varices, and ascites. Schistosome eggs, diverted to the lung through portal collaterals, may produce granulomatous pulmonary arteritis with intimal hyperplasia, progressive arterial obstruction, and ultimately heart failure (cor pulmonale). On histologic examination, arteries in the lungs show disruption of the elastic layer by granulomas and scars, luminal organizing thrombi, and angiomatoid lesions like those of idiopathic pulmonary hypertension (Chapter 15). Patients with hepatosplenic schistosomiasis also have an increased frequency of mesangioproliferative or membranous glomerulopathy (Chapter 20) in which glomeruli contain deposits of immunoglobulin and complement but rarely schistosome antigen.

In *S. haematobium* infection, inflammatory cystitis due to massive egg deposition and granulomas appears early, leading to mucosal erosions and hematuria. Later, the granulomas calcify and develop a sandy appearance, which, if severe, may line the wall of the bladder and cause a dense concentric rim (**calcified bladder**) on radiographic films. The most frequent complication of *S. haematobium* infection is inflammation and fibrosis of the ureteral walls, leading to obstruction, hydronephrosis, and chronic pyelonephritis. There is also an association between urinary schistosomiasis and squamous cell carcinoma of the bladder, typically preceded by the development of squamous metaplasia (Chapter 21).

Lymphatic Filariasis

Lymphatic filariasis is transmitted by mosquitoes and is caused by closely related nematodes, *Wuchereria bancrofti* (90% of total cases) and *Brugia* spp. (*B. malayi* or *B. timori*), which are responsible for 50 million infections worldwide. Mass drug administration programs have greatly reduced the number of cases in some nations. In endemic areas, which include parts of Latin America, sub-Saharan Africa, and Southeast Asia, filariasis causes a spectrum of diseases:

- Asymptomatic microfilaremia
- Recurrent lymphadenitis

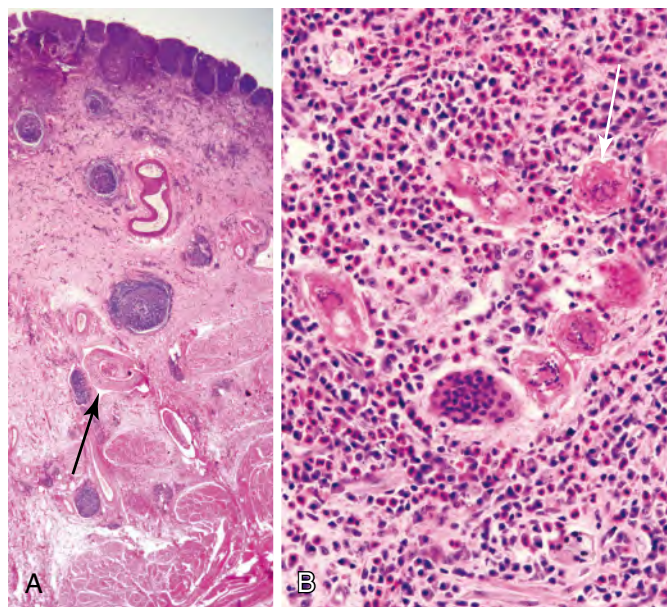


Fig. 8.60 (A) *Schistosoma haematobium* infection of the bladder showing dense fibrosis, scattered granulomas, and a cross-section of adult worms in a vessel (arrow). (B) High magnification demonstrates miracidium-containing eggs (arrow), prominent eosinophils, histiocytes, and giant cells.



Fig. 8.61 Pipestem fibrosis of the liver due to chronic *Schistosoma japonicum* infection.

- Chronic lymphadenitis with swelling of the dependent limb or scrotum (elephantiasis)
- Tropical pulmonary eosinophilia

Pathogenesis

Infective larvae released by mosquitoes into the tissues during a blood meal develop within lymphatic channels into adult males and females, which mate and release microfilariae that enter the bloodstream. Mosquitoes that bite infected individuals take up the microfilariae and can transmit the disease. The genomes of *W. bancrofti* and *B. malayi* have been sequenced, leading to the discovery of filarial molecules that may contribute to host invasion and enable these organisms to evade or inhibit immune defenses:

- *Elastases and trypsin-like proteases*, which facilitate invasion of host tissues
- *Several surface glycoproteins* with antioxidant function, which may protect the parasite from reactive oxygen species
- *Homologues of cystatins* (cysteine protease inhibitors), which can impair the MHC class II antigen-processing pathway
- *Serpins* (serine protease inhibitors), which can inhibit neutrophil proteases, critical inflammatory mediators
- *Homologues of host molecules*, such as TGF- β and macrophage migration inhibition factor, which could dampen the immune response

In addition, symbiotic *Wolbachia* bacteria infect filarial nematodes and contribute to pathogenesis of disease. *Wolbachia* spp. are required for nematode development and reproduction, and antibiotics that eradicate *Wolbachia* spp. impair nematode survival and fertility.

Immune responses to the filarial worms produce damage to the human host. As is the case with leprosy and leishmanial infections, some of the different disease manifestations caused by lymphatic filariae are likely related to variations in host T-cell responses to the parasites. In chronic lymphatic filariasis, damage to the lymphatics is caused directly by the adult parasites and by a Th1-mediated immune response, which stimulates the formation of granulomas around the adult parasites. Hypersensitivity to microfilaria in the lungs is thought to be associated with tropical pulmonary eosinophilia. IgE and eosinophils may be stimulated by IL-4 and IL-5, respectively, which are secreted by filaria-specific Th2 helper T cells.

MORPHOLOGY

Chronic filariasis is characterized by **persistent lymphedema** of the extremities, scrotum, penis, or vulva. Frequently there is hydrocele and lymph node enlargement. In severe and long-lasting infections, chylous weeping of the enlarged scrotum may ensue, or a chronically swollen leg may develop tough subcutaneous fibrosis and epithelial hyperkeratosis, termed **elephantiasis**. (Fig. 8.62) Elephantoid skin shows dilation of the dermal lymphatics, widespread lymphocytic infiltrates, and focal cholesterol deposits; the epidermis is thickened and hyperkeratotic. Adult filarial worms—live, dead, or calcified—are present in the draining lymphatics or lymph nodes, surrounded by (1) mild or no inflammation, (2) an intense eosinophilia with hemorrhage and fibrin (recurrent filarial funiculoepididymitis), or (3) granulomas. Over time, the dilated lymphatics develop polypoid infoldings. In the testis, hydrocele fluid, which often contains cholesterol crystals, red cells, and hemosiderin, induces thickening and calcification of the tunica vaginalis.

Lung involvement by microfilariae is marked by eosinophilia caused by Th2 responses and cytokine production (tropical eosinophilia) or by dead microfilariae surrounded by stellate, hyaline, eosinophilic precipitates embedded in small epithelioid granulomas (Meyers-Kouwenaar bodies). Typically, these patients lack other manifestations of filarial disease.

Onchocerciasis

***Onchocerca volvulus* is a filarial nematode that is the leading cause of preventable blindness in sub-Saharan Africa.** It is transmitted by black flies in Africa, South America, and Yemen. An aggressive campaign of mass ivermectin administration has dramatically reduced the incidence of *Onchocerca* spp. infections in Africa and South America. The WHO has declared that onchocerciasis has been eliminated in Colombia, Ecuador, Mexico, and Guatemala. Because the vector's habitat is near fast-moving water, there is higher incidence of human disease near rivers, accounting for the name *river blindness* given to this disease.

The disease attributable to onchocerciasis is primarily due to inflammation induced by microfilaria. Adult *O. volvulus* parasites mate in the dermis, where they are surrounded by a mixed infiltrate of host cells that produces a characteristic subcutaneous nodule (*onchocercoma*). Inseminated females produce microfilariae, which accumulate in the



Fig. 8.62 Massive edema and elephantiasis caused by filariasis of the leg. (Courtesy Dr. Willy Piessens, Harvard School of Public Health, Boston.)

skin and disseminate to the eye chambers. Ivermectin kills only immature worms, not adult worms, so parasites can repopulate the host, requiring retreatment every 3 to 6 months until symptoms are gone.

MORPHOLOGY

O. volvulus causes chronic, itchy dermatitis with focal darkening or loss of pigment and scaling, referred to as leopard, lizard, or elephant skin. Foci of epidermal atrophy and elastic fiber breakdown may alternate with areas of hyperkeratosis, hyperpigmentation with pigment incontinence, dermal atrophy, and fibrosis. The subcutaneous onchocercoma is composed of a fibrous capsule surrounding adult worms and a mixed chronic inflammatory infiltrate that includes fibrin, neutrophils, eosinophils, lymphocytes, and giant cells (Fig. 8.63). The progressive eye lesions begin with punctate keratitis along with small, fluffy opacities of the cornea caused by degenerating microfilariae, which evoke an eosinophilic infiltrate. This is followed by a sclerosing keratitis that opacifies the cornea, beginning at the scleral limbus. Keratitis is sometimes accentuated by treatment with antifilarial drugs (**Mazzotti reaction**). Microfilariae in the anterior chamber cause inflammation of the anterior chamber of the eye and ciliary body (iridocyclitis) and glaucoma, whereas involvement of the choroid and retina results in atrophy and loss of vision.

AGENTS OF BIOTERRORISM

Bioterrorism is the use of biologic or chemical agents as weapons, and microorganisms are classified based on an assessment of which pose the greatest danger. The CDC classifies potential agents of bioterrorism based on the risk they present and whether they can be easily disseminated.

- Category A agents pose the highest risk and can be readily disseminated or transmitted from person-to-person, can cause high mortality, might cause public panic, and might require public health preparedness. For example, smallpox is a category A agent because of its high transmissibility, case mortality rate of 30% or greater, and lack of effective antiviral therapy. Because vaccination ended in the United States in 1972 and immunity has waned, the population is highly susceptible to smallpox. Concern that smallpox could be used for bioterrorism has led to a return of vaccination for selected groups. Other category A agents include *B. anthracis*, *Yersinia pestis*, and viral hemorrhagic fever viruses.
- Category B agents are moderately easy to disseminate, produce moderate morbidity but low mortality, and require specific diagnostic and disease surveillance. Some of these agents are food-borne or water-borne. Examples include *Brucella* spp. and *V. cholerae*.
- Category C agents include emerging pathogens that could be engineered for mass dissemination because of availability, ease of

production and dissemination, potential for high morbidity and mortality, and great impact on health. Examples include hantavirus and Nipah virus.

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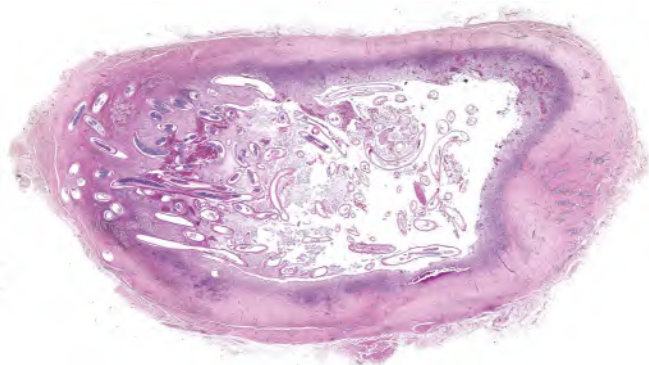


Fig. 8.63 Microfilaria-laden gravid female of *Onchocerca volvulus* in a subcutaneous fibrous nodule.